

## Every variable, in its own units

The three reports use these variables as model inputs, coefficients and residuals. This appendix shows what each one actually *is*: the real value, every year, for Greece, for the EU comparator, and for each of the other 26 member states. The country set is the 27 current EU members — the same panel the models are estimated on. Eurostat publishes many of these series for candidate and EFTA countries as well, and for the euro-area aggregates; those are excluded here so that what you see is the country set the analysis actually used.

**How this is organised.** Eight sections, one per stage of the technical report's own argument, in report order — not by subject, and not with the report's figures in one place and the evidence behind them in another. Open Stage 4 here and you find the report's own current-condition figures, the views it simplified out of them, the diagnostics that qualify them and the context the report discusses alongside them, together, followed by the underlying variables behind all of it. Inside each stage the order is always the same: the report's own figures first, then views it simplified away, then what it stated in prose rather than charted, then the technical diagnostics, then contextual evidence the report cannot use as a headline claim, and finally — **collapsed by default** — the underlying variables and tables in their own units. Open a group to read it, or follow a link straight to a chart and its group opens for you. Printing expands everything. Source and coverage notes, and the searchable glossary, follow the eighth stage. Charts marked **RELATIONSHIP** are cross-country scatters placed in the stage whose claim they illustrate. Version 2 adds the final fixed and nested model results, the complete multiple-testing screen, conditional diagnostics and country-level residual comparisons; detailed candidate trajectories are collapsed by default so the central evidence remains readable.

**How to read the charts.** Each line chart draws every EU member state as a faint grey line, with **Greece** in blue and the **EU comparator** as an orange dashed line. Hover anywhere to read the year, Greece's value, the EU value, and the value of whichever country's line your cursor is nearest — that line lights up in green. The small table under each chart gives the same figures at fixed years, so the numbers are readable without a mouse. On scatters, the dashed orange lines mark the EU value on each axis, so the plot divides into quadrants relative to the EU.

**Two cautions.** The EU line is Eurostat's own population-weighted EU27 aggregate where one is published, and an unweighted mean of member states where it is not; the difference is stated under every chart, and the two are not interchangeable. And a country line stopping early means that country stopped reporting, not that its value fell to zero.

- Stage 1 — The puzzle    Stage 2 — A broader measure, and a moving line
- Stage 3 — Is the hardship real?    Stage 4 — What current conditions explain
- Stage 5 — What accumulated history adds    Stage 6 — How much depends on the model
- Stage 7 — What this is not    Stage 8 — What the evidence supports    Source and coverage notes
- § Abbreviations

Stage 1

The puzzle

Are the two measures really describing different things, and is Greece unusual or merely extreme?

Greece reports far more hardship than countries with similar income poverty

**DESCRIPTIVE** Does official income poverty account for the hardship Greek households report?

**Read with this.** Descriptive comparison of two official measures. Reported hardship is answered by HOUSEHOLDS and income poverty counts PEOPLE, so the axis is labelled percent rather than either. In the second view the fitted line excludes Greece, describing the European pattern Greece is being judged against rather than one Greece helped set. The labelled point marks the median country on each measure taken SEPARATELY, so it is a reference point rather than an actual country: no member state necessarily sits there. Both views are country-level and say nothing about any individual household.

| Series                       | How Greece's hardship gap developed |      |      |      |      |      |      |      |      |      |
|------------------------------|-------------------------------------|------|------|------|------|------|------|------|------|------|
|                              | 2015                                | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: reported hardship    | 77.7                                | 76.8 | 77.2 | 74.1 | 71.0 | 69.8 | 71.0 | 68.4 | 67.0 | 66.7 |
| EU median: reported hardship | 28.9                                | 25.9 | 22.6 | 20.5 | 18.5 | 16.8 | 15.4 | 17.9 | 16.6 | 17.1 |
| Greece: income poverty       | 21.4                                | 21.2 | 20.2 | 18.5 | 17.9 | 17.7 | 19.6 | 18.8 | 18.9 | 19.6 |
| EU median: income poverty    | 16.3                                | 16.5 | 15.9 | 16.4 | 15.4 | 16.1 | 15.7 | 15.6 | 15.4 | 15.5 |

| Where countries stood in 2024 |                    |                       |
|-------------------------------|--------------------|-----------------------|
| Series                        | Income poverty (%) | Reported hardship (%) |
| Czechia                       | 9.5                | 14.2                  |
| Belgium                       | 11.4               | 17.1                  |
| Denmark                       | 11.6               | 12.8                  |
| Netherlands                   | 12.1               | 7.3                   |
| Finland                       | 12.6               | 8.9                   |
| Ireland                       | 12.7               | 16.9                  |
| Slovenia                      | 13.2               | 13.6                  |
| Poland                        | 13.8               | 11.5                  |
| Austria                       | 14.3               | 13.1                  |
| Hungary                       | 14.3               | 19.8                  |
| Slovakia                      | 14.5               | 28.7                  |
| Cyprus                        | 14.6               | 20.8                  |
| Sweden                        | 14.8               | 9.9                   |
| Germany                       | 15.5               | 7.3                   |
| France                        | 15.9               | 21.8                  |
| Portugal                      | 16.6               | 21.8                  |
| Malta                         | 16.9               | 17.3                  |
| Luxembourg                    | 18.1               | 8.5                   |
| Italy                         | 18.9               | 18.7                  |
| Romania                       | 19.0               | 23.8                  |
| Greece                        | 19.6               | 66.7                  |
| Spain                         | 19.7               | 21.9                  |
| Estonia                       | 20.2               | 9.9                   |
| Croatia                       | 20.3               | 19.8                  |
| Lithuania                     | 21.5               | 11.4                  |
| Latvia                        | 21.6               | 23.3                  |
| Bulgaria                      | 21.7               | 37.4                  |

Described in the report's prose rather than charted there.

## APPENDIX FIGURE

**In 2024 Greece stands far above the rest of the EU on reported hardship**

## DESCRIPTIVE

Is Greece unusual, or at one end of a continuum?

| Country     | Reported hardship (%) |
|-------------|-----------------------|
| Greece      | 66.7                  |
| Bulgaria    | 37.4                  |
| Slovakia    | 28.7                  |
| Romania     | 23.8                  |
| Latvia      | 23.3                  |
| Spain       | 21.9                  |
| France      | 21.8                  |
| Portugal    | 21.8                  |
| Cyprus      | 20.8                  |
| Croatia     | 19.8                  |
| Hungary     | 19.8                  |
| Italy       | 18.7                  |
| Malta       | 17.3                  |
| Belgium     | 17.1                  |
| Ireland     | 16.9                  |
| Czechia     | 14.2                  |
| Slovenia    | 13.6                  |
| Austria     | 13.1                  |
| Denmark     | 12.8                  |
| Poland      | 11.5                  |
| Lithuania   | 11.4                  |
| Estonia     | 9.9                   |
| Sweden      | 9.9                   |
| Finland     | 8.9                   |
| Luxembourg  | 8.5                   |
| Germany     | 7.3                   |
| Netherlands | 7.3                   |

Technical diagnostics: checks that qualify a result rather than establishing one.

APPENDIX FIGURE

Where every country stood, year by year

APPENDIX

**Read with this.** The report carries the latest year only. The axes here are fixed across every year, so switching year moves the countries and not the scale. The fitted line excludes Greece in each year.

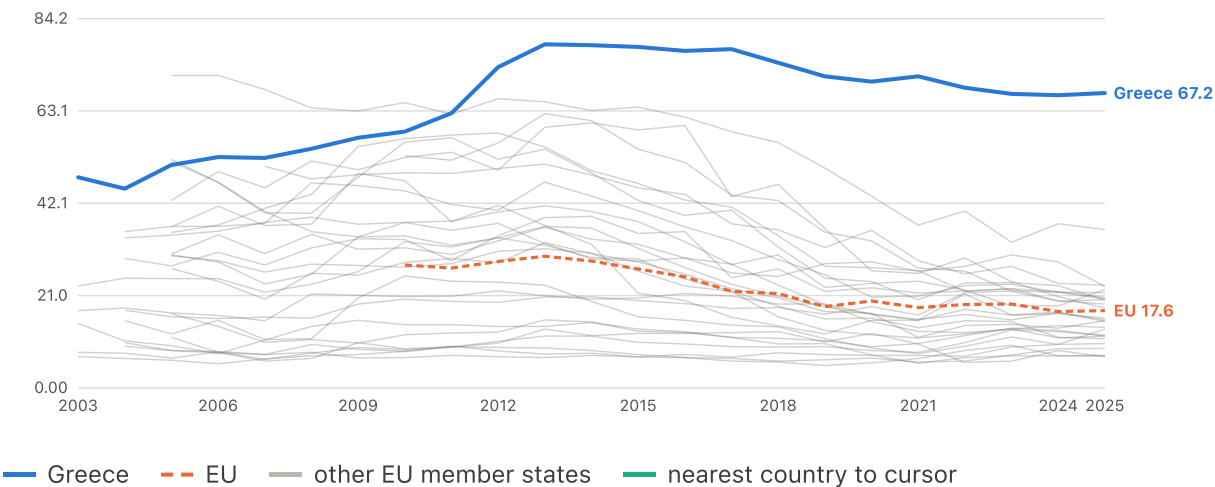
| Year | Greece: income poverty | Greece: reported hardship | Median country: income poverty | Median country: reported hardship |
|------|------------------------|---------------------------|--------------------------------|-----------------------------------|
| 2015 | 21.4                   | 77.7                      | 16.3                           | 28.9                              |
| 2016 | 21.2                   | 76.8                      | 16.5                           | 25.9                              |
| 2017 | 20.2                   | 77.2                      | 15.9                           | 22.6                              |
| 2018 | 18.5                   | 74.1                      | 16.4                           | 20.5                              |
| 2019 | 17.9                   | 71.0                      | 15.4                           | 18.5                              |
| 2020 | 17.7                   | 69.8                      | 16.1                           | 16.8                              |
| 2021 | 19.6                   | 71.0                      | 15.7                           | 15.4                              |
| 2022 | 18.8                   | 68.4                      | 15.6                           | 17.9                              |
| 2023 | 18.9                   | 67.0                      | 15.4                           | 16.6                              |
| 2024 | 19.6                   | 66.7                      | 15.5                           | 17.1                              |

Underlying variables and tables

4 charts

Subjective poverty (difficulty making ends meet)

% of households · up to 27 EU member states · 2003–2025



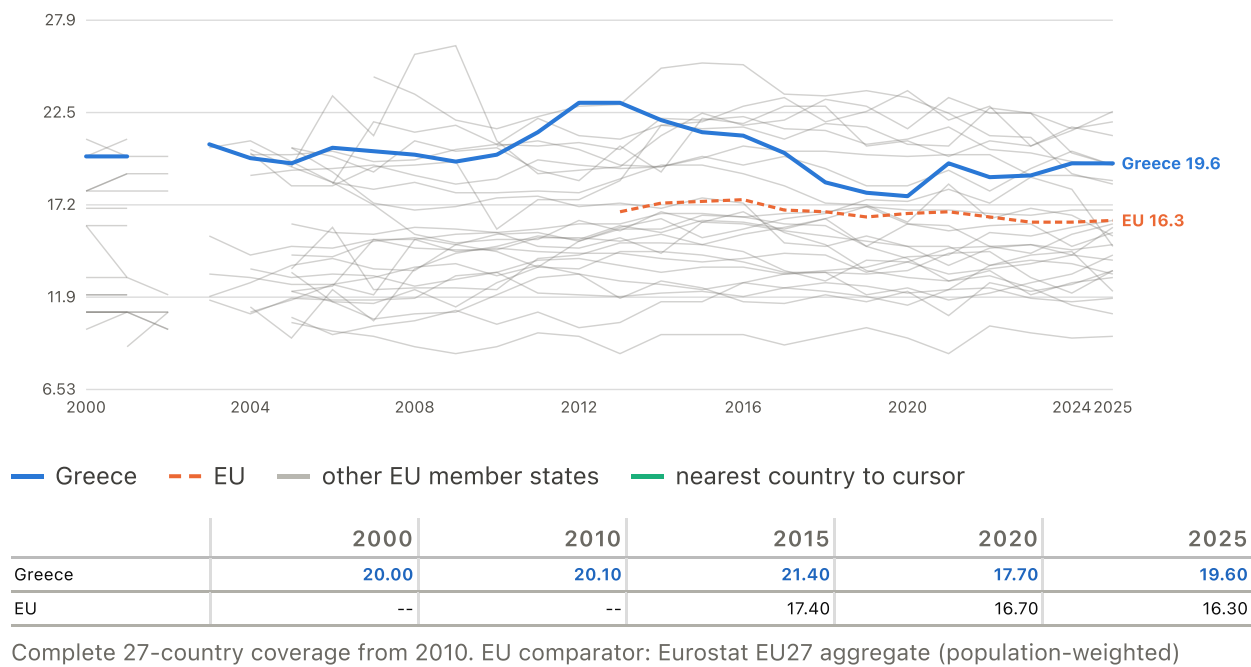
|        | 2003  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 48.00 | 58.40 | 77.70 | 69.80 | 67.20 |
| EU     | --    | 28.00 | 27.10 | 19.80 | 17.60 |

Sum of 'with difficulty' and 'with great difficulty'.

Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

AROP income poverty

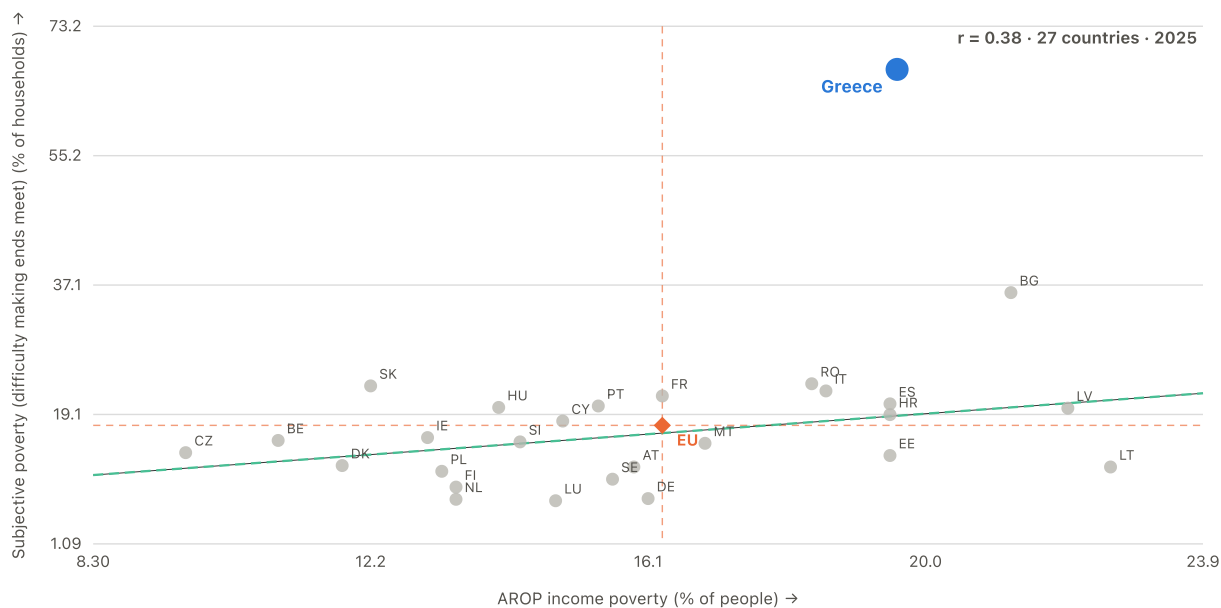
% of people · up to 27 EU member states · 2000–2025



RELATIONSHIP

Official income poverty against reported hardship

2025 · 27 member states ·  $r = +0.38$



The project's central paradox in one frame. Most countries cluster along a loose upward band; Greece sits far above it, reporting hardship no other country reports at a similar income-poverty rate.

EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

How wide the disadvantage is — descriptive, not a model input

The same 16 indicators in 2008 and 2024

position in the EU distribution

Every indicator with a valid EU position in both 2008 and 2024, selected without reference to whether it improved or deteriorated. Unemployment, youth unemployment and the employment rate are absent because comparable EU coverage for them begins in 2009; hours worked, pay per hour, the work-effort squeeze and real wages are all present, so the basket is not free of labour-market information.

Stage 2

A broader measure, and a moving line

Does the EU's broader poverty measure close the gap, and did the yardstick itself move?

Who counts as poor barely moved; what the poverty line buys fell by a fifth

DESCRIPTIVE

What happened to the line Greek poverty is measured against, and what does a fixed line show instead?

**Read with this.** The anchored series is an approximation built in this project and is labelled as such wherever it appears. The first view counts PEOPLE, the second counts EUROS. In the second, both lines are the same official threshold: one as published in each year's own money, the other converted into what it could buy in 2008. The cash line recovers and the purchasing-power line does not, and that difference is why the first view's two counts diverge. Both views are Greece only and support no cross-country statement.

| Who falls below a fixed line            |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
|-----------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|--|
| Series                                  | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |  |
| Greece: fixed 2008 threshold            | 25.6 | 22.8 | 20.7 | 21.5 | 21.2 | 19.7 | 17.0 | 18.0 | 24.6 | 33.3 | 39.7 | 40.6 | 39.8 | 39.8 | 38.3 | 37.3 | 34.6 | 30.8 | 32.7 | 33.6 | 32.0 | 30.2 | 28.5 |  |
| Greece: current-year threshold          | 20.7 | 19.9 | 19.6 | 20.5 | 20.3 | 20.1 | 19.7 | 20.1 | 21.4 | 23.1 | 23.1 | 22.1 | 21.4 | 21.2 | 20.2 | 18.5 | 17.9 | 17.7 | 19.6 | 18.8 | 18.9 | 19.6 | 19.6 |  |
| What the line itself is worth           |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |  |
| Series                                  | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |  |
| Threshold in cash terms                 | 4923 | 5306 | 5650 | 5910 | 6120 | 6480 | 6897 | 7178 | 6591 | 5708 | 5023 | 4608 | 4512 | 4500 | 4560 | 4718 | 4917 | 5269 | 5251 | 5712 | 6030 | 6510 | 7020 |  |
| Same threshold in 2008 purchasing power | 5819 | 6089 | 6264 | 6343 | 6377 | 6480 | 6808 | 6768 | 6027 | 5168 | 4589 | 4270 | 4228 | 4216 | 4226 | 4338 | 4498 | 4884 | 4838 | 4815 | 4878 | 5113 | 5358 |  |

AROPE sits between reported hardship and income poverty, and closes only about a fifth of the distance

**DESCRIPTIVE** What does AROPE add to AROP, and is that addition growing?

**Read with this.** AROPE components overlap and MAY NOT be added together: it is a union of income poverty, deprivation and low work intensity, and aggregate data do not reveal the overlaps The three measures are shown as they are reported, in shares of households, so the distance between them can be read directly. That distance is plotted on its own in the statistical appendix, where the points AROPE closes fall from 11.0 in 2015 to 7.3 in 2024.

| Series            | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|-------------------|------|------|------|------|------|------|------|------|------|------|
| Reported hardship | 77.7 | 76.8 | 77.2 | 74.1 | 71.0 | 69.8 | 71.0 | 68.4 | 67.0 | 66.7 |
| AROPE             | 32.4 | 32.6 | 32.2 | 30.3 | 29.0 | 27.4 | 28.3 | 26.3 | 26.1 | 26.9 |
| Income poverty    | 21.4 | 21.2 | 20.2 | 18.5 | 17.9 | 17.7 | 19.6 | 18.8 | 18.9 | 19.6 |

Where AROPE sits against the headline measures, what it's made of, and which groups moved

**DESCRIPTIVE**

Where does AROPE sit against income poverty and reported hardship, what is AROPE made of, and did every age and sex group move the same way?

**Read with this.** These are changes in group-level rates, not evidence about the same individuals over time. The 2024-2025 national increase was driven primarily by within-group changes, especially deterioration among people aged 65+, rather than population ageing. AROPE components are a UNION and may not be summed. Very low work intensity is part of AROPE, but the project does not hold a comparable national-total series for the Components view; its available age coverage uses a different population base, and aggregate component rates cannot reconstruct the AROPE union in any case. In all four views colour marks the measure or group and dash marks the country, so each Greek line is paired with the EU median for the same thing in the same colour. The per-component views with every other member state drawn behind are in the statistical appendix.

| Headline measures            |      |      |      |      |      |      |      |      |      |      |
|------------------------------|------|------|------|------|------|------|------|------|------|------|
| Series                       | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Income poverty: Greece       | 21.4 | 21.2 | 20.2 | 18.5 | 17.9 | 17.7 | 19.6 | 18.8 | 18.9 | 19.6 |
| Income poverty: EU median    | 16.3 | 16.5 | 15.9 | 16.4 | 15.4 | 16.1 | 15.7 | 15.6 | 15.4 | 15.5 |
| AROPE: Greece                | 32.4 | 32.6 | 32.2 | 30.3 | 29.0 | 27.4 | 28.3 | 26.3 | 26.1 | 26.9 |
| AROPE: EU median             | 22.5 | 22.2 | 21.4 | 20.1 | 20.1 | 19.9 | 19.6 | 19.6 | 19.8 | 19.6 |
| Reported hardship: Greece    | 77.7 | 76.8 | 77.2 | 74.1 | 71.0 | 69.8 | 71.0 | 68.4 | 67.0 | 66.7 |
| Reported hardship: EU median | 28.9 | 25.9 | 22.6 | 20.5 | 18.5 | 16.8 | 15.4 | 17.9 | 16.6 | 17.1 |



| Series                          | Components |      |      |      |      |      |      |      |      |      |      |
|---------------------------------|------------|------|------|------|------|------|------|------|------|------|------|
|                                 | 2015       | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |      |
| Income poverty: Greece          | 21.4       | 21.2 | 20.2 | 18.5 | 17.9 | 17.7 | 19.6 | 18.8 | 18.9 | 19.6 |      |
| Income poverty: EU median       | 16.3       | 16.5 | 15.9 | 16.4 | 15.4 | 16.1 | 15.7 | 15.6 | 15.4 | 15.5 |      |
| Material deprivation: Greece    | 17.6       | 18.4 | 18.3 | 16.1 | 15.8 | 14.9 | 13.9 | 13.9 | 13.5 | 14.0 |      |
| Material deprivation: EU median | 7.8        | 6.7  | 6.3  | 5.4  | 5.0  | 4.5  | 4.8  | 4.9  | 4.9  | 4.3  |      |
| AROPE: Greece                   | 32.4       | 32.6 | 32.2 | 30.3 | 29.0 | 27.4 | 28.3 | 26.3 | 26.1 | 26.9 |      |
| AROPE: EU median                | 22.5       | 22.2 | 21.4 | 20.1 | 20.1 | 19.9 | 19.6 | 19.6 | 19.8 | 19.6 |      |
| By age                          |            |      |      |      |      |      |      |      |      |      |      |
| Series                          | 2015       | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
| All ages                        | 32.4       | 32.6 | 32.2 | 30.3 | 29.0 | 27.4 | 28.3 | 26.3 | 26.1 | 26.9 | 27.5 |
| Under 18                        | 37.7       | 37.2 | 36.5 | 34.1 | 31.2 | 30.8 | 32.0 | 28.1 | 28.1 | 27.9 | 29.6 |
| 18-24                           | 43.6       | 44.6 | 42.1 | 39.0 | 37.5 | 35.0 | 37.5 | 33.4 | 33.1 | 34.5 | 33.2 |
| 25-49                           | 35.0       | 36.1 | 35.2 | 31.9 | 30.1 | 28.3 | 29.3 | 26.9 | 24.9 | 25.4 | 25.1 |
| 50-64                           | 34.8       | 35.3 | 34.9 | 33.4 | 31.9 | 29.2 | 30.5 | 27.6 | 26.3 | 28.2 | 27.5 |
| 65 and over                     | 17.5       | 16.6 | 18.4 | 19.2 | 20.5 | 19.4 | 19.3 | 21.0 | 23.9 | 24.9 | 27.7 |
| EU: all ages                    | 24.0       | 23.7 | 22.4 | 21.7 | 21.1 | 21.5 | 21.7 | 21.6 | 21.3 | 21.0 | 20.9 |
| EU: under 18                    | 27.4       | 27.1 | 25.1 | 23.9 | 22.8 | 24.1 | 24.5 | 24.8 | 24.9 | 24.2 | 24.3 |
| EU: 18-24                       | 29.7       | 29.8 | 28.1 | 27.2 | 26.6 | 27.8 | 27.3 | 26.5 | 26.0 | 26.3 | 26.3 |
| EU: 25-49                       | 23.3       | 22.8 | 21.3 | 20.3 | 19.3 | 19.6 | 20.2 | 20.0 | 19.6 | 19.2 | 19.1 |
| EU: 50-64                       | 25.6       | 24.4 | 23.3 | 22.4 | 22.1 | 21.5 | 21.9 | 21.0 | 20.8 | 20.7 | 20.8 |
| EU: 65 and over                 | 18.0       | 18.5 | 18.5 | 19.1 | 19.3 | 20.0 | 19.5 | 20.1 | 19.7 | 19.2 | 18.8 |
| By sex                          |            |      |      |      |      |      |      |      |      |      |      |
| Series                          | 2015       | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
| All                             | 32.4       | 32.6 | 32.2 | 30.3 | 29.0 | 27.4 | 28.3 | 26.3 | 26.1 | 26.9 | 27.5 |
| Women                           | 33.1       | 33.4 | 32.9 | 31.3 | 30.1 | 28.6 | 29.2 | 27.4 | 27.3 | 27.9 | 29.1 |
| Men                             | 31.6       | 31.7 | 31.6 | 29.2 | 28.0 | 26.1 | 27.3 | 25.2 | 24.8 | 25.9 | 25.9 |
| EU: all                         | 24.0       | 23.7 | 22.4 | 21.7 | 21.1 | 21.5 | 21.7 | 21.6 | 21.3 | 21.0 | 20.9 |
| EU: women                       | 24.9       | 24.7 | 23.4 | 22.7 | 22.1 | 22.5 | 22.6 | 22.7 | 22.3 | 21.9 | 21.9 |
| EU: men                         | 23.1       | 22.5 | 21.4 | 20.6 | 20.0 | 20.4 | 20.7 | 20.4 | 20.3 | 20.0 | 19.8 |

## On the same 16 indicators, Greece went from the EU's worst fifth on 4 in 2008 to 11 in 2024

**DESCRIPTIVE** Did Greek disadvantage deepen on a few measures, or spread across many?

**Read with this.** Descriptive summary of the condition, not a predictor of it. DESCRIPTIVE ONLY. Breadth was tested as a predictor of reported hardship in P3a and does not survive: on its own it is not significant ( $p = 0.12$ ), and once the other accumulated measures enter the model its coefficient reverses sign. It summarises the condition rather than explaining it. THE BASKET IS FIXED: these are the 16 indicators with a valid EU position in both 2008 and 2024, chosen without reference to whether they improved or deteriorated, so the same things are counted at both ends and in every year between. Unemployment, youth unemployment and the employment rate are absent because comparable EU coverage for them begins in 2009; hours worked, pay per hour, the work-effort squeeze and real wages are all present, so this is not a basket without labour-market information. The outcome and every model covariate are excluded. THE EU-COUNTRY MEDIAN is the median of the 27 member states' own shares on this basket - a median across countries, not Eurostat's population-weighted EU aggregate -- named accordingly rather than simply 'EU'. Croatia never reaches full 16-indicator coverage in any basket year and does not appear as a line for that reason. The 'Which measures' view's axis is POSITION, not value: the indicators

have no common unit, so 0 is the best place in the Union to be on that indicator and 100 the worst, whichever direction 'worse' runs in.

Of 16 indicators, **4 were already in the EU's worst fifth in 2008, 7 entered it by 2024, and none left.**

|                                          |               | Which measures |      |      |      |      |               |      |      |      |      |      |                     |      |      |      |      |  |
|------------------------------------------|---------------|----------------|------|------|------|------|---------------|------|------|------|------|------|---------------------|------|------|------|------|--|
| Series                                   | 2008 position |                |      |      |      |      | 2024 position |      |      |      |      |      | Transition          |      |      |      |      |  |
| Hours worked against hourly pay          | 59.3          |                |      |      |      |      | 100.0         |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| Prices measured against wages            | 48.1          |                |      |      |      |      | 100.0         |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| Pay per hour worked                      | 55.6          |                |      |      |      |      | 100.0         |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| Household income after inflation         | 51.9          |                |      |      |      |      | 100.0         |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| Wages after inflation                    | 51.9          |                |      |      |      |      | 100.0         |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| The poverty line after inflation         | 51.9          |                |      |      |      |      | 100.0         |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| Keeping the home warm                    | 73.1          |                |      |      |      |      | 98.1          |      |      |      |      |      | entered worst fifth |      |      |      |      |  |
| Hours worked each week                   | 94.4          |                |      |      |      |      | 100.0         |      |      |      |      |      | already worst fifth |      |      |      |      |  |
| What households expect of the year ahead | 100.0         |                |      |      |      |      | 100.0         |      |      |      |      |      | already worst fifth |      |      |      |      |  |
| Citizens leaving the country             | 86.4          |                |      |      |      |      | 96.2          |      |      |      |      |      | already worst fifth |      |      |      |      |  |
| Income inequality                        | 80.8          |                |      |      |      |      | 81.5          |      |      |      |      |      | already worst fifth |      |      |      |      |  |
| Food prices                              | 14.8          |                |      |      |      |      | 77.8          |      |      |      |      |      | outside both years  |      |      |      |      |  |
| Economic output per person               | 55.6          |                |      |      |      |      | 77.8          |      |      |      |      |      | outside both years  |      |      |      |      |  |
| Prices overall                           | 53.7          |                |      |      |      |      | 70.4          |      |      |      |      |      | outside both years  |      |      |      |      |  |
| Housing and energy prices                | 63.0          |                |      |      |      |      | 59.3          |      |      |      |      |      | outside both years  |      |      |      |      |  |
| What households actually spend           | 48.1          |                |      |      |      |      | 48.1          |      |      |      |      |      | outside both years  |      |      |      |      |  |
| How many measures                        |               |                |      |      |      |      |               |      |      |      |      |      |                     |      |      |      |      |  |
| Series                                   | 2008          | 2009           | 2010 | 2011 | 2012 | 2013 | 2014          | 2015 | 2016 | 2017 | 2018 | 2019 | 2020                | 2021 | 2022 | 2023 | 2024 |  |
| Greece                                   | 25.0          | 18.8           | 43.8 | 50.0 | 50.0 | 50.0 | 43.8          | 50.0 | 50.0 | 62.5 | 56.2 | 56.2 | 56.2                | 56.2 | 56.2 | 62.5 | 68.8 |  |
| EU-country median                        | 6.2           | 12.5           | 9.4  | 12.5 | 18.8 | 18.8 | 15.6          | 12.5 | 18.8 | 18.8 | 12.5 | 12.5 | 18.8                | 12.5 | 12.5 | 18.8 | 12.5 |  |

Views the report carried and then simplified away -- removed to leave one question per figure, not because the numbers stopped mattering.

## SIMPLIFIED OUT OF THE REPORT

## What AROPE actually closes, and what stays open

## APPENDIX

How much of the distance does the broader measure actually close, and is that share growing?

**Read with this.** This is the distance the report's AROPE figure shows as three levels, plotted as a subtraction. It is the same quantity seen a second way, not a second result: the points AROPE closes fall from 11.0 in 2015 to 7.3 in 2024 while the gap it leaves open grows.

| Series                     | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|
| Points closed by AROPE     | 11.0 | 11.4 | 12.0 | 11.8 | 11.1 | 9.7  | 8.7  | 7.5  | 7.2  | 7.3  |
| Gap still open after AROPE | 45.3 | 44.2 | 45.0 | 43.8 | 42.0 | 42.4 | 42.7 | 42.1 | 40.9 | 39.8 |

## SIMPLIFIED OUT OF THE REPORT

## Each AROPE component on its own, against every member state

## APPENDIX

Where does Greece sit against every other member state on each AROPE component separately?

**Read with this.** The report merges these three into one view so the components can be compared directly; that merge cannot carry the faint per-country layer, which is what these keep. Very low work intensity is missing for the reason the report's figure states, and aggregate component rates cannot reconstruct the AROPE union in any case.

| Income poverty            |      |      |      |      |      |      |      |      |      |      |
|---------------------------|------|------|------|------|------|------|------|------|------|------|
| Series                    | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: income poverty    | 21.4 | 21.2 | 20.2 | 18.5 | 17.9 | 17.7 | 19.6 | 18.8 | 18.9 | 19.6 |
| EU median: income poverty | 16.3 | 16.5 | 15.9 | 16.4 | 15.4 | 16.1 | 15.7 | 15.6 | 15.4 | 15.5 |

| Material deprivation            |      |      |      |      |      |      |      |      |      |      |
|---------------------------------|------|------|------|------|------|------|------|------|------|------|
| Series                          | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: material deprivation    | 17.6 | 18.4 | 18.3 | 16.1 | 15.8 | 14.9 | 13.9 | 13.9 | 13.5 | 14.0 |
| EU median: material deprivation | 7.8  | 6.7  | 6.3  | 5.4  | 5.0  | 4.5  | 4.8  | 4.9  | 4.9  | 4.3  |

| AROPE            |      |      |      |      |      |      |      |      |      |      |
|------------------|------|------|------|------|------|------|------|------|------|------|
| Series           | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: AROPE    | 32.4 | 32.6 | 32.2 | 30.3 | 29.0 | 27.4 | 28.3 | 26.3 | 26.1 | 26.9 |
| EU median: AROPE | 22.5 | 22.2 | 21.4 | 20.1 | 20.1 | 19.9 | 19.6 | 19.6 | 19.8 | 19.6 |

Described in the report's prose rather than charted there.

## APPENDIX FIGURE

## The most recent rise came from rates within age groups, not from the changing size of those groups

## DESCRIPTIVE

What drove the most recent rise in AROPE: rates within age groups, or the changing size of those groups?

**Read with this.** These are exact decomposition terms, not estimates: they carry no uncertainty and no interval is drawn. Within-group means the rate changed inside an age group; composition means the size of the group changed.

| Age group   | Within-group (pp) | Composition (pp) |
|-------------|-------------------|------------------|
| Under 18    | 0.283             | -0.065           |
| 18-24       | -0.088            | 0.080            |
| 25-49       | -0.094            | -0.154           |
| 50-64       | -0.154            | 0.050            |
| 65 and over | 0.651             | 0.107            |

Evidence that cannot support a headline claim by design, and is recorded in the report's context register.

## DESCRIPTIVE CORROBORATION

### Anchored poverty, corroborated by independent microdata

Same paper cited above for the crisis and adjustment-policy background, read here for a different figure. Andriopoulou, Kanavitsa & Tsakloglou (2020) construct an anchored FGT0 poverty rate on a 2007 base from EU-SILC/ELSTAT microdata and reach 48% at the 2013 income-year peak. This project's own anchored series, built independently on a 2008 base from country-level EU-SILC releases rather than household microdata, puts the comparable income year at 40.6%. No test links the two figures; the comparison is descriptive only.

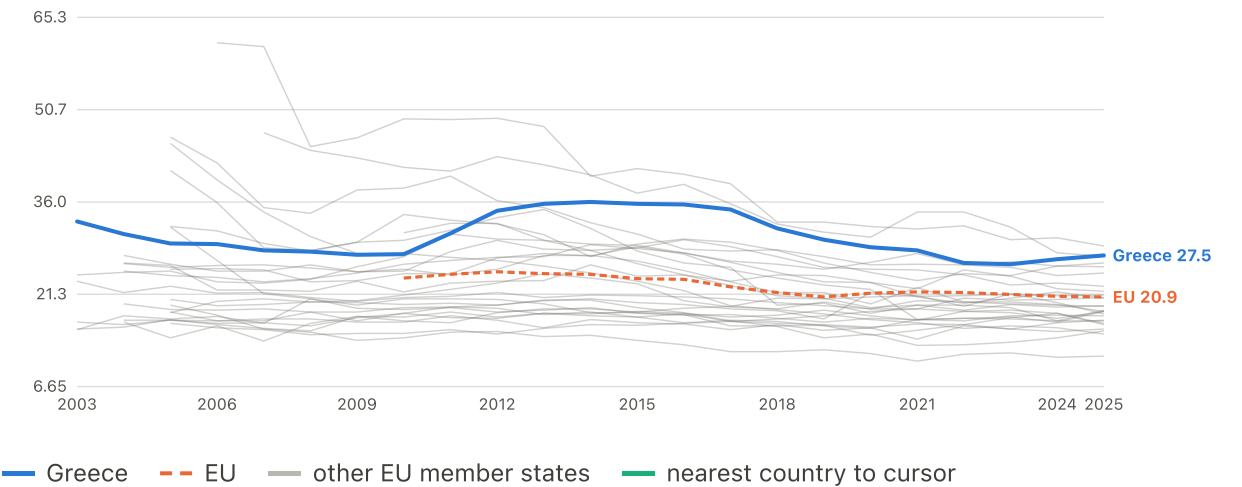
**What may be concluded.** Independent microdata evidence points in the same direction: a fixed pre-crisis poverty line shows much larger crisis-era poverty than contemporaneous AROP.

**Limitation.** Treating 48% against 40.6% as a direct validation test, a bias estimate, or proof that this project's reconstruction is too low by seven to eight points -- the two differ in data, anchor year, price adjustment and vintage.

**Source.** Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. EU-SILC microdata via ELSTAT, analysed independently of this project. Anchored FGT0 poverty on a 2007 base reaches 48% at the 2013 income-year peak, against this project's own 40.6% for the same income year. source

AROPE (poverty or social exclusion)

% of people · up to 27 EU member states · 2003–2025

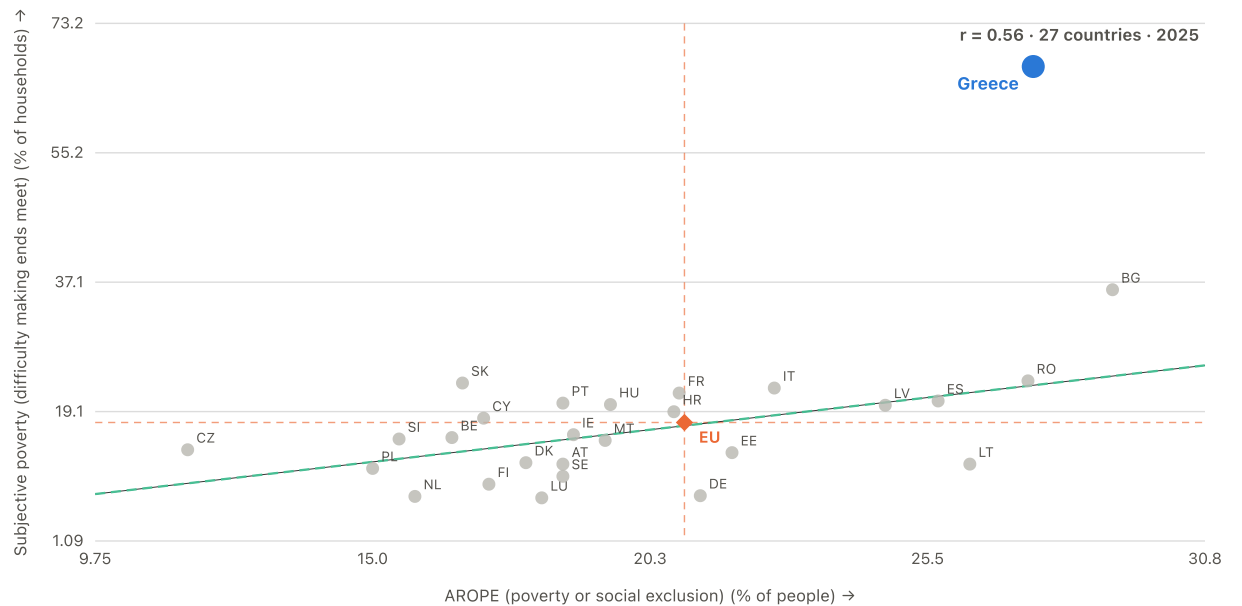


|        | 2003  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 32.90 | 27.70 | 35.70 | 28.80 | 27.50 |
| EU     | --    | 23.90 | 23.80 | 21.50 | 20.90 |

Legacy ilc\_peps01 to 2020 spliced with revised ilc\_peps01n from 2021.  
Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

RELATIONSHIP AROPE against reported hardship

2025 · 27 member states ·  $r = +0.56$

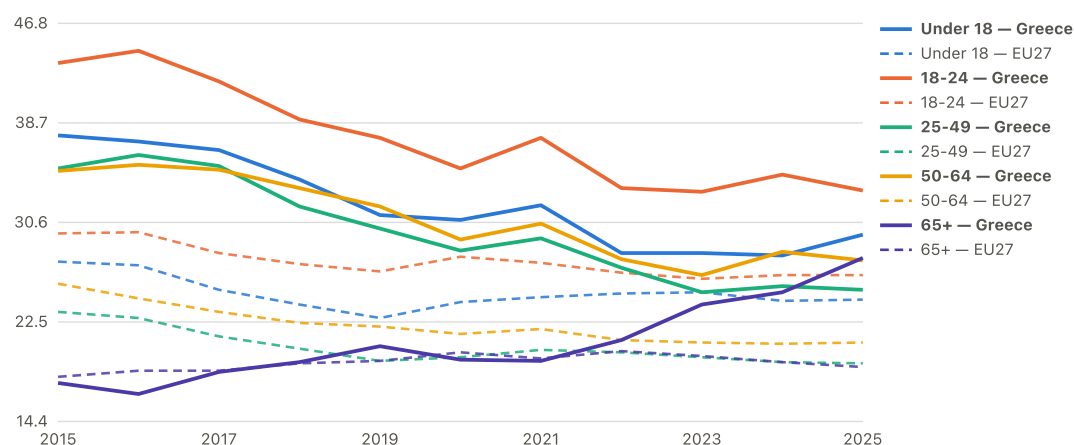


The official broader measure moves closer to reported hardship than AROP does, but Greece still sits far above countries with a similar AROPE rate. This is the bridge from Part I to the decomposed hardship models in Part II.

EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

## AROPE by age group: Greece vs EU27

% of people

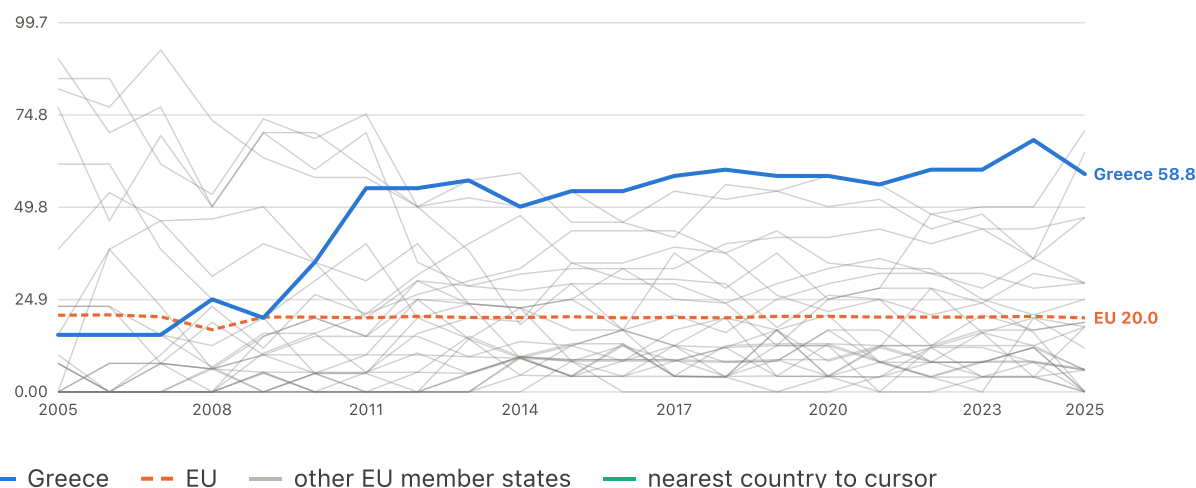


Solid = Greece, dashed = EU27. Every working-age group improved while 65+ reversed.

EU comparator: Eurostat EU27 aggregate (population-weighted)

## Breadth of disadvantage: share of indicators placing the country in the EU's worst fifth

% of indicators · up to 27 EU member states · 2005–2025



— Greece — EU — other EU member states — nearest country to cursor

|        | 2005  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 15.40 | 35.00 | 54.20 | 58.30 | 58.80 |
| EU     | 20.70 | 20.20 | 20.30 | 20.40 | 20.00 |

Across 25 indicators with an unambiguous worse-direction, the share that put this country in the EU's bottom quintile that year. The outcome (subjective poverty) and every model covariate are excluded, so this cannot restate what the reports set out to explain. Years with fewer than 10 reporting indicators are dropped. DESCRIPTIVE ONLY: tested as a predictor of reported hardship in P3a. On its own it is not significant (coefficient +5.94,  $p = 0.12$ ), and once the other accumulated measures enter the model the

coefficient reverses sign (-2.17). A quantity whose sign depends on what else is in the model cannot carry an explanatory reading. It summarises the condition rather than explaining it.

Complete 27-country coverage from 2005. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Every indicator behind the breadth measure: earliest available year versus latest year

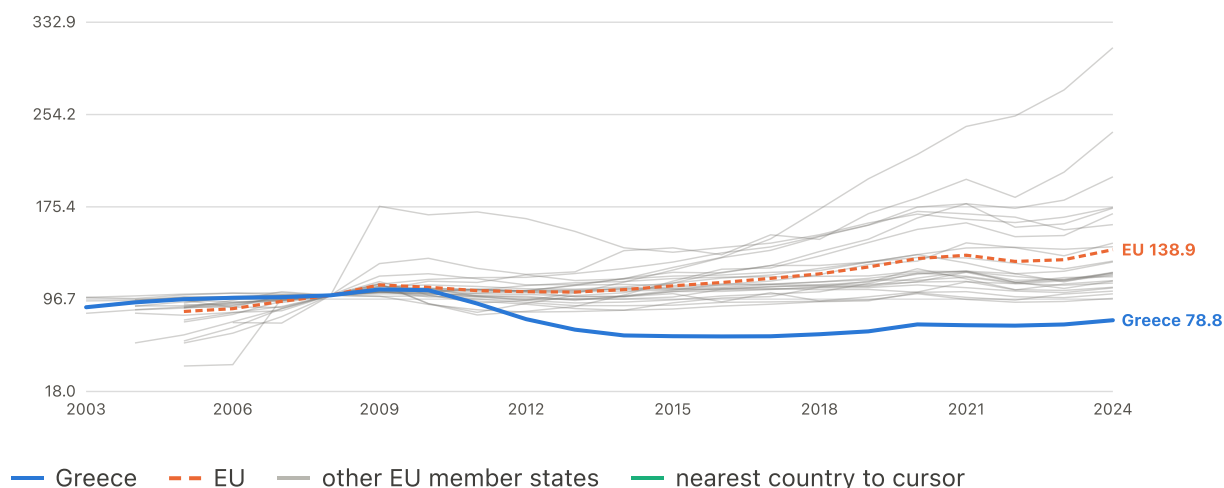
position in the EU distribution



One row per contributing indicator. The hollow dot is Greece's position at that indicator's earliest usable year, the solid dot its latest; the shaded band is the EU's worst fifth, which is what the breadth measure counts. The axis is position, not value — 0 is the best place in the Union to be on that indicator and 100 the worst, whichever direction 'worse' happens to run in, because the 25 indicators have no common unit. Hover any row for the underlying figures in their own units. The diamond is the EU's own position, shown only for the 17 indicators where Eurostat publishes a population-weighted aggregate; for the other 8 the EU line is an unweighted mean of member states, which sits mid-distribution by construction and would read as a finding when it is only arithmetic.

## Real AROP poverty threshold (own 2008 = 100)

index, 2008=100 · up to 26 EU member states · 2003–2024



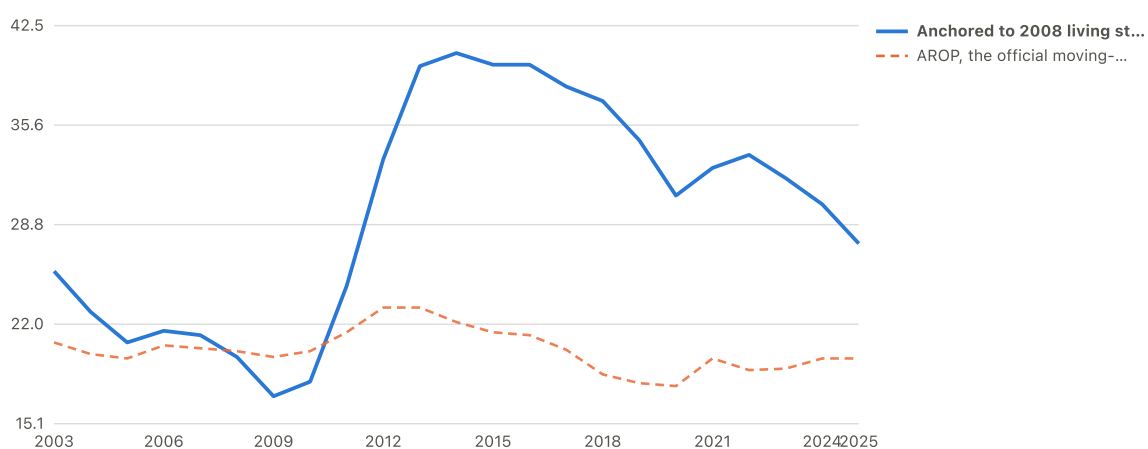
|        | 2003  | 2010  | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|-------|-------|
| Greece | 89.83 | 104.4 | 65.14 | 75.19 | 78.77 |
| EU     | --    | 106.9 | 108.1 | 131.3 | 138.9 |

The poverty line itself in real terms -- the 'shrinking ruler'.

Complete 26-country coverage from 2007. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Greece: fixed-yardstick (anchored) poverty vs AROP

% of people

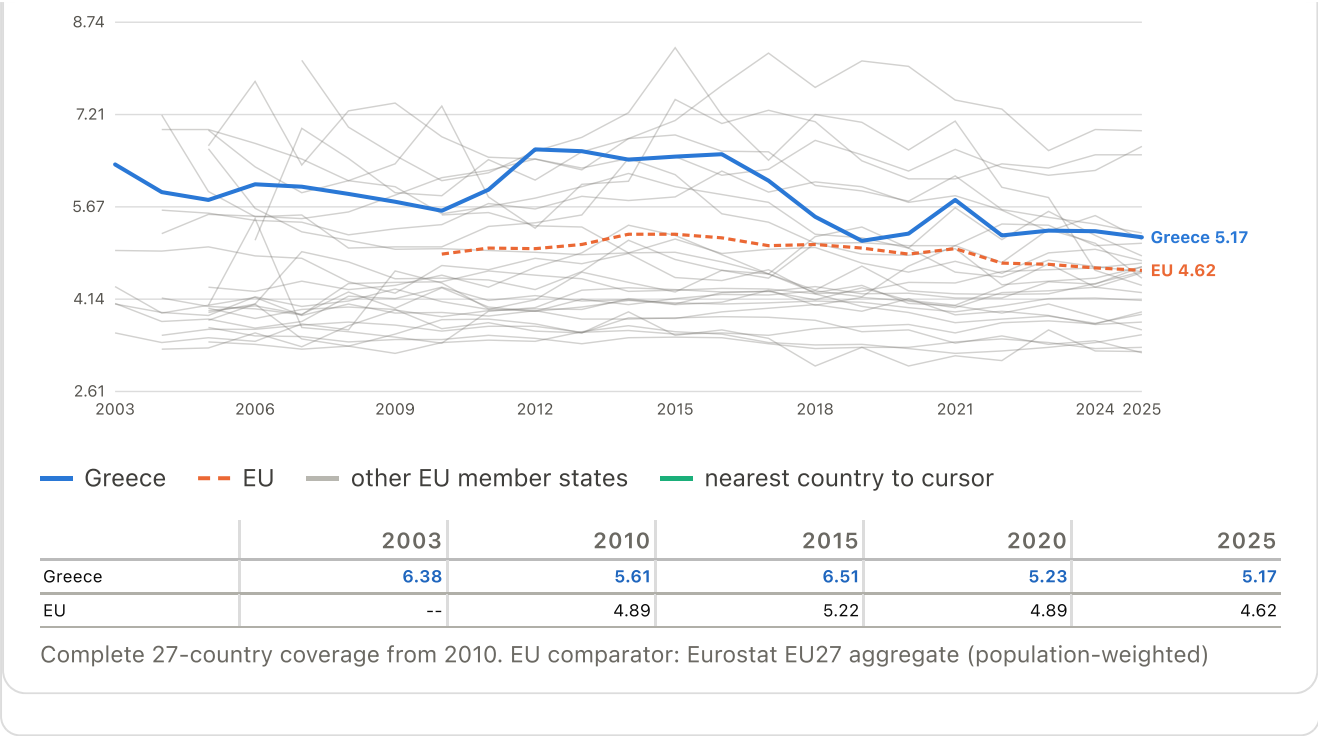


Greece only. The anchored series holds the poverty line at its 2008 real value; AROP lets it move with the collapsing median. Anchored figures are an approximation from four published summary points, least certain in 2012-2014.

## Income inequality (S80/S20 ratio)

ratio · up to 27 EU member states · 2003–2025





Stage 3

# Is the hardship real?

Does reported difficulty track concrete affordability failure, or does it float free of material circumstances?

Some gaps narrowed, especially long-term unemployment; wage, resource and affordability gaps widened

**DESCRIPTIVE** Which measures converged toward the EU and which diverged?

**Read with this.** A **NARROWING GAP DOES NOT MEAN GREECE IMPROVED**. A gap can close because Greece caught up, because the rest of the EU deteriorated, or because everyone moved in the same direction at different speeds. This chart measures **RELATIVE CONVERGENCE**, not national recovery, which is why both endpoints for Greece and for the EU median are in the tooltip and the table. "71% of the gap closed" and "conditions improved by 71%" are entirely different claims. The plotted quantity is dimensionless because the underlying gaps are measured in percentage points, index points and PPS per head and cannot share an axis.

| Measure                             | Share of 2015 gap closed | Greece 2015 | Greece 2024 | EU median 2015 | EU median 2024 | Gap 2015 | Gap 2024 |
|-------------------------------------|--------------------------|-------------|-------------|----------------|----------------|----------|----------|
| Long-term unemployment              | 0.71                     | 16.40       | 5.40        | 3.60           | 1.70           | 12.80    | 3.70     |
| Share below own GDP peak            | 0.55                     | 24.84       | 11.20       | 0.00           | 0.00           | 24.84    | 11.20    |
| Housing-cost overburden             | 0.40                     | 45.50       | 28.90       | 8.70           | 6.70           | 36.80    | 22.20    |
| AROPE                               | 0.26                     | 32.40       | 26.90       | 22.50          | 19.60          | 9.90     | 7.30     |
| Income poverty (AROP)               | 0.20                     | 21.40       | 19.60       | 16.30          | 15.50          | 5.10     | 4.10     |
| Real poverty threshold              | 0.02                     | 65.14       | 78.77       | 106.45         | 119.11         | -41.31   | -40.34   |
| Material deprivation                | 0.01                     | 17.60       | 14.00       | 7.80           | 4.30           | 9.80     | 9.70     |
| Reported hardship                   | -0.02                    | 77.70       | 66.70       | 28.90          | 17.10          | 48.80    | 49.60    |
| Real household income               | -0.06                    | 67.79       | 84.13       | 102.25         | 120.59         | -34.46   | -36.46   |
| Hardship gap against income poverty | -0.06                    | 56.30       | 47.10       | 13.10          | 1.20           | 43.20    | 45.90    |
| Hardship gap against AROPE          | -0.08                    | 45.30       | 39.80       | 6.40           | -2.30          | 38.90    | 42.10    |
| Real wages                          | -0.78                    | 76.91       | 68.18       | 101.43         | 111.85         | -24.52   | -43.67   |
| Material resources                  | -0.93                    | 14769.70    | 21309.90    | 16230.40       | 24129.10       | -1460.70 | -2819.20 |
| Wage-adjusted affordability         | -2.55                    | 141.89      | 173.09      | 127.22         | 120.97         | 14.67    | 52.12    |

Three affordability measures closely track hardship in Greece; falling behind on bills tracks it much less closely

**DESCRIPTIVE CORROBORATION**

Did concrete affordability difficulties rise and fall with what Greek households reported?

**Read with this.** Same-survey corroboration, not independent validation or causal evidence. Greek correlations with reported hardship: unexpected expenses 0.92 · material deprivation 0.94 · keeping the home warm 0.87 · falling behind on bills 0.37. Every line is a distance from its own 2015-2024 average, including the European median, so three series on different scales can share one axis: the shapes may be compared, the levels may not. The scale is the same in all four tabs. European median hardship is deliberately absent, since the question here is whether the concrete difficulty moved with Greek reports, not how Greece compares.

| Unexpected expenses                |      |      |      |      |      |      |      |      |      |      |
|------------------------------------|------|------|------|------|------|------|------|------|------|------|
| Year                               | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: reported hardship          | 5.7  | 4.8  | 5.2  | 2.1  | -1.0 | -2.2 | -1.0 | -3.6 | -5.0 | -5.3 |
| Greece: unexpected expenses        | 4.7  | 4.9  | 4.0  | 1.7  | -0.9 | 2.0  | -2.4 | -5.1 | -4.4 | -4.8 |
| EU median: unexpected expenses     | 7.5  | 5.6  | 2.5  | -0.2 | -0.9 | -1.8 | -4.7 | -1.1 | -2.9 | -3.6 |
| Material deprivation               |      |      |      |      |      |      |      |      |      |      |
| Year                               | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: reported hardship          | 5.7  | 4.8  | 5.2  | 2.1  | -1.0 | -2.2 | -1.0 | -3.6 | -5.0 | -5.3 |
| Greece: material deprivation       | 2.0  | 2.8  | 2.7  | 0.5  | 0.2  | -0.7 | -1.7 | -1.7 | -2.1 | -1.6 |
| EU median: material deprivation    | 2.3  | 1.2  | 0.8  | -0.1 | -0.5 | -1.0 | -0.7 | -0.6 | -0.6 | -1.2 |
| Keeping the home warm              |      |      |      |      |      |      |      |      |      |      |
| Year                               | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: reported hardship          | 5.7  | 4.8  | 5.2  | 2.1  | -1.0 | -2.2 | -1.0 | -3.6 | -5.0 | -5.3 |
| Greece: keeping the home warm      | 7.6  | 7.5  | 4.1  | 1.1  | -3.7 | -4.5 | -4.1 | -2.9 | -2.4 | -2.6 |
| EU median: keeping the home warm   | 1.5  | -0.1 | -0.2 | -0.9 | -0.6 | -0.3 | -1.1 | 0.8  | 1.1  | -0.0 |
| Falling behind on bills            |      |      |      |      |      |      |      |      |      |      |
| Year                               | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece: reported hardship          | 5.7  | 4.8  | 5.2  | 2.1  | -1.0 | -2.2 | -1.0 | -3.6 | -5.0 | -5.3 |
| Greece: falling behind on bills    | 5.8  | 4.4  | 1.4  | -0.5 | -2.1 | -6.6 | -7.1 | 2.0  | 3.8  | -0.7 |
| EU median: falling behind on bills | 1.9  | 2.2  | 0.3  | 0.9  | -0.1 | -1.3 | -1.2 | -1.5 | -0.5 | -0.9 |

The real poverty threshold shows the only material sign reversal when the comparison moves within countries; the rest either hold their sign or start from a between-country correlation too close to zero for a reversal to be meaningful

#### DESCRIPTIVE

Which relationships with reported hardship change when the comparison moves from between countries to within them?

**Read with this.** Correlations identify duplication and sign reversals. They do NOT select variables, and a reversal is a fact about the two scopes rather than evidence about mechanism.

| Measure                     | Hardship correlations |                  | Change |
|-----------------------------|-----------------------|------------------|--------|
|                             | Between countries     | Within countries |        |
| Real poverty threshold      | 0.095                 | -0.699           | -0.794 |
| Real wages                  | 0.005                 | -0.654           | -0.659 |
| Real household income       | -0.197                | -0.755           | -0.558 |
| Share below own GDP peak    | 0.645                 | 0.231            | -0.413 |
| Inflation                   | -0.047                | -0.439           | -0.393 |
| Wage-adjusted affordability | 0.709                 | 0.525            | -0.185 |
| AROPE                       | 0.617                 | 0.772            | 0.155  |
| Material deprivation        | 0.679                 | 0.786            | 0.107  |
| Long-term unemployment      | 0.788                 | 0.689            | -0.099 |
| Housing-cost overburden     | 0.573                 | 0.491            | -0.081 |
| Income poverty (AROP)       | 0.446                 | 0.427            | -0.019 |
| Material resources          | -0.587                | -0.593           | -0.006 |

Technical diagnostics: checks that qualify a result rather than establishing one.

#### APPENDIX FIGURE

The same pair can point one way across countries and the other way within them

#### DESCRIPTIVE

Do relationships between the candidate variables hold across countries and within them, or do they change?

**Read with this.** Correlations identify duplication and sign reversals. They do NOT select variables. AROPE contains AROP and material deprivation, so those correlations are partly mechanical and must not be interpreted as independent relationships; the affected cells are outlined. This view carries the outcomes and the 10 construct representatives only, because at full size a matrix contains everything and shows almost nothing. The complete 35-variable matrix IS available: the statistical appendix carries it three times, within countries, between countries and pooled.

| Between countries           |                   |                       |        |                    |                        |            |                       |                        |                          |                             |           |                         |                      |
|-----------------------------|-------------------|-----------------------|--------|--------------------|------------------------|------------|-----------------------|------------------------|--------------------------|-----------------------------|-----------|-------------------------|----------------------|
| Measure                     | Reported hardship | Income poverty (AROP) | AROPE  | Material resources | Long-term unemployment | Real wages | Real household income | Real poverty threshold | Share below own GDP peak | Wage-adjusted affordability | Inflation | Housing-cost overburden | Material deprivation |
| Reported hardship           | 1.000             | 0.446                 | 0.617  | -0.587             | 0.788                  | 0.005      | -0.197                | 0.095                  | 0.645                    | 0.709                       | -0.047    | 0.573                   | 0.679                |
| Income poverty (AROP)       | 0.446             | 1.000                 | 0.882  | -0.368             | 0.393                  | 0.279      | 0.104                 | 0.396                  | 0.153                    | 0.450                       | 0.151     | 0.178                   | 0.587                |
| AROPE                       | 0.617             | 0.882                 | 1.000  | -0.423             | 0.441                  | 0.276      | 0.153                 | 0.442                  | 0.219                    | 0.538                       | 0.145     | 0.364                   | 0.879                |
| Material resources          | -0.587            | -0.368                | -0.423 | 1.000              | -0.285                 | -0.259     | -0.398                | -0.496                 | -0.017                   | -0.887                      | -0.507    | -0.035                  | -0.536               |
| Long-term unemployment      | 0.788             | 0.393                 | 0.441  | -0.285             | 1.000                  | -0.228     | -0.572                | -0.364                 | 0.797                    | 0.335                       | -0.323    | 0.611                   | 0.358                |
| Real wages                  | 0.005             | 0.279                 | 0.276  | -0.259             | -0.228                 | 1.000      | 0.475                 | 0.521                  | -0.416                   | 0.286                       | 0.175     | -0.127                  | 0.315                |
| Real household income       | -0.197            | 0.104                 | 0.153  | -0.398             | -0.572                 | 0.475      | 1.000                 | 0.744                  | -0.639                   | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Real poverty threshold      | 0.095             | 0.396                 | 0.442  | -0.496             | -0.364                 | 0.521      | 0.744                 | 0.744                  | -0.639                   | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Share below own GDP peak    | 0.645             | 0.153                 | 0.219  | -0.017             | 0.797                  | -0.416     | -0.639                | -0.639                 | 0.399                    | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Wage-adjusted affordability | 0.709             | 0.450                 | 0.538  | -0.887             | 0.335                  | 0.286      | 0.399                 | 0.399                  | 0.399                    | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Inflation                   | -0.047            | 0.151                 | 0.145  | -0.507             | -0.323                 | 0.175      | 0.648                 | 0.648                  | -0.639                   | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Housing-cost overburden     | 0.573             | 0.178                 | 0.364  | -0.035             | 0.611                  | -0.127     | -0.349                | -0.349                 | -0.639                   | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Material deprivation        | 0.679             | 0.587                 | 0.879  | -0.536             | 0.358                  | 0.315      | 0.294                 | 0.294                  | -0.639                   | 0.399                       | 0.648     | -0.349                  | 0.294                |
| Within countries            |                   |                       |        |                    |                        |            |                       |                        |                          |                             |           |                         |                      |
| Measure                     | Reported hardship | Income poverty (AROP) | AROPE  | Material resources | Long-term unemployment | Real wages | Real household income | Real poverty threshold | Share below own GDP peak | Wage-adjusted affordability | Inflation | Housing-cost overburden | Material deprivation |
| Reported hardship           | 1.000             | 0.427                 | 0.772  | -0.593             | 0.689                  | -0.654     | -0.755                | -0.699                 | 0.231                    | 0.525                       | -0.439    | 0.491                   | 0.786                |
| Income poverty (AROP)       | 0.427             | 1.000                 | 0.668  | -0.304             | 0.361                  | -0.314     | -0.374                | -0.491                 | 0.091                    | 0.277                       | -0.150    | 0.386                   | 0.423                |
| AROPE                       | 0.772             | 0.668                 | 1.000  | -0.497             | 0.586                  | -0.625     | -0.738                | -0.778                 | 0.175                    | 0.573                       | -0.365    | 0.577                   | 0.912                |
| Material resources          | -0.593            | -0.304                | -0.497 | 1.000              | -0.565                 | 0.488      | 0.724                 | 0.540                  | -0.217                   | -0.397                      | 0.602     | -0.300                  | -0.478               |
| Long-term unemployment      | 0.689             | 0.361                 | 0.586  | -0.565             | 1.000                  | -0.385     | -0.560                | -0.391                 | 0.362                    | 0.143                       | -0.377    | 0.602                   | 0.510                |
| Real wages                  | -0.654            | -0.314                | -0.625 | 0.488              | -0.385                 | 1.000      | 0.715                 | 0.761                  | 0.007                    | -0.818                      | 0.273     | -0.415                  | -0.734               |
| Real household income       | -0.755            | -0.374                | -0.738 | 0.724              | -0.560                 | 0.715      | 1.000                 | 0.813                  | -0.132                   | -0.626                      | 0.512     | -0.413                  | -0.763               |
| Real poverty threshold      | -0.699            | -0.491                | -0.778 | 0.540              | -0.391                 | 0.761      | 0.813                 | 1.000                  | -0.009                   | -0.771                      | 0.330     | -0.519                  | -0.787               |
| Share below own GDP peak    | 0.231             | 0.091                 | 0.175  | -0.217             | 0.362                  | 0.007      | -0.132                | -0.009                 | 1.000                    | -0.085                      | -0.220    | 0.211                   | 0.134                |
| Wage-adjusted affordability | 0.525             | 0.277                 | 0.573  | -0.397             | 0.143                  | -0.818     | -0.626                | -0.771                 | -0.085                   | 1.000                       | -0.285    | 0.292                   | 0.678                |
| Inflation                   | -0.439            | -0.150                | -0.365 | 0.602              | -0.377                 | 0.273      | 0.512                 | 0.330                  | -0.220                   | -0.285                      | 1.000     | -0.252                  | -0.392               |
| Housing-cost overburden     | 0.491             | 0.386                 | 0.577  | -0.300             | 0.602                  | -0.415     | -0.413                | -0.519                 | 0.211                    | 0.292                       | -0.252    | 1.000                   | 0.530                |
| Material deprivation        | 0.786             | 0.423                 | 0.912  | -0.478             | 0.510                  | -0.734     | -0.763                | -0.787                 | 0.134                    | 0.678                       | -0.392    | 0.530                   | 1.000                |

| Measure                     | All country-years |                       |        |                    |                        |            |                       |                        |                          |                             |           |                         |                      |
|-----------------------------|-------------------|-----------------------|--------|--------------------|------------------------|------------|-----------------------|------------------------|--------------------------|-----------------------------|-----------|-------------------------|----------------------|
|                             | Reported hardship | Income poverty (AROP) | AROPE  | Material resources | Long-term unemployment | Real wages | Real household income | Real poverty threshold | Share below own GDP peak | Wage-adjusted affordability | Inflation | Housing-cost overburden | Material deprivation |
| Reported hardship           | 1.000             | 0.439                 | 0.636  | -0.585             | 0.758                  | -0.093     | -0.312                | -0.053                 | 0.553                    | 0.684                       | -0.176    | 0.561                   | 0.694                |
| Income poverty (AROP)       | 0.439             | 1.000                 | 0.859  | -0.350             | 0.370                  | 0.220      | 0.029                 | 0.272                  | 0.138                    | 0.438                       | 0.006     | 0.193                   | 0.567                |
| AROPE                       | 0.636             | 0.859                 | 1.000  | -0.431             | 0.458                  | 0.157      | -0.015                | 0.228                  | 0.207                    | 0.540                       | -0.081    | 0.387                   | 0.883                |
| Material resources          | -0.585            | -0.350                | -0.431 | 1.000              | -0.354                 | -0.123     | -0.127                | -0.254                 | -0.066                   | -0.805                      | 0.145     | -0.073                  | -0.522               |
| Long-term unemployment      | 0.758             | 0.370                 | 0.458  | -0.354             | 1.000                  | -0.257     | -0.569                | -0.371                 | 0.676                    | 0.295                       | -0.268    | 0.592                   | 0.382                |
| Real wages                  | -0.093            | 0.220                 | 0.157  | -0.123             | -0.257                 | 1.000      | 0.525                 | 0.563                  | -0.326                   | 0.172                       | 0.147     | -0.162                  | 0.164                |
| Real household income       | -0.312            | 0.029                 | -0.015 | -0.127             | -0.569                 | 0.525      | 1.000                 | 0.758                  | -0.502                   | 0.243                       | 0.404     | -0.353                  | 0.081                |
| Real poverty threshold      | -0.053            | 0.272                 | 0.228  | -0.254             | -0.371                 | 0.563      | 0.758                 | 1.000                  | -0.337                   | 0.341                       | 0.293     | -0.326                  | 0.257                |
| Share below own GDP peak    | 0.553             | 0.138                 | 0.207  | -0.066             | 0.676                  | -0.326     | -0.502                | -0.337                 | 1.000                    | 0.133                       | -0.202    | 0.653                   | 0.156                |
| Wage-adjusted affordability | 0.684             | 0.438                 | 0.540  | -0.805             | 0.295                  | 0.172      | 0.243                 | 0.341                  | 0.133                    | 1.000                       | 0.058     | 0.223                   | 0.643                |
| Inflation                   | -0.176            | 0.006                 | -0.081 | 0.145              | -0.268                 | 0.147      | 0.404                 | 0.293                  | -0.202                   | 0.058                       | 1.000     | -0.122                  | -0.071               |
| Housing-cost overburden     | 0.561             | 0.193                 | 0.387  | -0.073             | 0.592                  | -0.162     | -0.353                | -0.326                 | 0.653                    | 0.223                       | -0.122    | 1.000                   | 0.423                |
| Material deprivation        | 0.694             | 0.567                 | 0.883  | -0.522             | 0.382                  | 0.164      | 0.081                 | 0.257                  | 0.156                    | 0.643                       | -0.071    | 0.423                   | 1.000                |

APPENDIX FIGURE

The baseline model expects Greece to report 25% hardship; Greek households report 72%

DESCRIPTIVE CORROBORATION

What does the model predict Greek households should report, and what do they report?

**Read with this.** Absorption is not explanation. The distance from a prediction to what Greece reports is the part the model does not account for: 46.92 points on income poverty and year alone, 13.74 after the four items enter, so 33.18 points, or 71%, is absorbed. These four items are answered by the same households in the same interview as the outcome, so part of what they take with them is the interview rather than the world.

| Model                                          | Percent of households |
|------------------------------------------------|-----------------------|
| Prediction from income poverty and year alone  | 25.05                 |
| Prediction including four related survey items | 58.23                 |
| What Greek households actually report          | 71.97                 |

## APPENDIX FIGURE

## Every country-year behind the binned European view

## APPENDIX

**Read with this.** These are the observations the report's second view bins. Both axes are in standard deviations from each country's own average. Greek observations are marked. At this density the cloud shows spread rather than shape, which is why the report bins it.

| Measure                           | Country-years | Pooled correlation |
|-----------------------------------|---------------|--------------------|
| Cannot meet an unexpected expense | 270.000       | 0.780              |
| Material deprivation              | 270.000       | 0.786              |
| Arrears on bills                  | 268.000       | 0.797              |
| Cannot keep the home warm         | 270.000       | 0.626              |

## APPENDIX FIGURE

## The European relationship, as binned country-year averages

## APPENDIX

**Read with this.** Both axes are in standard deviations from each country's own average, so one step means the same thing in all four panels. The line is fitted through the nine plotted bins while the correlation in each title is computed from all country-years: they are not the same calculation.

| Measure                           | Bins  | Country-years | Pooled correlation |
|-----------------------------------|-------|---------------|--------------------|
| Cannot meet an unexpected expense | 9.000 | 270.000       | 0.780              |
| Material deprivation              | 9.000 | 270.000       | 0.786              |
| Arrears on bills                  | 9.000 | 268.000       | 0.797              |
| Cannot keep the home warm         | 9.000 | 270.000       | 0.626              |

## APPENDIX FIGURE

## The full 35-variable correlation matrix, within countries

## APPENDIX

**Read with this.** The report shows ten construct representatives because at this size a matrix contains everything and shows almost nothing. It is here for anyone checking which variables duplicate which.

| Variable                            | Material resources | GDP per head | Real GDP per head | Consumption per head | Hourly compensation | Household saving rate | Unemployment |
|-------------------------------------|--------------------|--------------|-------------------|----------------------|---------------------|-----------------------|--------------|
| Material resources                  | 1.000              | 0.929        | 0.592             | 0.831                | 0.932               | 0.117                 |              |
| GDP per head                        | 0.929              | 1.000        | 0.809             | 0.776                | 0.885               | 0.170                 |              |
| Real GDP per head                   | 0.592              | 0.809        | 1.000             | 0.585                | 0.545               | 0.144                 |              |
| Consumption per head                | 0.831              | 0.776        | 0.585             | 1.000                | 0.656               | -0.109                |              |
| Hourly compensation                 | 0.932              | 0.885        | 0.545             | 0.656                | 1.000               | 0.322                 |              |
| Household saving rate               | 0.117              | 0.170        | 0.144             | -0.109               | 0.322               | 1.000                 |              |
| Unemployment                        | -0.585             | -0.536       | -0.458            | -0.638               | -0.474              | -0.144                |              |
| Long-term unemployment              | -0.565             | -0.523       | -0.443            | -0.580               | -0.500              | -0.238                |              |
| Youth unemployment                  | -0.456             | -0.415       | -0.383            | -0.530               | -0.315              | -0.042                |              |
| Employment rate                     | 0.841              | 0.791        | 0.611             | 0.830                | 0.739               | 0.220                 |              |
| Weekly working hours                | -0.626             | -0.586       | -0.256            | -0.356               | -0.722              | -0.268                |              |
| Real wages                          | 0.488              | 0.452        | 0.343             | 0.480                | 0.533               | 0.268                 |              |
| Real household income               | 0.724              | 0.663        | 0.493             | 0.730                | 0.706               | 0.424                 |              |
| Real poverty threshold              | 0.540              | 0.491        | 0.338             | 0.531                | 0.549               | 0.336                 |              |
| Share below own GDP peak            | -0.217             | -0.166       | -0.209            | -0.395               | 0.004               | 0.171                 |              |
| Inflation                           | 0.602              | 0.546        | 0.401             | 0.562                | 0.503               | -0.067                |              |
| Food inflation                      | 0.566              | 0.514        | 0.355             | 0.492                | 0.488               | -0.050                |              |
| Housing inflation                   | 0.329              | 0.307        | 0.277             | 0.364                | 0.248               | -0.095                |              |
| Wage-adjusted affordability         | -0.397             | -0.318       | -0.175            | -0.382               | -0.440              | -0.167                |              |
| Work-effort squeeze                 | -0.491             | -0.410       | -0.238            | -0.436               | -0.558              | -0.171                |              |
| Material deprivation                | -0.478             | -0.434       | -0.347            | -0.530               | -0.441              | -0.267                |              |
| Housing-cost overburden             | -0.300             | -0.246       | -0.195            | -0.319               | -0.282              | -0.227                |              |
| Household debt to income            | -0.553             | -0.646       | -0.652            | -0.478               | -0.487              | -0.191                |              |
| Income inequality (S80/S20)         | -0.375             | -0.359       | -0.291            | -0.357               | -0.362              | -0.140                |              |
| Net migration                       | -0.351             | -0.293       | -0.209            | -0.344               | -0.361              | -0.311                |              |
| Falling behind on bills             | -0.444             | -0.406       | -0.369            | -0.518               | -0.388              | -0.182                |              |
| Unexpected expenses                 | -0.552             | -0.540       | -0.510            | -0.593               | -0.524              | -0.310                |              |
| Keeping the home warm               | -0.179             | -0.162       | -0.179            | -0.308               | -0.127              | -0.145                |              |
| Income poverty (AROP)               | -0.304             | -0.344       | -0.393            | -0.280               | -0.311              | -0.210                |              |
| arop_before_transfers               | -0.485             | -0.514       | -0.395            | -0.539               | -0.415              | -0.209                |              |
| transfer_effect                     | -0.353             | -0.363       | -0.210            | -0.422               | -0.275              | -0.111                |              |
| Reported hardship                   | -0.593             | -0.557       | -0.450            | -0.626               | -0.549              | -0.309                |              |
| AROPE                               | -0.497             | -0.479       | -0.425            | -0.548               | -0.455              | -0.291                |              |
| Hardship gap against income poverty | -0.577             | -0.533       | -0.411            | -0.616               | -0.529              | -0.291                |              |
| Hardship gap against AROPE          | -0.548             | -0.510       | -0.395            | -0.568               | -0.510              | -0.270                |              |



## APPENDIX FIGURE

## The full 35-variable correlation matrix, between countries

## APPENDIX

**Read with this.** The report shows ten construct representatives because at this size a matrix contains everything and shows almost nothing. It is here for anyone checking which variables duplicate which.

| Variable                            | Material resources | GDP per head | Real GDP per head | Consumption per head | Hourly compensation | Household saving rate | Unemployment |
|-------------------------------------|--------------------|--------------|-------------------|----------------------|---------------------|-----------------------|--------------|
| Material resources                  | 1.000              | 0.864        | 0.921             | 0.972                | 0.915               | 0.550                 |              |
| GDP per head                        | 0.864              | 1.000        | 0.980             | 0.823                | 0.767               | 0.514                 |              |
| Real GDP per head                   | 0.921              | 0.980        | 1.000             | 0.906                | 0.835               | 0.532                 |              |
| Consumption per head                | 0.972              | 0.823        | 0.906             | 1.000                | 0.889               | 0.540                 |              |
| Hourly compensation                 | 0.915              | 0.767        | 0.835             | 0.889                | 1.000               | 0.634                 |              |
| Household saving rate               | 0.550              | 0.514        | 0.532             | 0.540                | 0.634               | 1.000                 |              |
| Unemployment                        | -0.119             | -0.223       | -0.169            | -0.008               | -0.161              | -0.401                |              |
| Long-term unemployment              | -0.285             | -0.318       | -0.302            | -0.192               | -0.302              | -0.476                |              |
| Youth unemployment                  | -0.139             | -0.221       | -0.183            | -0.045               | -0.197              | -0.439                |              |
| Employment rate                     | 0.197              | 0.202        | 0.219             | 0.165                | 0.179               | 0.509                 |              |
| Weekly working hours                | -0.750             | -0.541       | -0.637            | -0.742               | -0.825              | -0.665                |              |
| Real wages                          | -0.259             | -0.113       | -0.145            | -0.309               | -0.195              | -0.281                |              |
| Real household income               | -0.398             | -0.219       | -0.291            | -0.479               | -0.385              | -0.142                |              |
| Real poverty threshold              | -0.496             | -0.365       | -0.445            | -0.582               | -0.527              | -0.541                |              |
| Share below own GDP peak            | -0.017             | -0.079       | -0.049            | 0.077                | -0.138              | -0.348                |              |
| Inflation                           | -0.507             | -0.378       | -0.460            | -0.622               | -0.451              | -0.143                |              |
| Food inflation                      | -0.602             | -0.535       | -0.589            | -0.672               | -0.555              | -0.239                |              |
| Housing inflation                   | -0.327             | -0.145       | -0.235            | -0.436               | -0.265              | -0.166                |              |
| Wage-adjusted affordability         | -0.887             | -0.712       | -0.772            | -0.855               | -0.942              | -0.696                |              |
| Work-effort squeeze                 | -0.876             | -0.708       | -0.784            | -0.870               | -0.955              | -0.736                |              |
| Material deprivation                | -0.536             | -0.421       | -0.462            | -0.532               | -0.484              | -0.738                |              |
| Housing-cost overburden             | -0.035             | -0.125       | -0.069            | 0.042                | -0.066              | -0.411                |              |
| Household debt to income            | 0.765              | 0.630        | 0.725             | 0.810                | 0.665               | 0.388                 |              |
| Income inequality (S80/S20)         | -0.380             | -0.329       | -0.362            | -0.403               | -0.445              | -0.686                |              |
| Net migration                       | -0.106             | -0.099       | -0.124            | -0.171               | -0.106              | -0.216                |              |
| Falling behind on bills             | -0.416             | -0.330       | -0.335            | -0.311               | -0.461              | -0.656                |              |
| Unexpected expenses                 | -0.625             | -0.497       | -0.549            | -0.584               | -0.632              | -0.692                |              |
| Keeping the home warm               | -0.440             | -0.427       | -0.458            | -0.415               | -0.519              | -0.688                |              |
| Income poverty (AROP)               | -0.368             | -0.307       | -0.345            | -0.399               | -0.432              | -0.637                |              |
| arop_before_transfers               | 0.053              | -0.042       | 0.003             | 0.096                | 0.051               | -0.343                |              |
| transfer_effect                     | 0.402              | 0.261        | 0.337             | 0.468                | 0.462               | 0.321                 |              |
| Reported hardship                   | -0.587             | -0.490       | -0.521            | -0.507               | -0.607              | -0.697                |              |
| AROPE                               | -0.423             | -0.334       | -0.368            | -0.426               | -0.433              | -0.756                |              |
| Hardship gap against income poverty | -0.536             | -0.448       | -0.471            | -0.440               | -0.540              | -0.581                |              |
| Hardship gap against AROPE          | -0.509             | -0.435       | -0.455            | -0.409               | -0.529              | -0.479                |              |

## APPENDIX FIGURE

## The full 35-variable correlation matrix, all country-years

## APPENDIX

**Read with this.** The report shows ten construct representatives because at this size a matrix contains everything and shows almost nothing. It is here for anyone checking which variables duplicate which.

| Variable                            | Material resources | GDP per head | Real GDP per head | Consumption per head | Hourly compensation | Household saving rate | Unemployment |
|-------------------------------------|--------------------|--------------|-------------------|----------------------|---------------------|-----------------------|--------------|
| Material resources                  | 1.000              | 0.866        | 0.842             | 0.899                | 0.903               | 0.455                 |              |
| GDP per head                        | 0.866              | 1.000        | 0.950             | 0.801                | 0.779               | 0.454                 |              |
| Real GDP per head                   | 0.842              | 0.950        | 1.000             | 0.901                | 0.811               | 0.471                 |              |
| Consumption per head                | 0.899              | 0.801        | 0.901             | 1.000                | 0.866               | 0.461                 |              |
| Hourly compensation                 | 0.903              | 0.779        | 0.811             | 0.866                | 1.000               | 0.580                 |              |
| Household saving rate               | 0.455              | 0.454        | 0.471             | 0.461                | 0.580               | 1.000                 |              |
| Unemployment                        | -0.230             | -0.273       | -0.174            | -0.046               | -0.204              | -0.356                |              |
| Long-term unemployment              | -0.354             | -0.348       | -0.285            | -0.200               | -0.324              | -0.425                |              |
| Youth unemployment                  | -0.205             | -0.249       | -0.185            | -0.069               | -0.210              | -0.371                |              |
| Employment rate                     | 0.340              | 0.294        | 0.228             | 0.192                | 0.255               | 0.458                 |              |
| Weekly working hours                | -0.719             | -0.545       | -0.614            | -0.717               | -0.816              | -0.597                |              |
| Real wages                          | -0.123             | -0.039       | -0.116            | -0.261               | -0.109              | -0.130                |              |
| Real household income               | -0.127             | -0.059       | -0.209            | -0.356               | -0.204              | 0.007                 |              |
| Real poverty threshold              | -0.254             | -0.215       | -0.360            | -0.468               | -0.352              | -0.317                |              |
| Share below own GDP peak            | -0.066             | -0.092       | -0.056            | 0.039                | -0.112              | -0.231                |              |
| Inflation                           | 0.145              | 0.077        | -0.079            | -0.108               | 0.024               | -0.057                |              |
| Food inflation                      | 0.126              | 0.044        | -0.101            | -0.107               | 0.011               | -0.065                |              |
| Housing inflation                   | 0.090              | 0.073        | -0.014            | -0.048               | 0.020               | -0.073                |              |
| Wage-adjusted affordability         | -0.805             | -0.675       | -0.744            | -0.831               | -0.901              | -0.611                |              |
| Work-effort squeeze                 | -0.804             | -0.680       | -0.768            | -0.857               | -0.926              | -0.651                |              |
| Material deprivation                | -0.522             | -0.423       | -0.442            | -0.514               | -0.478              | -0.645                |              |
| Housing-cost overburden             | -0.073             | -0.138       | -0.073            | 0.028                | -0.085              | -0.380                |              |
| Household debt to income            | 0.605              | 0.532        | 0.682             | 0.771                | 0.587               | 0.310                 |              |
| Income inequality (S80/S20)         | -0.372             | -0.332       | -0.354            | -0.395               | -0.438              | -0.592                |              |
| Net migration                       | -0.183             | -0.136       | -0.103            | -0.145               | -0.150              | -0.250                |              |
| Falling behind on bills             | -0.416             | -0.339       | -0.330            | -0.309               | -0.452              | -0.572                |              |
| Unexpected expenses                 | -0.609             | -0.500       | -0.516            | -0.550               | -0.609              | -0.608                |              |
| Keeping the home warm               | -0.394             | -0.397       | -0.435            | -0.401               | -0.479              | -0.589                |              |
| Income poverty (AROP)               | -0.350             | -0.309       | -0.344            | -0.392               | -0.423              | -0.565                |              |
| arop_before_transfers               | -0.072             | -0.121       | -0.023            | 0.050                | -0.020              | -0.303                |              |
| transfer_effect                     | 0.253              | 0.171        | 0.291             | 0.400                | 0.365               | 0.237                 |              |
| Reported hardship                   | -0.585             | -0.498       | -0.498            | -0.493               | -0.598              | -0.623                |              |
| AROPE                               | -0.431             | -0.351       | -0.362            | -0.420               | -0.435              | -0.670                |              |
| Hardship gap against income poverty | -0.542             | -0.459       | -0.450            | -0.431               | -0.536              | -0.525                |              |
| Hardship gap against AROPE          | -0.512             | -0.444       | -0.439            | -0.403               | -0.525              | -0.442                |              |

## APPENDIX FIGURE

## Falling behind on bills against reported hardship, year by year

## APPENDIX

**Read with this.** SAME-INSTRUMENT CORROBORATION, NOT EXPLANATION. Both axes are EU-SILC self-reports collected from the same household in the same interview, so agreement between them is partly the interview. Arrears also depend on HAVING credit and obligations to fall behind on: a household that has lost access to credit, or never had it, can be in severe difficulty and register no arrears at all, which is why this is the weakest of the four affordability items in the report's own tracking figure. The fitted line excludes Greece.

| Year | Greece: arrears | Greece: reported hardship | Pooled correlation | Countries |
|------|-----------------|---------------------------|--------------------|-----------|
| 2015 | 49.30           | 77.70                     | 0.90               | 27.00     |
| 2016 | 47.90           | 76.80                     | 0.90               | 27.00     |
| 2017 | 44.90           | 77.20                     | 0.90               | 27.00     |
| 2018 | 43.00           | 74.10                     | 0.90               | 26.00     |
| 2019 | 41.40           | 71.00                     | 0.91               | 26.00     |
| 2020 | 36.90           | 69.80                     | 0.89               | 27.00     |
| 2021 | 36.40           | 71.00                     | 0.89               | 27.00     |
| 2022 | 45.50           | 68.40                     | 0.89               | 27.00     |
| 2023 | 47.30           | 67.00                     | 0.88               | 27.00     |
| 2024 | 42.80           | 66.70                     | 0.88               | 27.00     |

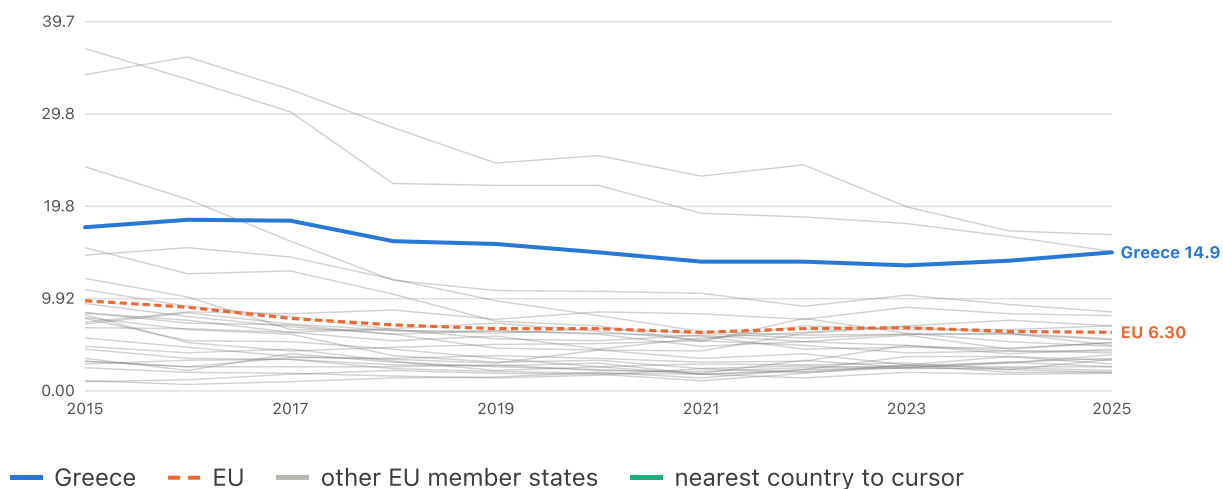
## Underlying variables and tables

6 charts

## Material deprivation

## Severe material &amp; social deprivation (13-item, 2015-)

% of people · up to 27 EU member states · 2015–2025

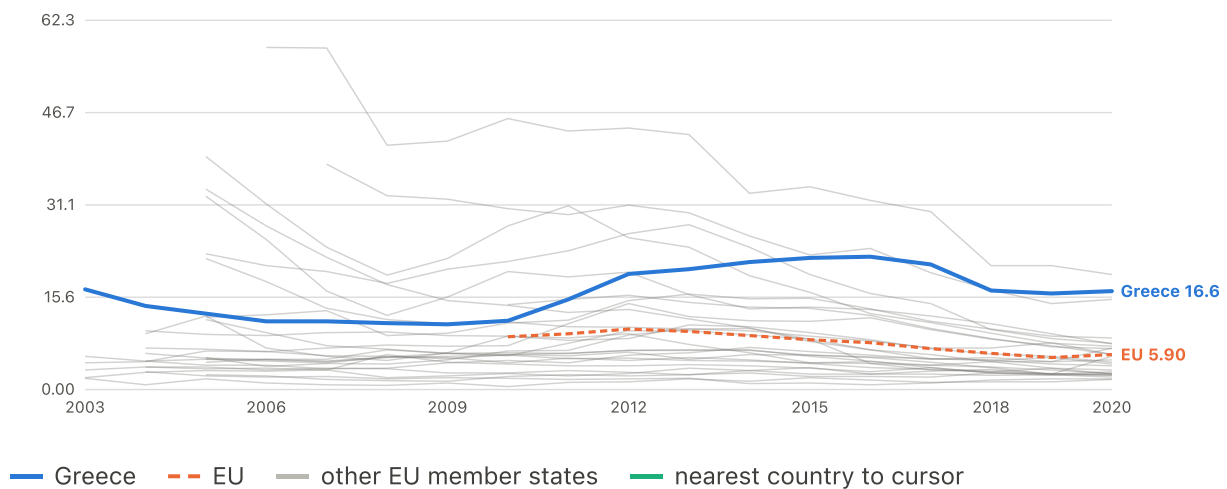


|        | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|
| Greece | 17.60 | 14.90 | 14.90 |
| EU     | 9.70  | 6.70  | 6.30  |

Complete 27-country coverage from 2015. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Severe material deprivation (legacy 9-item, to 2020)

% of people · up to 27 EU member states · 2003–2020



|        | 2003 | 2010 | 2015 | 2020 |
|--------|------|------|------|------|
| Greece | 15.6 | 16.6 | 16.6 | 16.6 |
| EU     | --   | 8.90 | 8.40 | 5.90 |

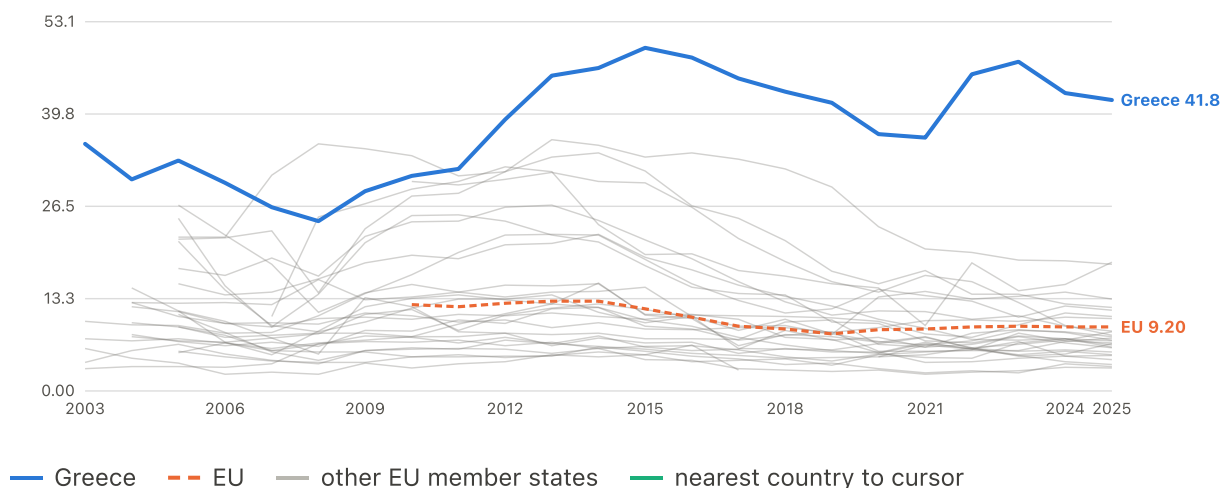
Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Housing

### Cash-flow strain — the two measures closest to the outcome itself

## Households in arrears (bills, rent or loans)

% of people · up to 27 EU member states · 2003–2025

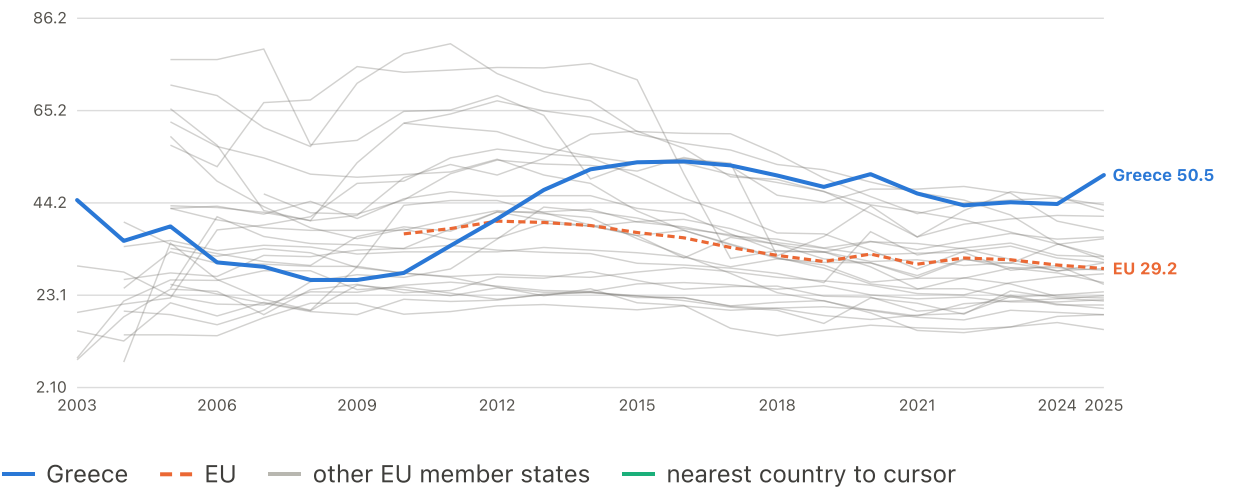


|        | 2003  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 35.50 | 30.90 | 49.30 | 36.90 | 41.80 |
| EU     | --    | 12.40 | 11.80 | 8.80  | 9.20  |

Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

Cannot face an unexpected expense

% of people · up to 27 EU member states · 2003–2025

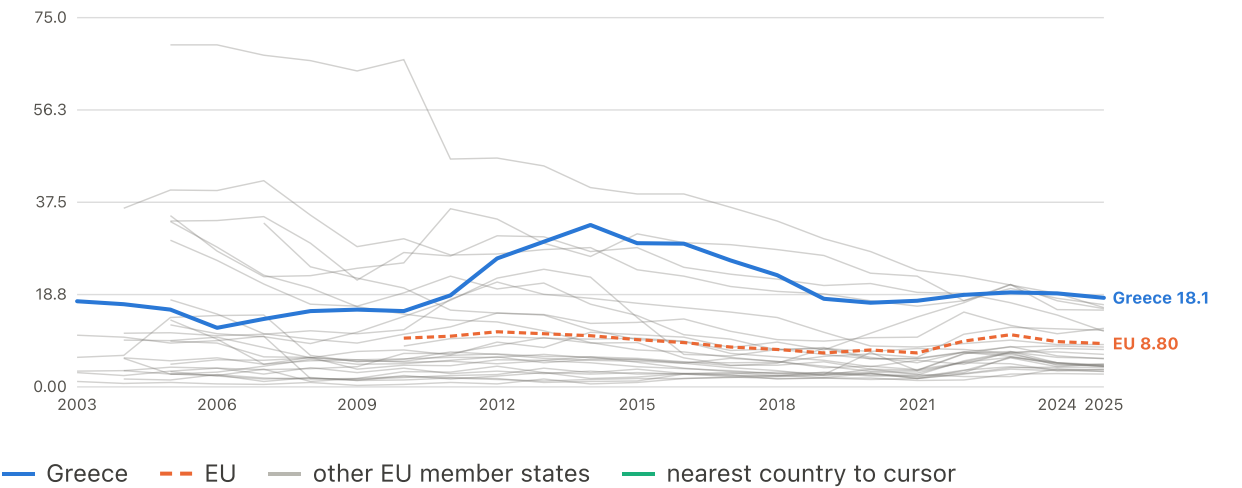


|        | 2003  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 44.80 | 28.20 | 53.40 | 50.70 | 50.50 |
| EU     | --    | 37.10 | 37.40 | 32.50 | 29.20 |

Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

Cannot keep home adequately warm

% of people · up to 27 EU member states · 2003–2025



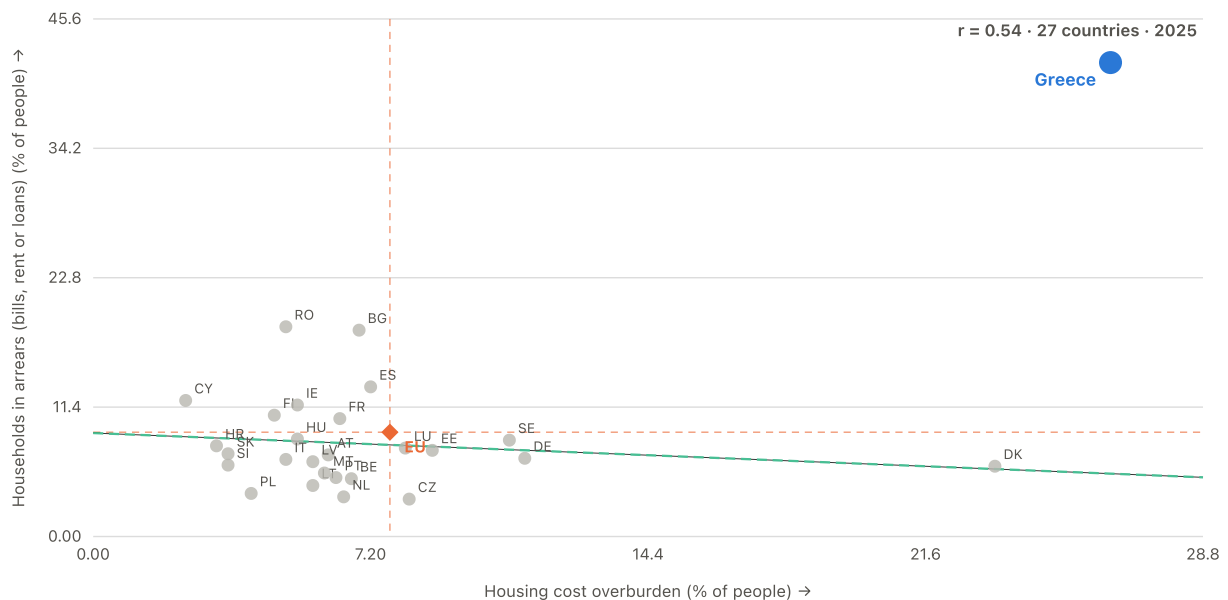
|        | 2003  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 17.40 | 15.40 | 29.20 | 17.10 | 18.10 |
| EU     | --    | 9.90  | 9.60  | 7.50  | 8.80  |

Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

RELATIONSHIP

Housing cost overburden against arrears

2025 · 27 member states ·  $r = +0.54$



Two different cash-flow pressures that tend to travel together. Greece is extreme on both.

EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

### Saving and debt — the counter-narrative pair

## Stage 4

## What current conditions explain

Which present-day conditions predict hardship beyond income poverty, and which merely could not be resolved?

Three current-condition constructs survive the full testing sequence

PRE-PLANNED CONFIRMATORY

Which current-level constructs survive every pre-registered condition, and which gate stopped the rest?

**Read with this.** Inconclusive is not evidence of absence. Bars are CLUSTER-ROBUST 95% confidence intervals, not bootstrap intervals: the wild cluster bootstrap determines the final support status, and two constructs whose intervals exclude zero here still fail it.

| Construct                   | Effect (SD) | CI low | CI high | p     | p FDR | Bootstrap p |
|-----------------------------|-------------|--------|---------|-------|-------|-------------|
| Long-term unemployment      | 0.808       | 0.523  | 1.093   | 0.000 | 0.000 | 0.009       |
| Material deprivation        | 0.720       | 0.239  | 1.202   | 0.003 | —     | —           |
| Wage-adjusted affordability | 0.687       | 0.386  | 0.988   | 0.000 | 0.000 | 0.001       |
| Share below own GDP peak    | 0.620       | 0.343  | 0.897   | 0.000 | 0.000 | 0.401       |
| Material resources          | 0.590       | 0.330  | 0.851   | 0.000 | 0.000 | 0.005       |
| Housing-cost overburden     | 0.545       | 0.141  | 0.949   | 0.008 | 0.015 | 0.553       |
| Real household income       | 0.315       | -0.279 | 0.908   | 0.299 | 0.413 | —           |
| Inflation                   | -0.274      | -0.817 | 0.268   | 0.321 | 0.413 | —           |
| Real wages                  | 0.174       | -0.501 | 0.850   | 0.613 | 0.653 | —           |
| Real poverty threshold      | 0.138       | -0.464 | 0.740   | 0.653 | 0.653 | —           |

What the three supported constructs look like

PRE-PLANNED CONFIRMATORY

What do the three supported constructs actually look like for Greece against the EU?

| Long-term unemployment |      |      |      |      |      |      |      |      |      |      |
|------------------------|------|------|------|------|------|------|------|------|------|------|
| Series                 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
| Greece                 | 16.4 | 15.4 | 14.3 | 12.5 | 11.3 | 10.5 | 9.2  | 7.7  | 6.2  | 5.4  |
| EU median              | 3.6  | 3.1  | 2.7  | 2.1  | 1.9  | 1.8  | 2.0  | 1.9  | 1.7  | 1.7  |

| Material resources |       |       |       |       |       |       |       |       |       |       |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Series             | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
| Greece             | 14770 | 14786 | 15372 | 15833 | 16372 | 15475 | 16772 | 19402 | 20701 | 21310 |
| EU median          | 16230 | 16494 | 17038 | 18006 | 18707 | 17386 | 19479 | 21317 | 23055 | 24129 |

| Wage-adjusted affordability |       |       |       |       |       |       |       |       |       |       |
|-----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Series                      | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
| Greece                      | 141.9 | 148.6 | 148.2 | 156.8 | 162.2 | 163.3 | 165.6 | 171.2 | 173.1 | 173.1 |
| EU median                   | 127.2 | 124.7 | 124.6 | 119.8 | 118.4 | 117.7 | 120.2 | 120.7 | 122.3 | 121.0 |

Evidence that cannot support a headline claim by design, and is recorded in the report's context register.

DESCRIPTIVE CORROBORATION

## Independent descriptive corroboration (Greece in Figures)

Greece in Figures analysis article, drawing on Eurostat's 2025 AIC release, Eurostat's 2025 price-level release, the Eurostat working-week comparison and a 2026 ELSTAT resident-travel bulletin -- four external releases, none fetched or checked by this project's own pipeline. The article's central figures were checked by hand against those four releases and reproduce; two of its claims (a vehicle-stock figure read as new-car purchases, and resident trip growth generalised to "all Greeks") do not, and are named here rather than repeated. No test was run by this project on any of the article's own comparisons.

**What may be concluded.** The independent picture -- Greece not last on actual consumption, long hours for comparatively low hourly reward, high relative prices in food and information/communication -- is consistent with and external corroboration for this project's own material-resources and wage-adjusted-affordability findings. Its central descriptive numbers reproduce against the primary Eurostat and ELSTAT releases they draw on.

**Limitation.** Treating the article's own comparisons, or its constructed AIC-per-hour ratio, as inferential evidence: it applies no bootstrap, no FDR correction, no within/between decomposition and no model comparison, and it does not test whether these factors account for reported hardship. Merging its 2025/2026-vintage figures into this project's frozen 2015-2024 panel. Repeating its car-ownership sentence, which cites a vehicle-STOCK figure as evidence of new purchases, or its 'all Greeks travelled' generalisation, which the underlying ELSTAT survey does not support. Reading its AIC rank, hours rank or price-level figure as contradicting this project's own numbers without first noting the different vintage or aggregate behind each.

**Source.** Greece in Figures, "Γιατί οι Έλληνες νιώθουν τόσο φτωχοί" [Why Greeks feel so poor]. Constructs an AIC-per-hour ratio from aggregate actual individual consumption and aggregate annual hours worked; cites 2025-vintage Eurostat AIC and price-level releases and a 2026 ELSTAT resident travel bulletin. source

### Underlying variables and tables

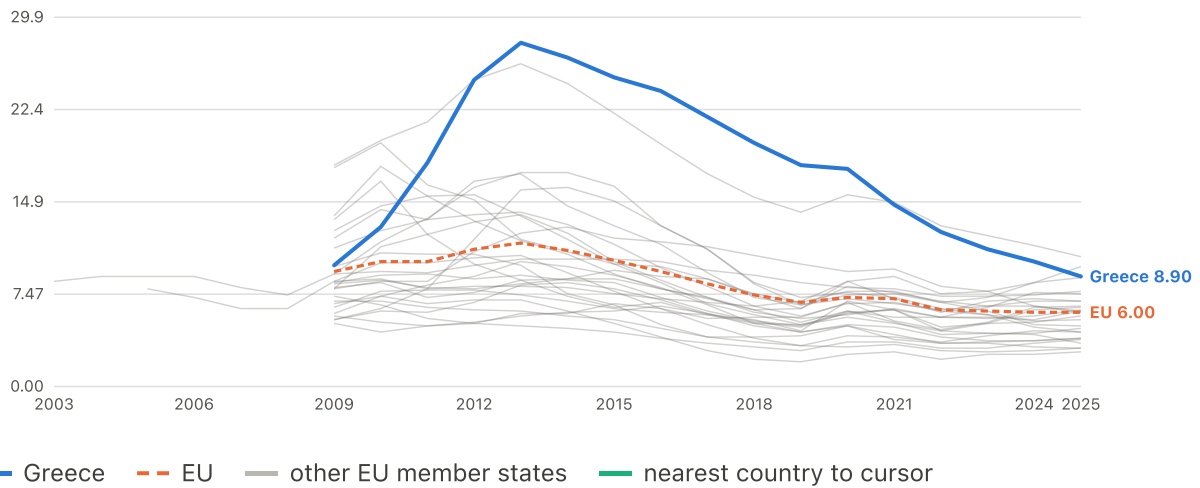
43 charts

#### How much unemployment, and how long it lasts

##### Unemployment rate

% of active pop. · up to 27 EU member states · 2003–2025



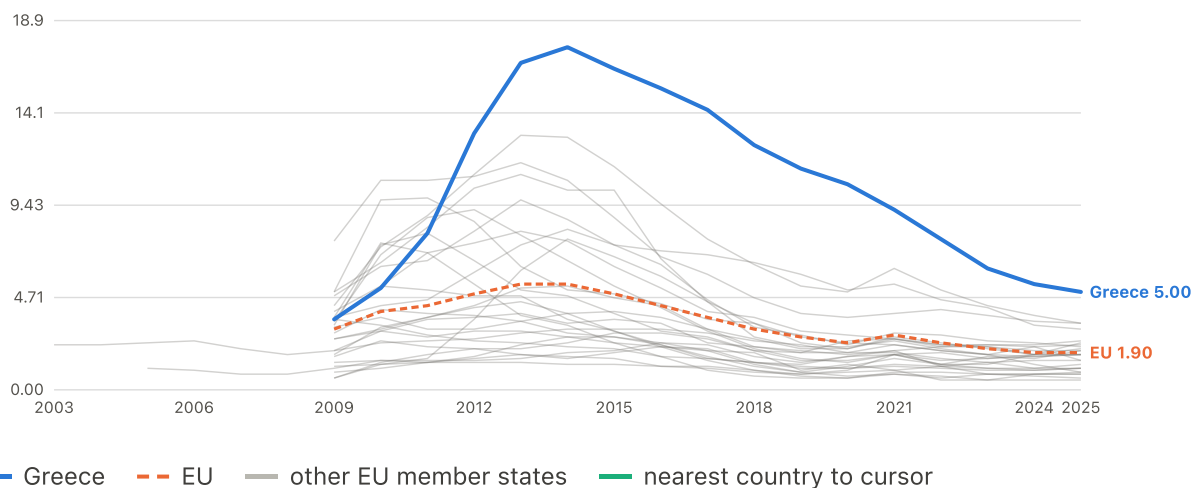


|        | 2009 | 2010  | 2015  | 2020  | 2025 |
|--------|------|-------|-------|-------|------|
| Greece | 9.80 | 12.90 | 25.00 | 17.60 | 8.90 |
| EU     | 9.30 | 10.10 | 10.20 | 7.20  | 6.00 |

Complete 27-country coverage from 2009. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Long-term unemployment (12 months or more)

% of active pop. · up to 27 EU member states · 2003–2025

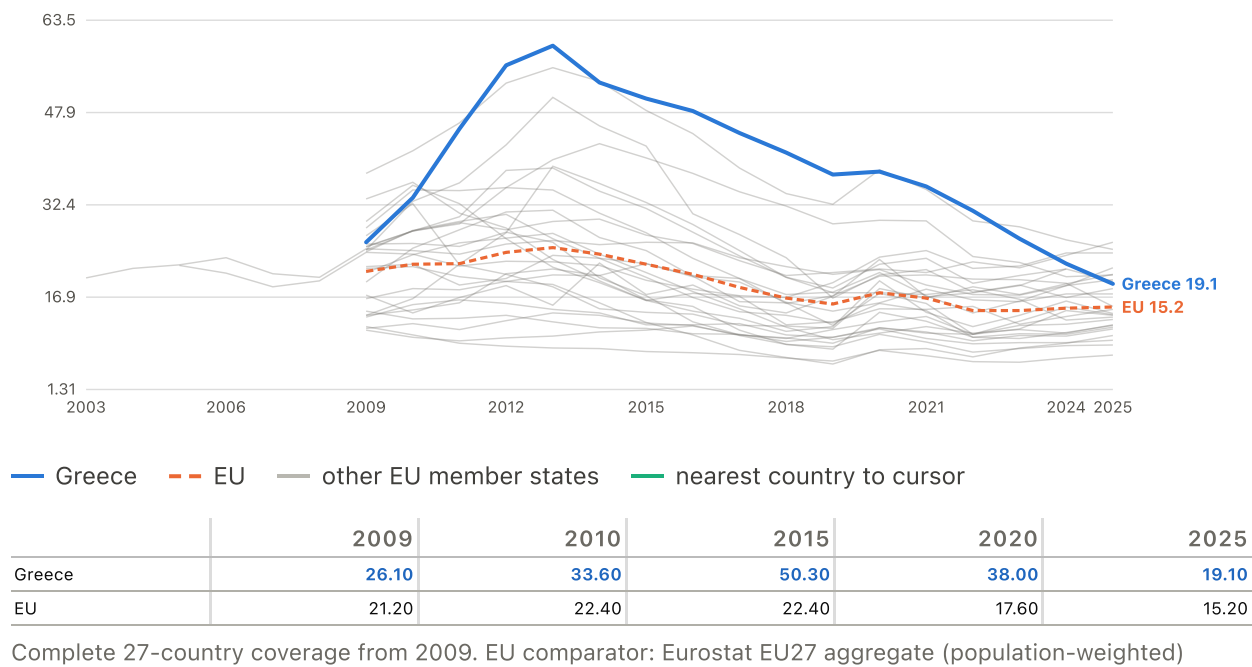


|        | 2009 | 2010 | 2015  | 2020  | 2025 |
|--------|------|------|-------|-------|------|
| Greece | 3.60 | 5.20 | 16.40 | 10.50 | 5.00 |
| EU     | 3.10 | 4.00 | 4.90  | 2.40  | 1.90 |

Complete 27-country coverage from 2009. EU comparator: Eurostat EU27 aggregate (population-weighted)

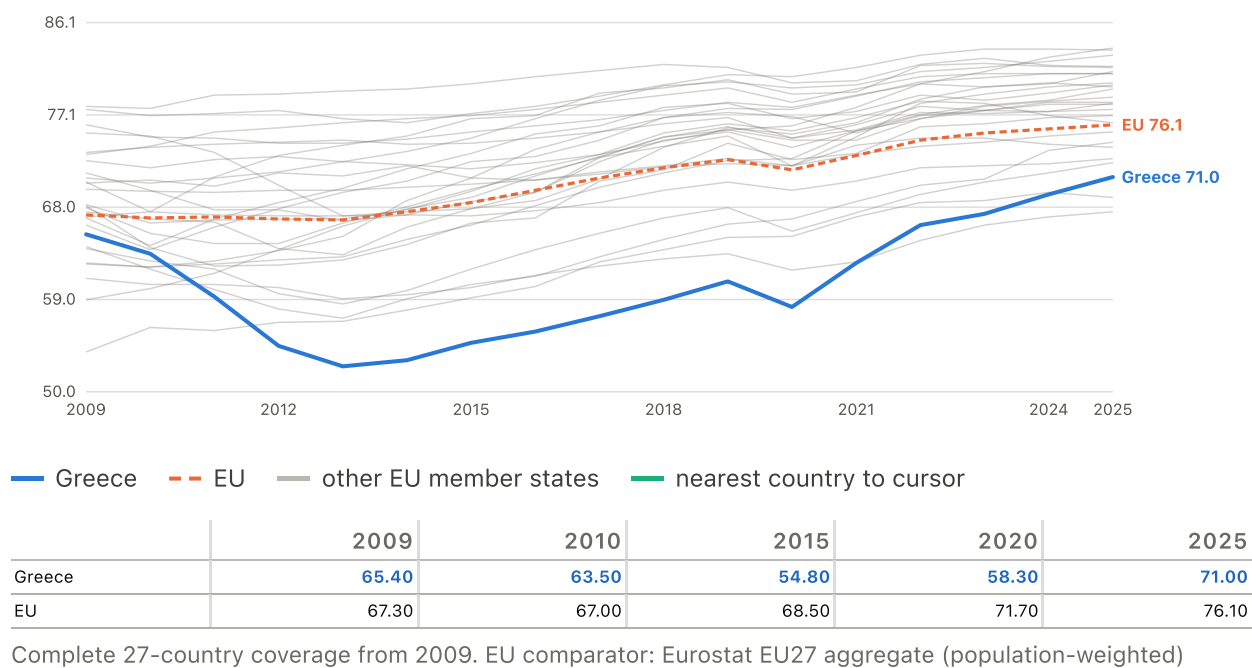
## Youth unemployment (ages 15-24)

% of active pop. · up to 27 EU member states · 2003–2025



## Employment rate (ages 20-64)

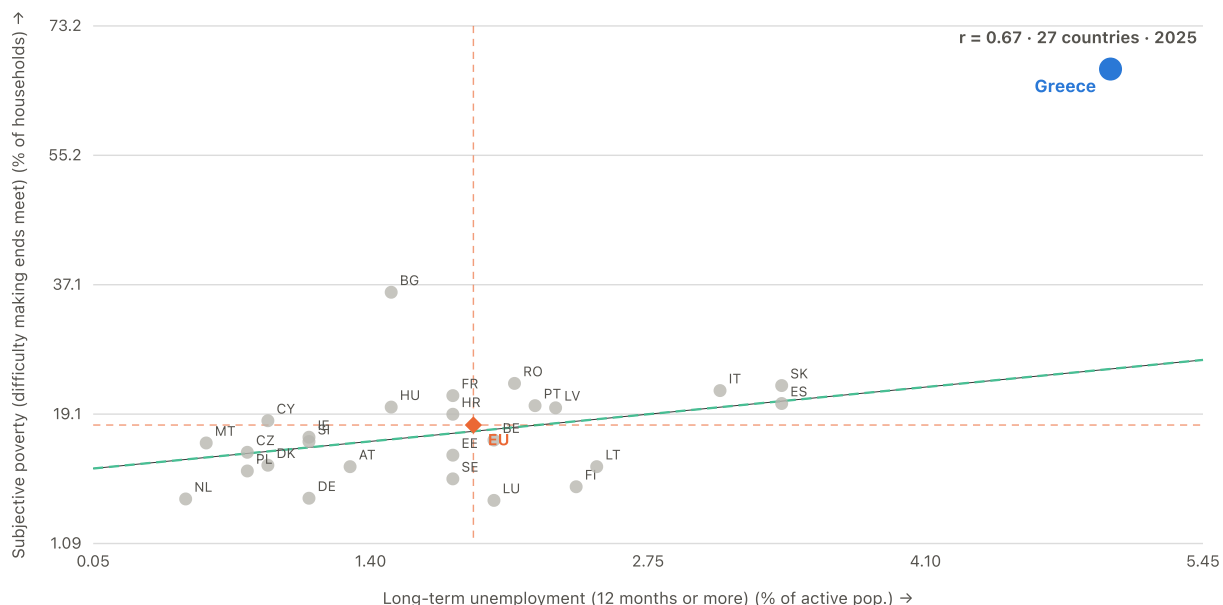
% of population · up to 27 EU member states · 2009–2025



### RELATIONSHIP

## Long-term unemployment against reported hardship

2025 · 27 member states ·  $r = +0.67$



Long-term unemployment is the single strongest correlate of Greek subjective poverty in the whole project. Across countries the relationship is real but far from deterministic.

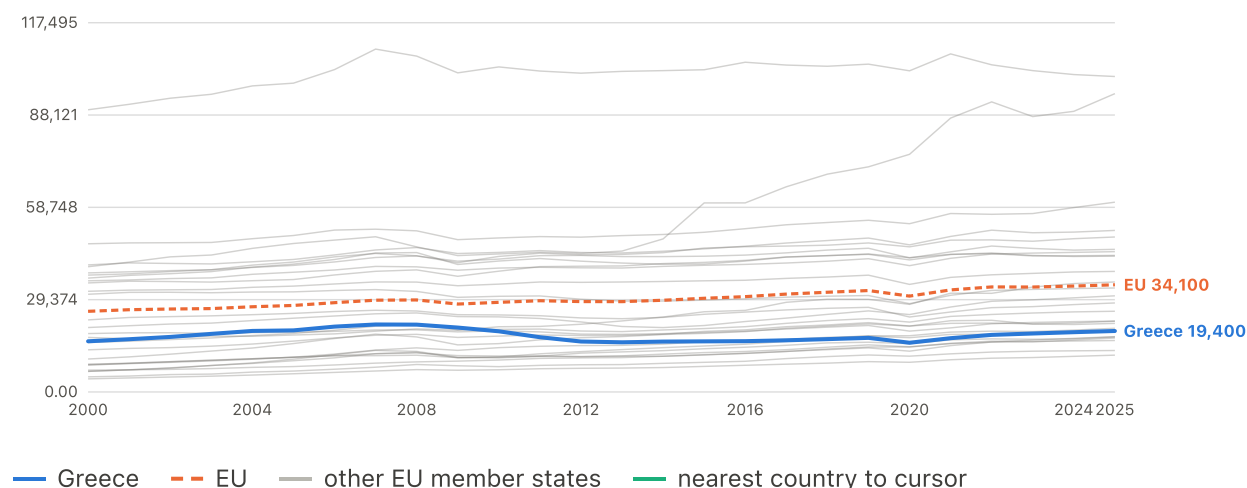
EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

## Accumulated over time, not just this year

## Output, and how far below its own peak

### Real GDP per capita

chain-linked EUR · up to 27 EU member states · 2000–2025

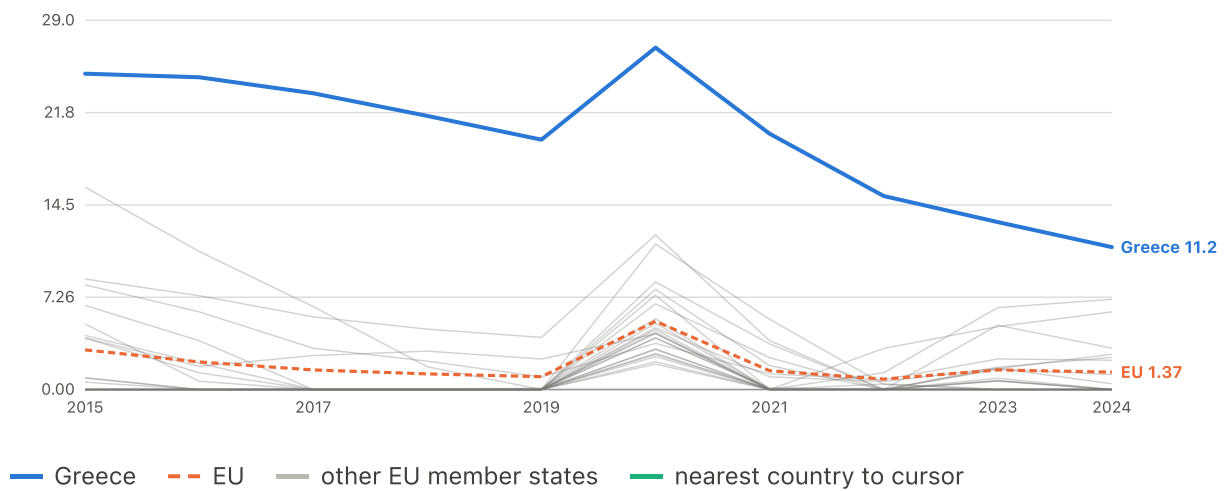


|        | 2000   | 2010   | 2015   | 2020   | 2025   |
|--------|--------|--------|--------|--------|--------|
| Greece | 16,120 | 19,300 | 16,100 | 15,660 | 19,400 |
| EU     | 25,660 | 28,520 | 29,810 | 30,500 | 34,100 |

Complete 27-country coverage from 2000. EU comparator: Eurostat EU27 aggregate (population-weighted)

## % below own GDP peak (scarring stock)

% below peak · up to 27 EU member states · 2015–2024



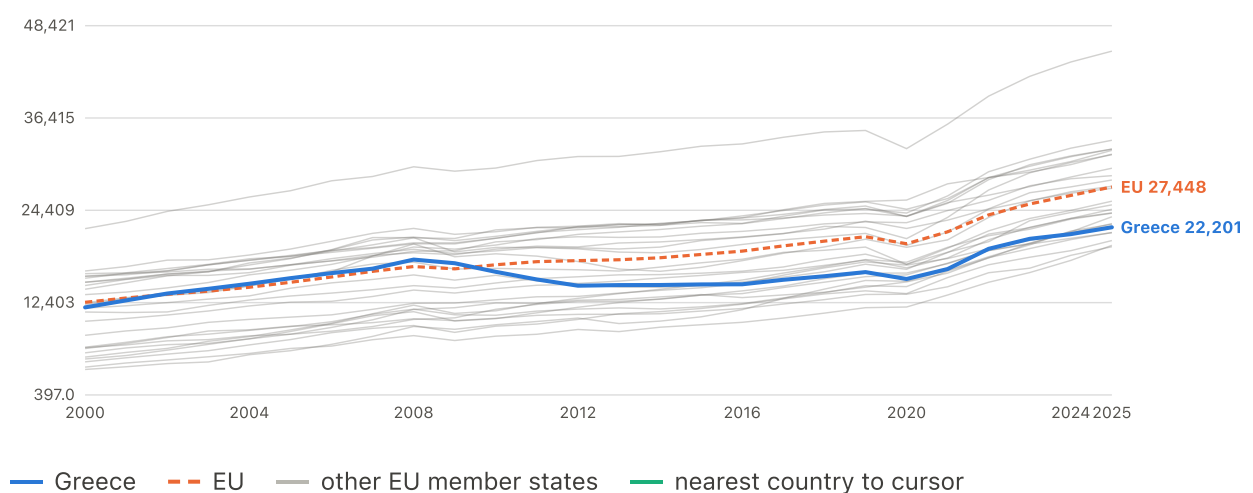
|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 24.84 | 26.89 | 11.20 |
| EU     | 3.12  | 5.37  | 1.37  |

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## What households actually receive and spend

### Actual individual consumption (PPS per capita)

PPS per capita · up to 27 EU member states · 2000–2025

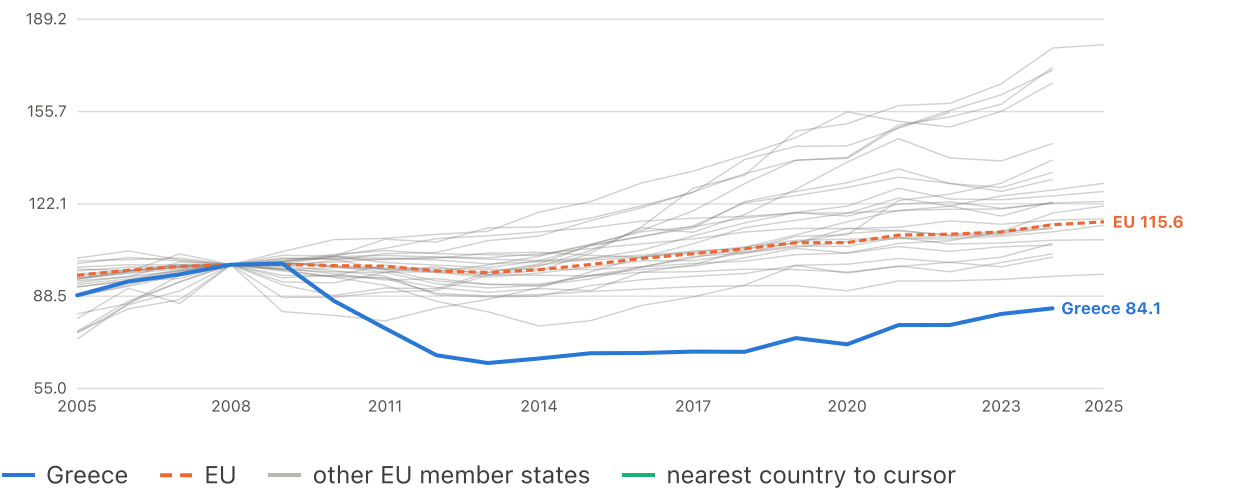


|        | 2000   | 2010   | 2015   | 2020   | 2025   |
|--------|--------|--------|--------|--------|--------|
| Greece | 11,782 | 16,406 | 14,770 | 15,475 | 22,201 |
| EU     | 12,444 | 17,327 | 18,666 | 20,062 | 27,448 |

Complete 27-country coverage from 2000. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Real household disposable income (2008 = 100)

index, 2008=100 · up to 27 EU member states · 2005–2025

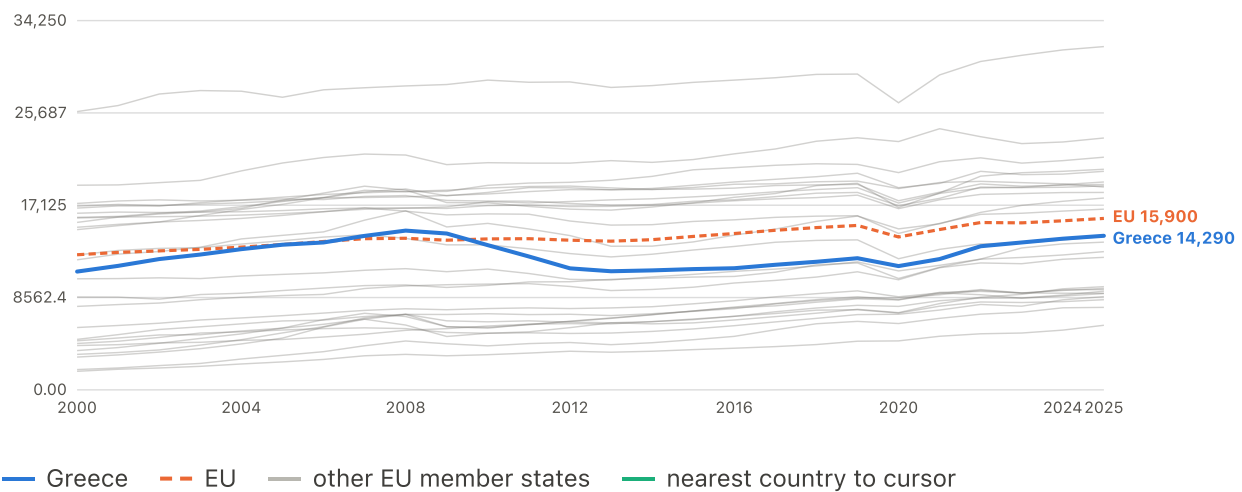


|        | 2005  | 2010  | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|-------|-------|
| Greece | 88.85 | 86.87 | 67.79 | 71.04 | 84.13 |
| EU     | 96.10 | 99.65 | 100.1 | 108.0 | 114.5 |

Complete 27-country coverage from 2005. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Real household consumption per capita

chain-linked EUR · up to 27 EU member states · 2000–2025



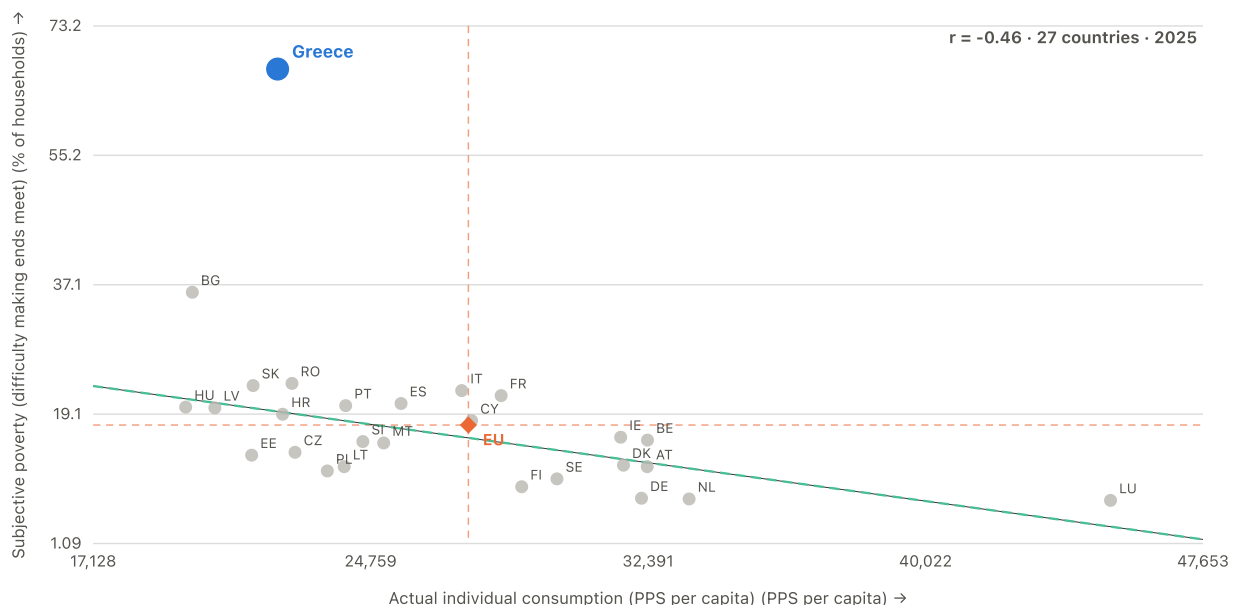
|        | 2000   | 2010   | 2015   | 2020   | 2025   |
|--------|--------|--------|--------|--------|--------|
| Greece | 10,970 | 13,420 | 11,200 | 11,480 | 14,290 |
| EU     | 12,530 | 14,000 | 14,230 | 14,180 | 15,900 |

Complete 27-country coverage from 2000. EU comparator: Eurostat EU27 aggregate (population-weighted)

### RELATIONSHIP

### Actual individual consumption against reported hardship

2025 · 27 member states ·  $r = -0.46$

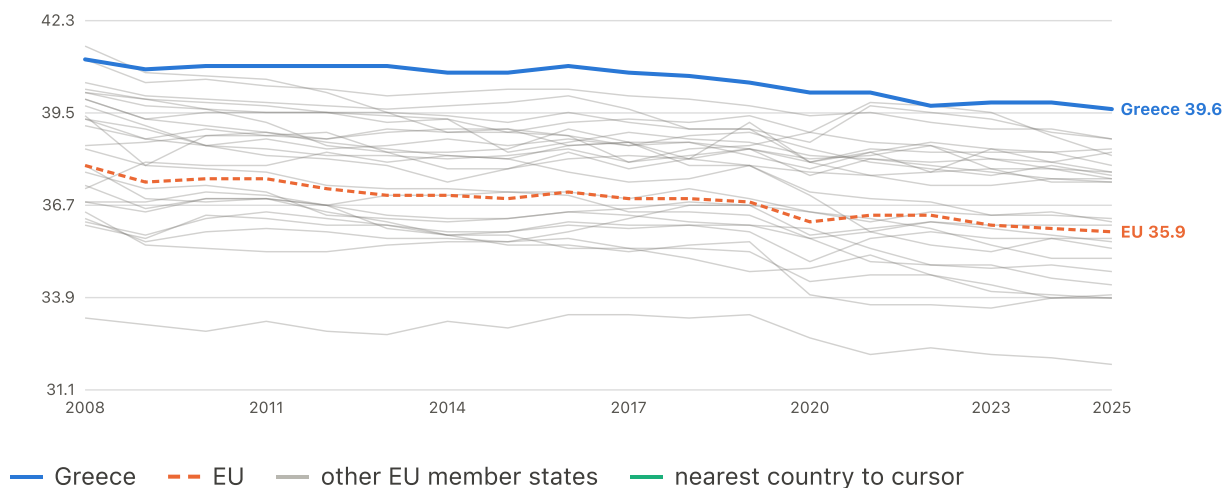


Countries with higher actual individual consumption generally report less hardship. This is a consumption-based welfare measure, not household income; Greece reports far more hardship than its consumption level alone would predict.

EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

## Actual weekly working hours (main job)

hours/week · up to 27 EU member states · 2008–2025

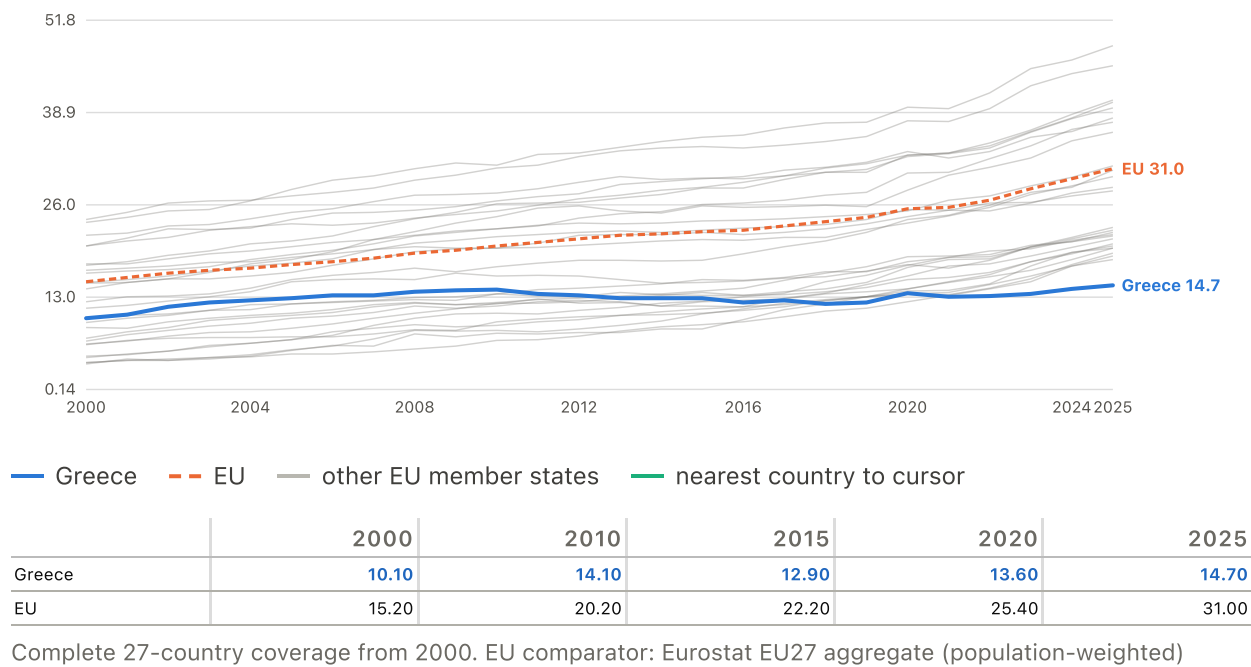


|        | 2008  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 41.10 | 40.90 | 40.70 | 40.10 | 39.60 |
| EU     | 37.90 | 37.50 | 36.90 | 36.20 | 35.90 |

Complete 27-country coverage from 2008. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Compensation per hour worked (PPS)

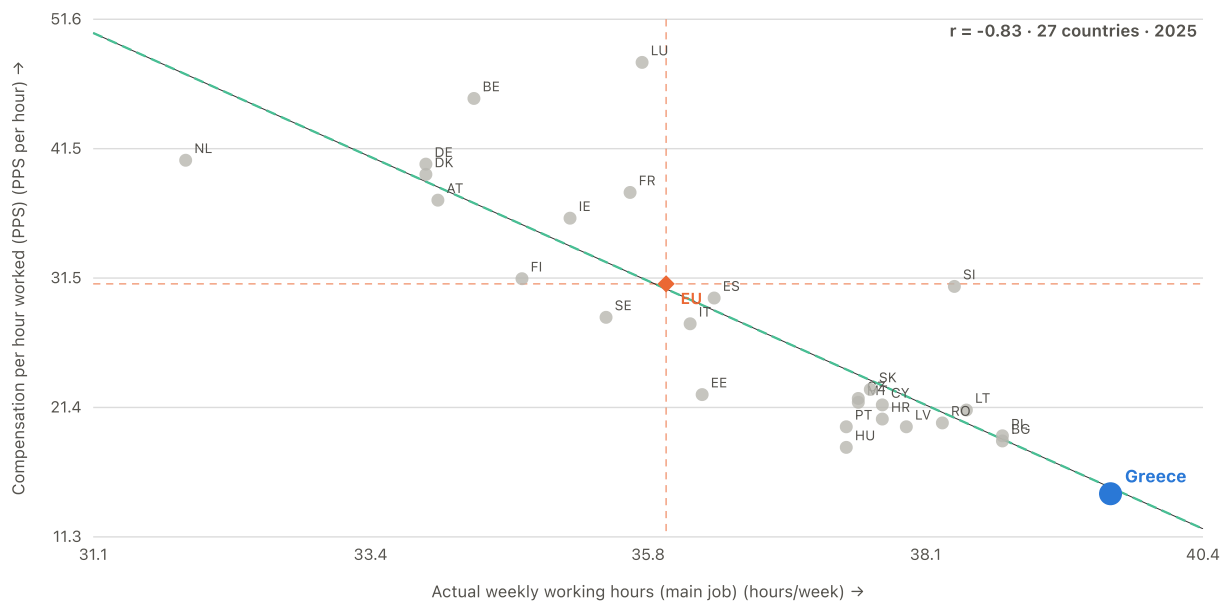
PPS per hour · up to 27 EU member states · 2000–2025



RELATIONSHIP

Hours worked against what an hour is paid

2025 · 27 member states ·  $r = -0.83$

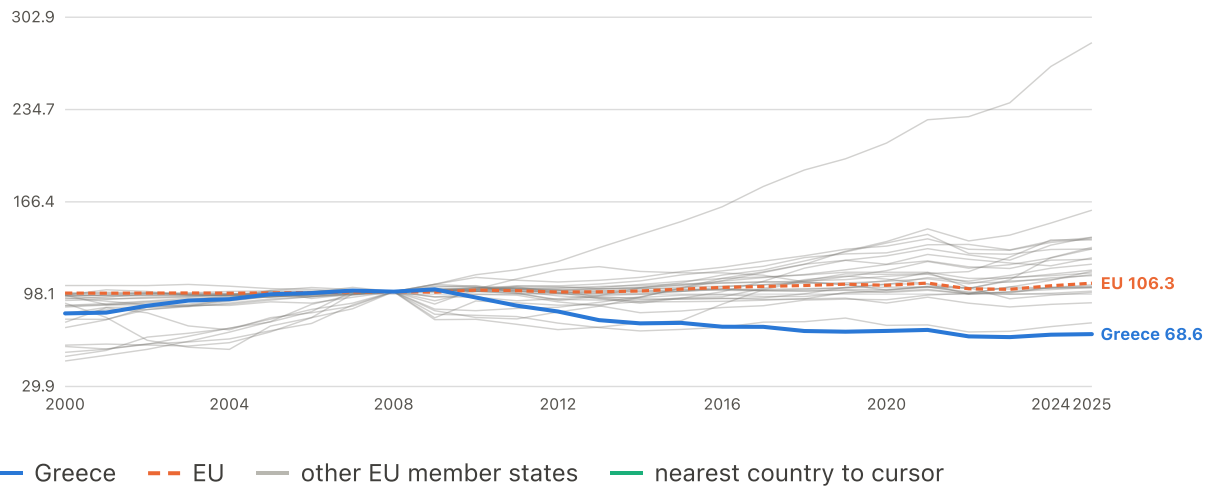


Greece sits in the corner no country wants: the longest working week in the Union at the lowest hourly compensation. This is the work-effort squeeze before it is compressed into a single index.

EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

Real wages, compensation per employee (own 2008 = 100)

index, 2008=100 · up to 27 EU member states · 2000–2025



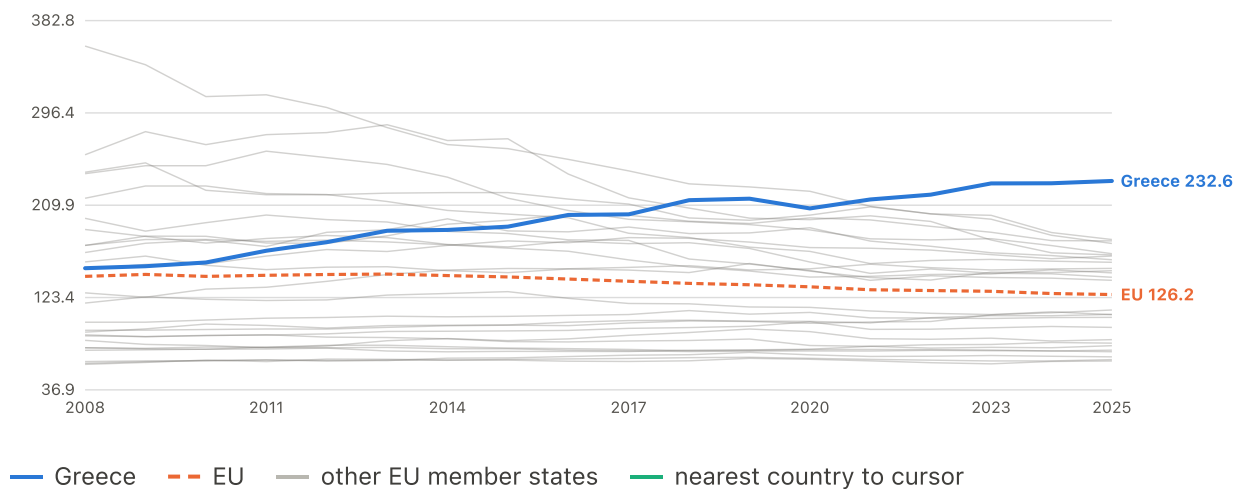
|        | 2000  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 83.92 | 95.68 | 76.91 | 70.97 | 68.61 |
| EU     | 98.95 | 101.2 | 101.8 | 104.7 | 106.3 |

Nominal compensation per employee deflated by each country's own HICP.

Complete 27-country coverage from 2000. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Work-effort squeeze (hours vs hourly pay, EU = 100)

index, EU=100 · up to 27 EU member states · 2008–2025



|        | 2008  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 150.9 | 156.2 | 189.8 | 206.9 | 232.6 |
| EU     | 143.2 | 143.2 | 142.7 | 133.4 | 126.2 |

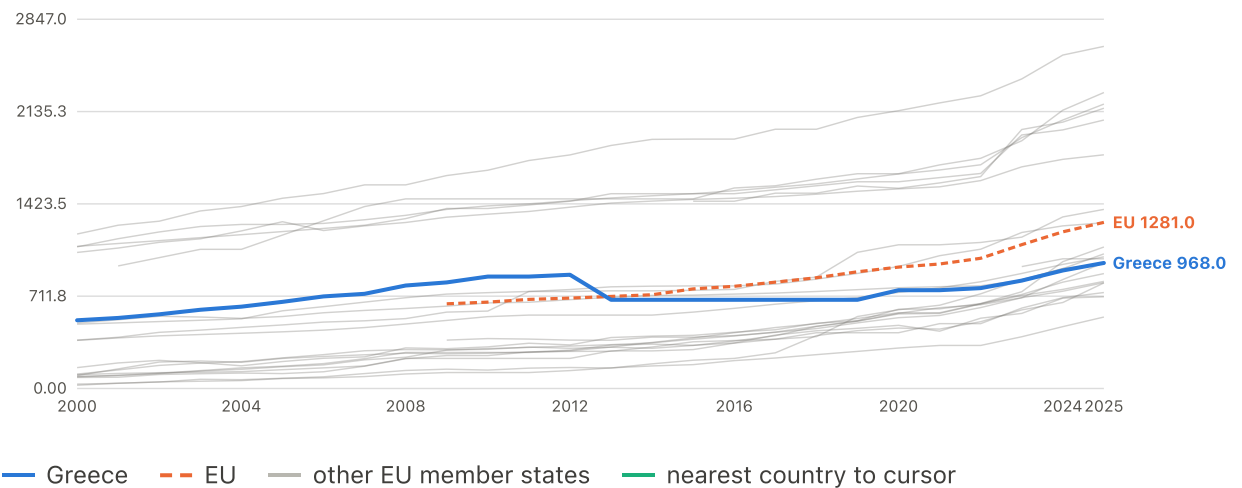
Relative weekly hours divided by relative PPS hourly compensation. The index is normalised so that the population-weighted EU27 aggregate equals 100. The plotted EU line is instead the unweighted mean across member states, which sits well above 100: most member states have below-average hourly pay, and the weighted aggregate is pulled up by the large, high-wage economies. Compare Greece against the individual country lines rather than against 100.

Complete 27-country coverage from 2008. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Minimum wage (first semester of each year)



EUR per month · up to 22 EU member states · 2000–2025



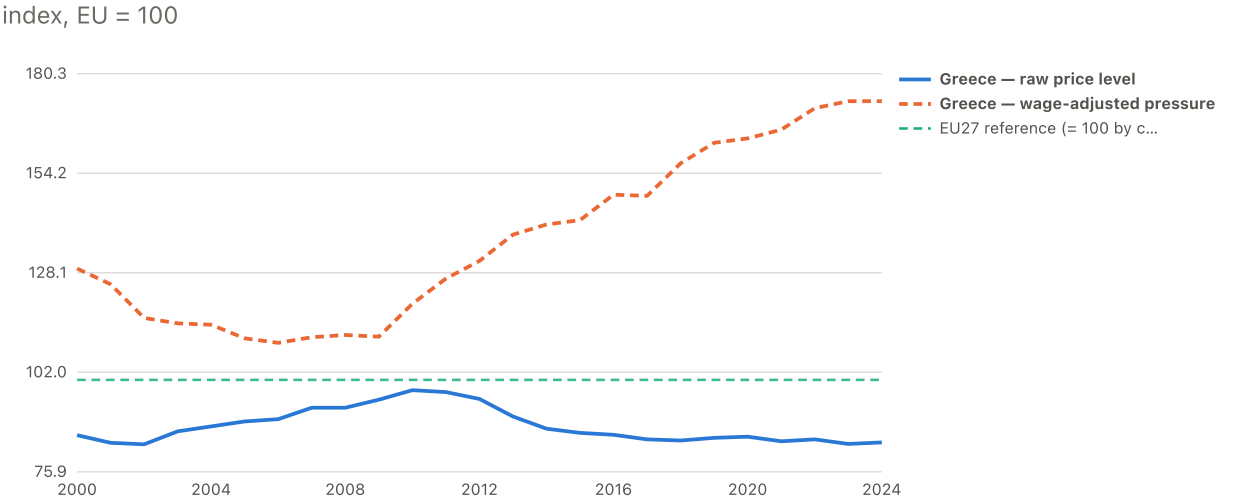
|        | 2000  | 2010  | 2015  | 2020  | 2025   |
|--------|-------|-------|-------|-------|--------|
| Greece | 526.0 | 863.0 | 684.0 | 758.0 | 968.0  |
| EU     | --    | 666.0 | 768.0 | 936.0 | 1281.0 |

Semi-annual series; S1 of each year taken as the annual reference point. Not every member state has a statutory minimum wage.

Complete 22-country coverage from 2023. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

The overall picture

Price level vs wage-adjusted price pressure: Greece vs EU27

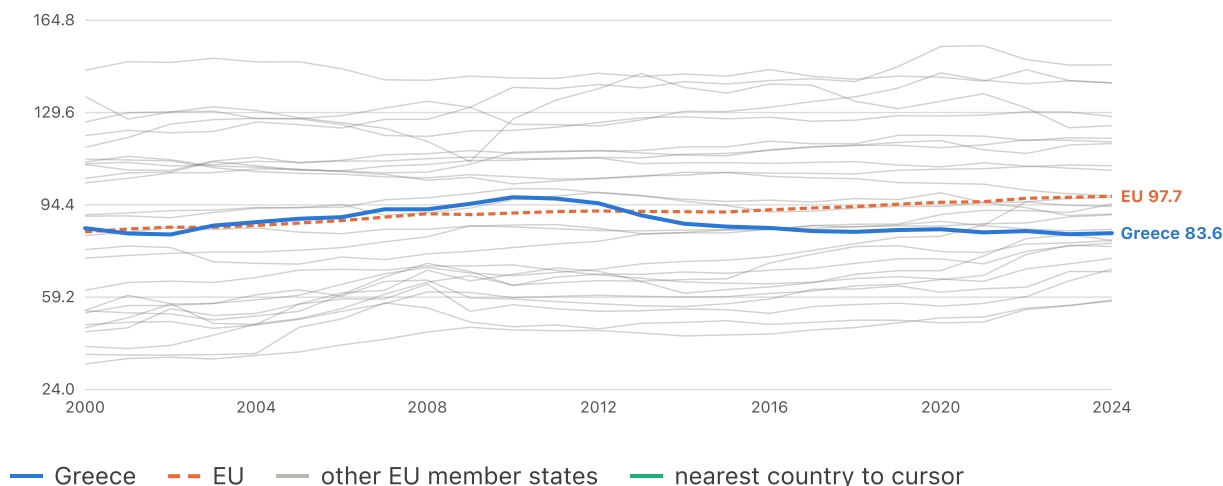


Solid = raw price level (what things cost). Dashed = that price level divided by the country's own wage level (what they cost relative to a local paycheck). Greece sits below the EU on the first and far above it on the second.

EU comparator: Eurostat EU27 aggregate (population-weighted)

Price level — Overall household consumption

index, EU = 100 · up to 27 EU member states · 2000–2024

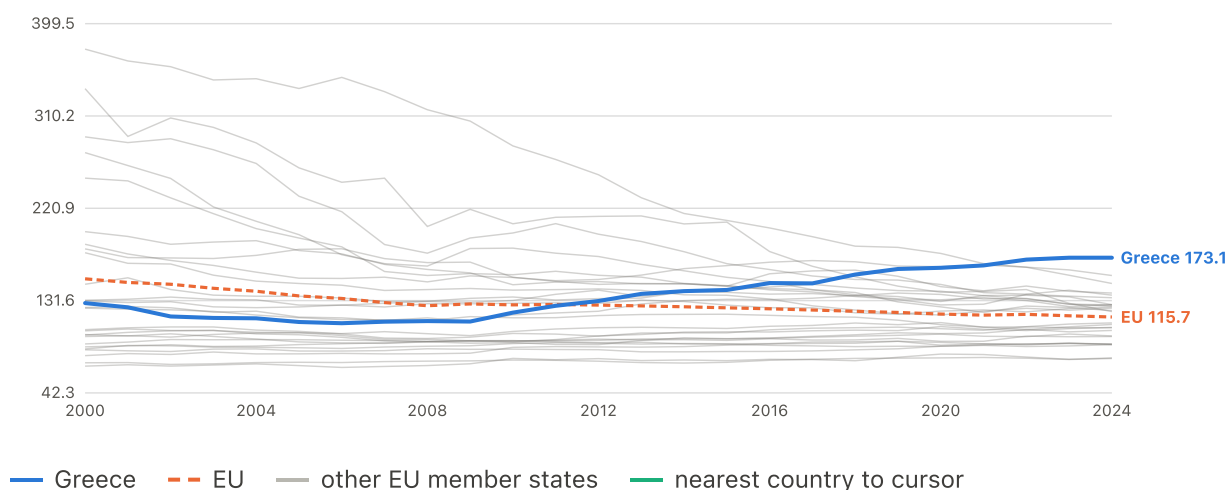


|        | 2000  | 2010  | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|-------|-------|
| Greece | 85.50 | 97.30 | 86.10 | 85.10 | 83.60 |
| EU     | 84.13 | 91.27 | 91.67 | 95.37 | 97.69 |

Comparative price level index, EU27 = 100. Categories other than the overall basket are published for 2022 onward only. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean. Complete 27-country coverage from 2000. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Wage-adjusted price pressure — Overall household consumption

index, EU = 100 · up to 27 EU member states · 2000–2024



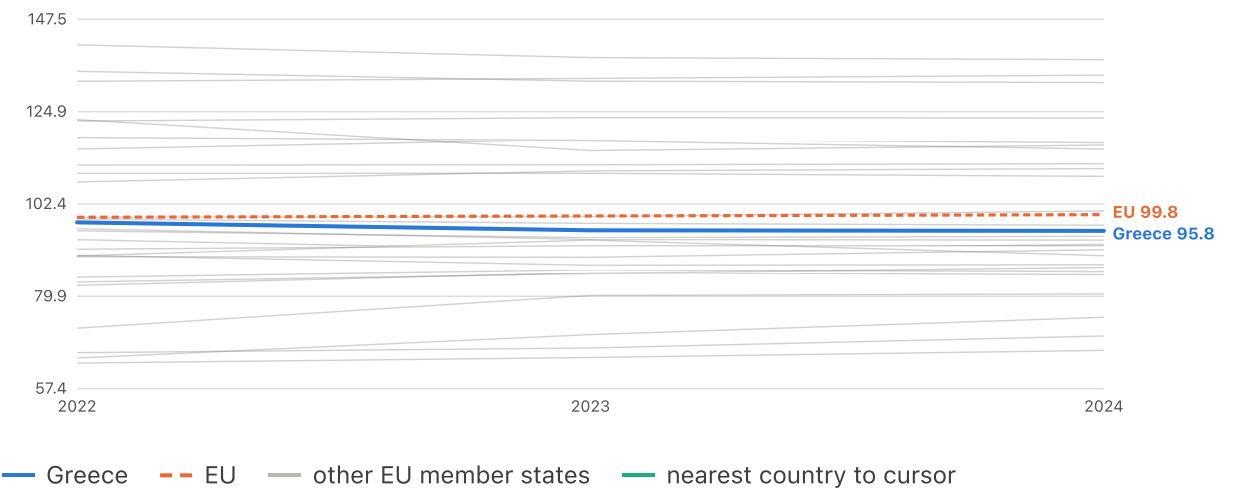
|        | 2000  | 2010  | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|-------|-------|
| Greece | 129.2 | 120.0 | 141.9 | 163.3 | 173.1 |
| EU     | 152.7 | 127.5 | 124.6 | 118.6 | 115.7 |

Price level divided by the country's wage level (EU27 = 100 on both). Above 100 = this category costs proportionally more of a local paycheck than of an EU-average one. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2000. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

### Average price level across consumption categories

index, EU = 100 · up to 27 EU member states · 2022–2024



|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 97.92 | 95.84 |
| EU     | 99.16 | 99.80 |

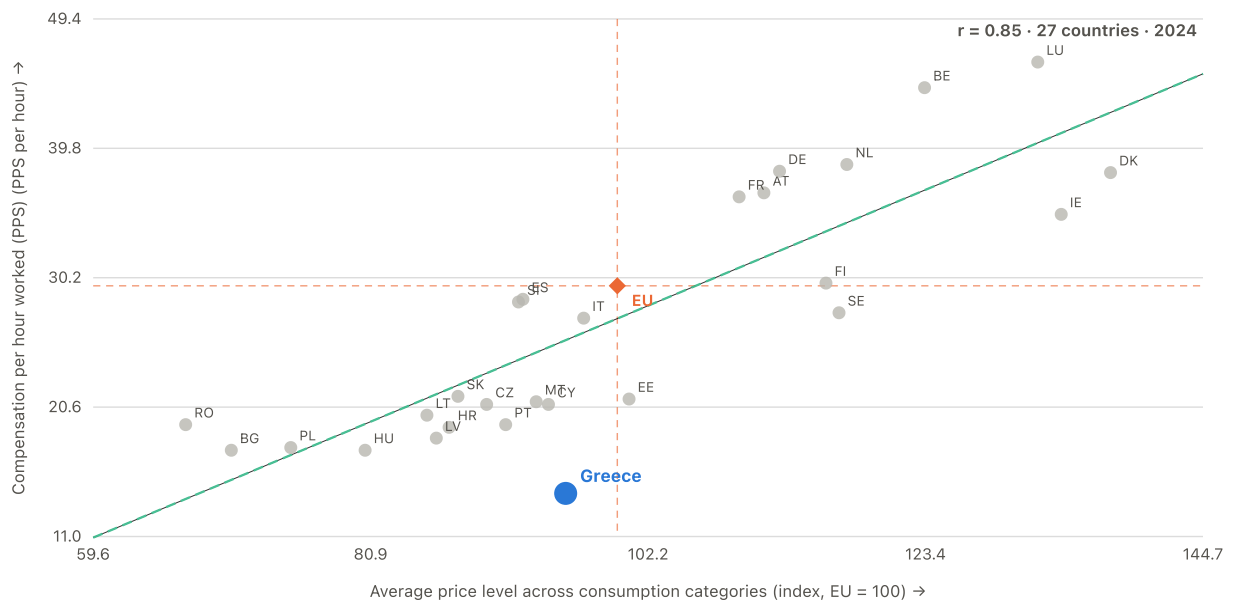
Unweighted mean of the 5 specific categories (food, housing & energy, transport, communications, restaurants), published 2022 onward. Eurostat's weighted overall index is the separate 'Overall household consumption' chart. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

RELATIONSHIP

### What things cost against what an hour of work pays

2024 · 27 member states ·  $r = +0.85$



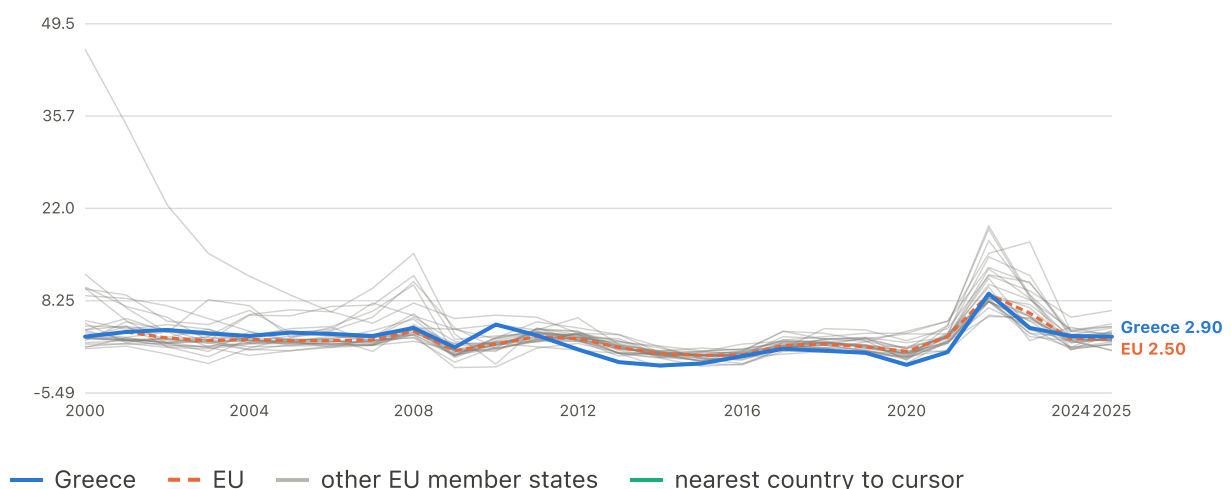
The cost-of-living squeeze as a single frame. The horizontal axis is the average price level across consumption categories, EU27 = 100; the vertical axis is compensation per hour worked in PPS. Richer countries are expensive because they pay well, so the cloud slopes upward. What matters is distance from that line: a country below it faces prices its hourly pay does not support. Greece is mid-pack on prices -- 15th of 27, close to the EU average -- and last of 27 on hourly pay. That combination puts it further below the fitted line than any other member state, and by a wide margin: its shortfall is roughly twice the next country's. This is the same squeeze the wage-adjusted price charts show category by category, here in one picture.

EU marker: unweighted mean of member states (years with <20 reporting dropped) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

## Inflation

### HICP inflation, headline

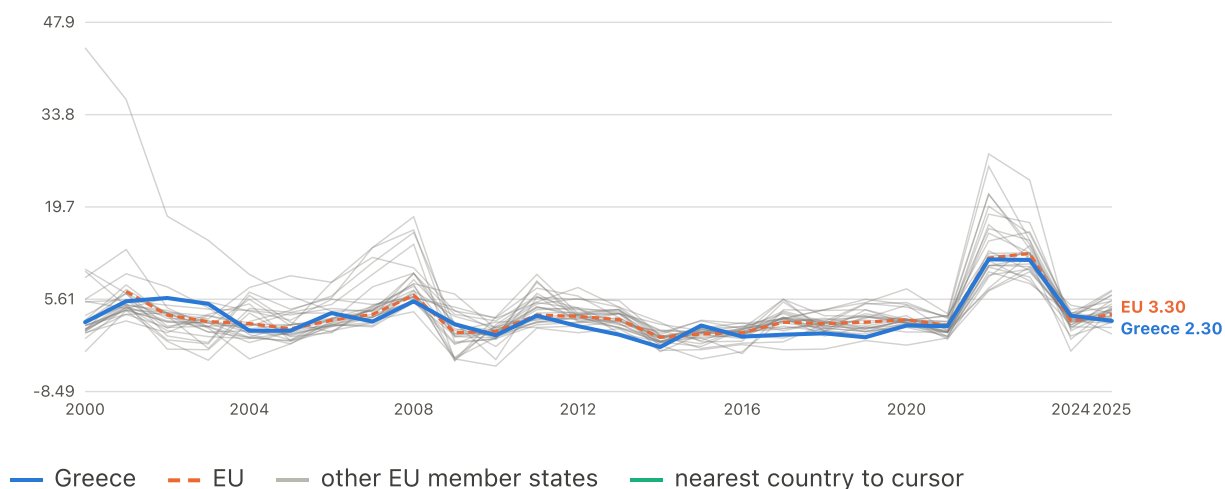
% annual change · up to 27 EU member states · 2000–2025



|        | 2000 | 2010 | 2015  | 2020  | 2025 |
|--------|------|------|-------|-------|------|
| Greece | 2.90 | 4.70 | -1.10 | -1.30 | 2.90 |
| EU     | --   | 1.80 | 0.10  | 0.70  | 2.50 |

## HICP inflation, food & non-alcoholic beverages

% annual change · up to 27 EU member states · 2000–2025

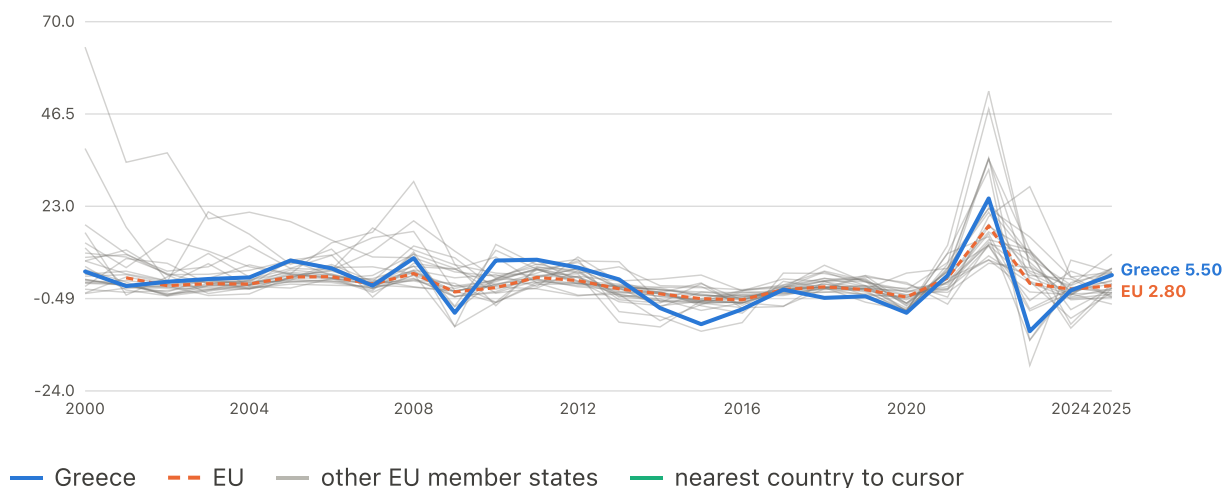


|        | 2000 | 2010 | 2015 | 2020 | 2025 |
|--------|------|------|------|------|------|
| Greece | 2.10 | 0.10 | 1.60 | 1.60 | 2.30 |
| EU     | --   | 0.70 | 0.30 | 2.50 | 3.30 |

Complete 27-country coverage from 2000. EU comparator: Eurostat EU27 aggregate (population-weighted)

## HICP inflation, housing & energy

% annual change · up to 27 EU member states · 2000–2025



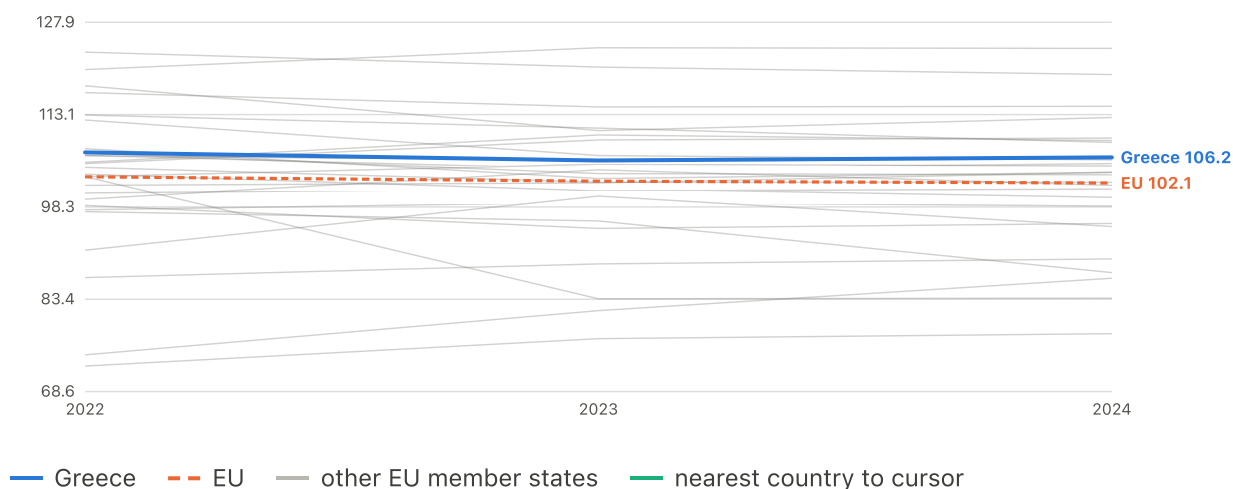
|        | 2000 | 2010 | 2015  | 2020  | 2025 |
|--------|------|------|-------|-------|------|
| Greece | 6.40 | 9.20 | -7.00 | -4.10 | 5.50 |
| EU     | --   | 2.30 | -0.50 | -0.10 | 2.80 |

Complete 27-country coverage from 2000. EU comparator: Eurostat EU27 aggregate (population-weighted)

**Category detail — published for 2022 onward only. Each category appears twice: the raw price level, then the same prices divided by the country's own wage level.**

## Price level — Food & non-alcoholic beverages

index, EU = 100 · up to 27 EU member states · 2022–2024



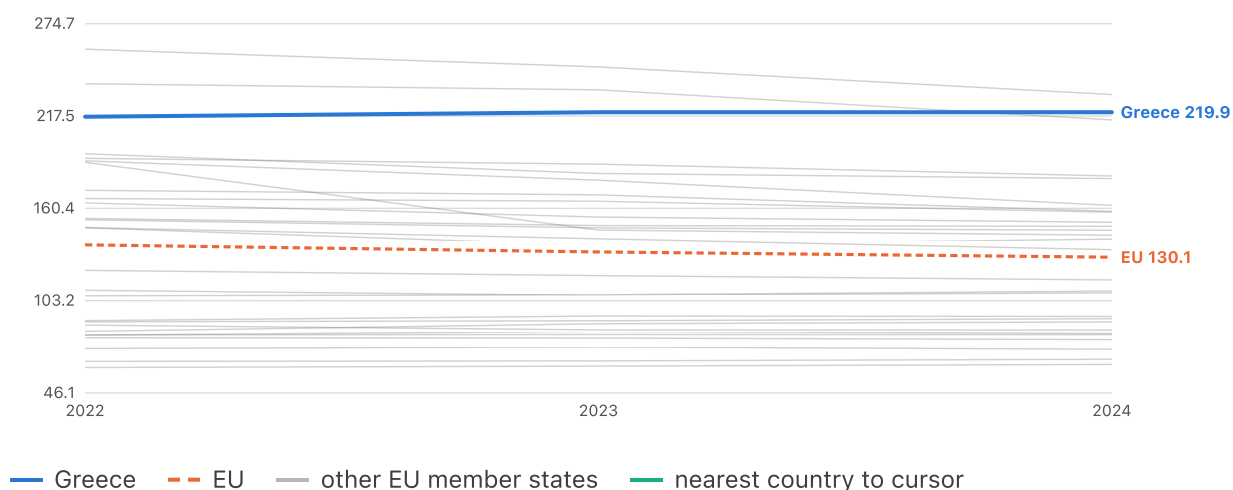
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 107.0 | 106.2 |
| EU     | 103.1 | 102.1 |

Comparative price level index, EU27 = 100. Categories other than the overall basket are published for 2022 onward only. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Wage-adjusted price pressure — Food & non-alcoholic beverages

index, EU = 100 · up to 27 EU member states · 2022–2024



|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 217.1 | 219.9 |
| EU     | 137.8 | 130.1 |

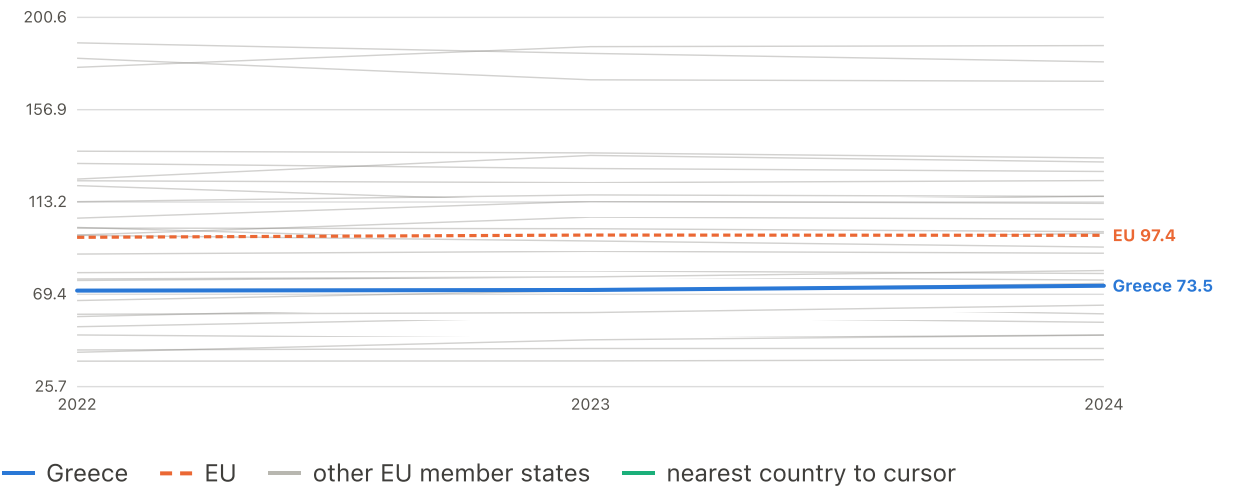
Price level divided by the country's wage level (EU27 = 100 on both). Above 100 = this category costs proportionally more of a local paycheck than of an EU-average one. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-

weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

Price level — Housing, water, electricity, gas & fuels

index, EU = 100 · up to 27 EU member states · 2022–2024



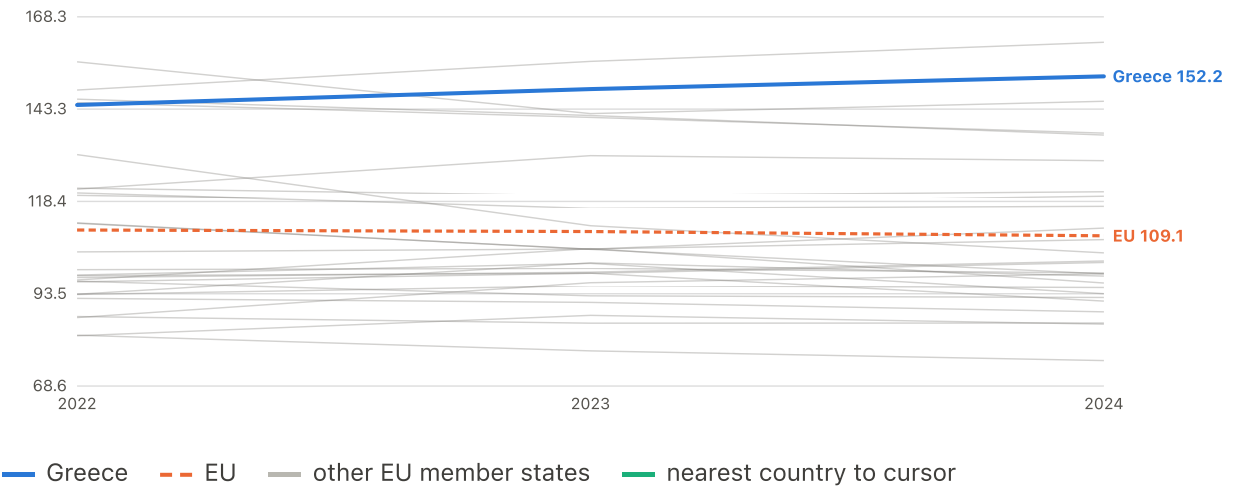
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 71.20 | 73.50 |
| EU     | 96.43 | 97.37 |

Comparative price level index, EU27 = 100. Categories other than the overall basket are published for 2022 onward only. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

Wage-adjusted price pressure — Housing, water, electricity, gas & fuels

index, EU = 100 · up to 27 EU member states · 2022–2024



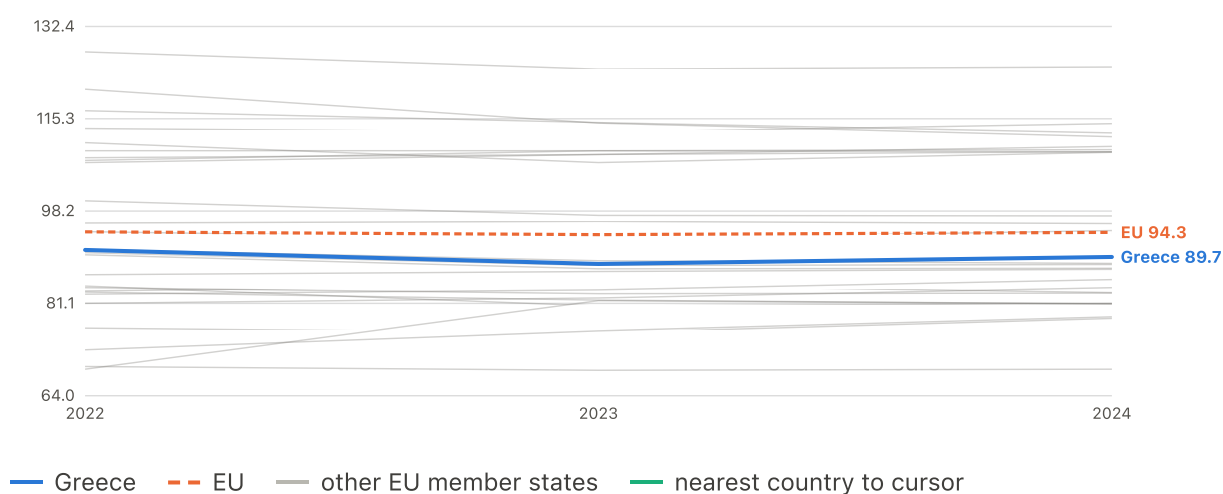
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 144.4 | 152.2 |
| EU     | 110.7 | 109.1 |

Price level divided by the country's wage level (EU27 = 100 on both). Above 100 = this category costs proportionally more of a local paycheck than of an EU-average one. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Price level — Transport

index, EU = 100 · up to 27 EU member states · 2022–2024



|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 91.00 | 89.70 |
| EU     | 94.36 | 94.26 |

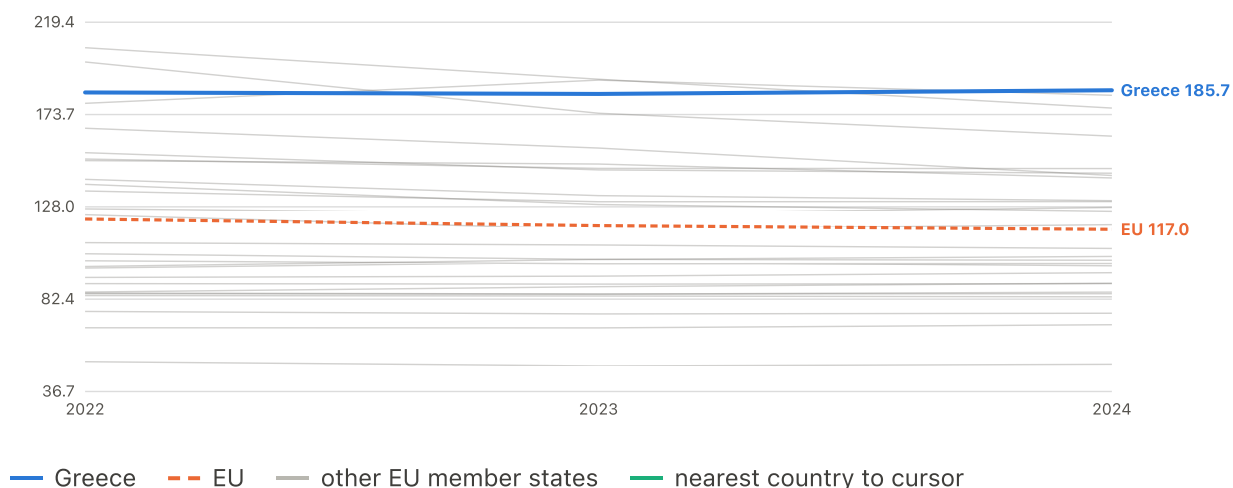
Comparative price level index, EU27 = 100. Categories other than the overall basket are published for 2022 onward only. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Wage-adjusted price pressure — Transport

index, EU = 100 · up to 27 EU member states · 2022–2024





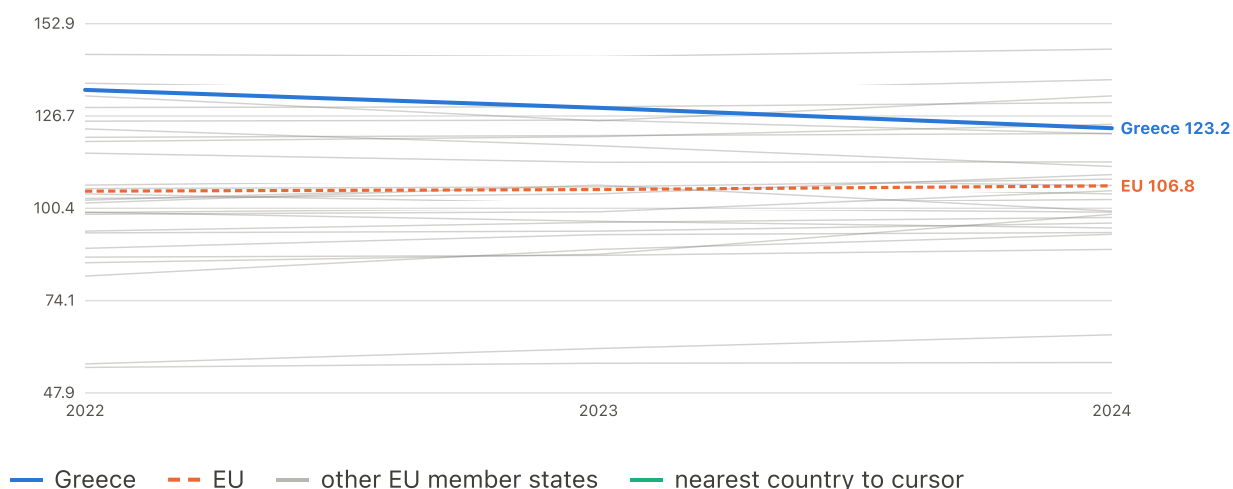
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 184.6 | 185.7 |
| EU     | 122.0 | 117.0 |

Price level divided by the country's wage level (EU27 = 100 on both). Above 100 = this category costs proportionally more of a local paycheck than of an EU-average one. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Price level — Information & communication

index, EU = 100 · up to 27 EU member states · 2022–2024



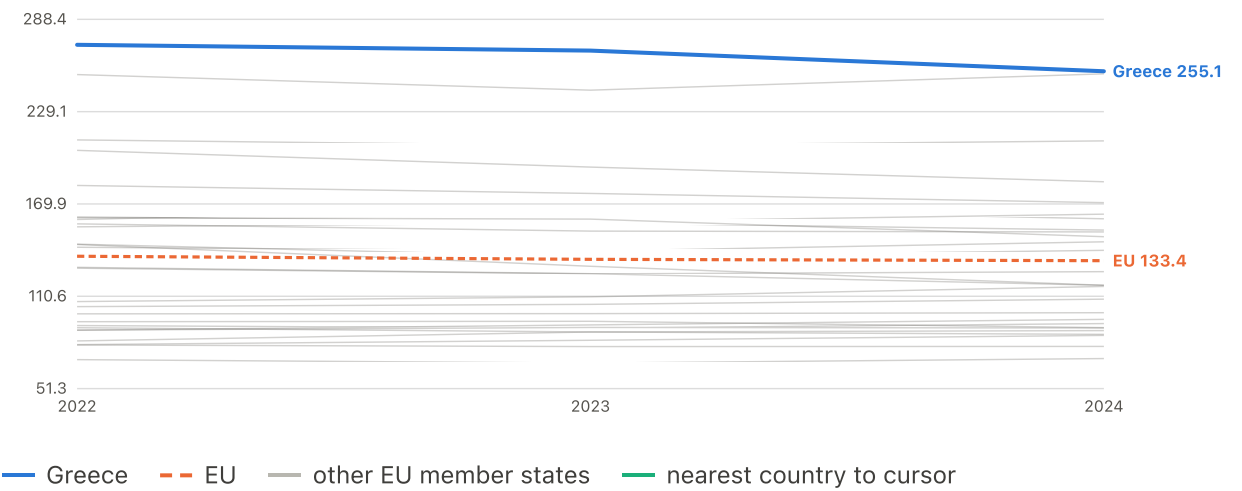
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 134.1 | 123.2 |
| EU     | 105.3 | 106.8 |

Comparative price level index, EU27 = 100. Categories other than the overall basket are published for 2022 onward only. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

Wage-adjusted price pressure — Information & communication

index, EU = 100 · up to 27 EU member states · 2022–2024



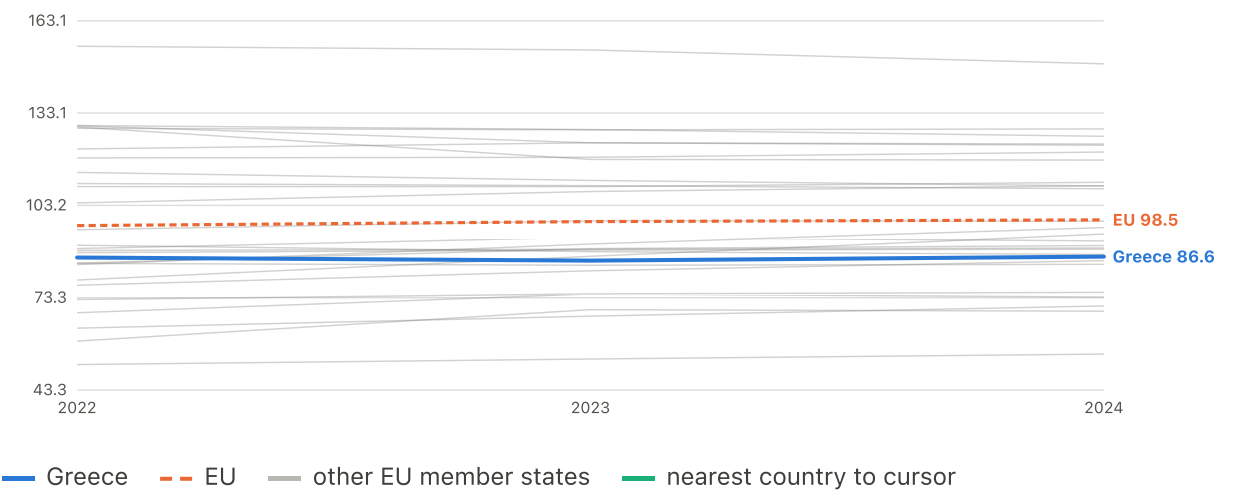
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 272.1 | 255.1 |
| EU     | 136.3 | 133.4 |

Price level divided by the country's wage level (EU27 = 100 on both). Above 100 = this category costs proportionally more of a local paycheck than of an EU-average one. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

Price level — Restaurants & accommodation

index, EU = 100 · up to 27 EU member states · 2022–2024



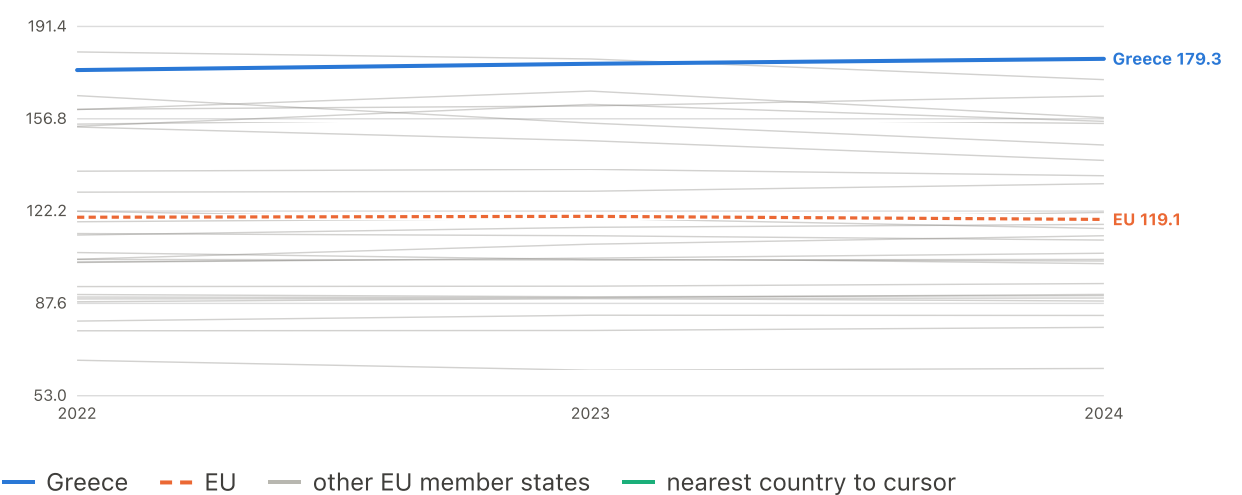
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 86.30 | 86.60 |
| EU     | 96.64 | 98.51 |

Comparative price level index, EU27 = 100. Categories other than the overall basket are published for 2022 onward only. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

### Wage-adjusted price pressure — Restaurants & accommodation

index, EU = 100 · up to 27 EU member states · 2022–2024



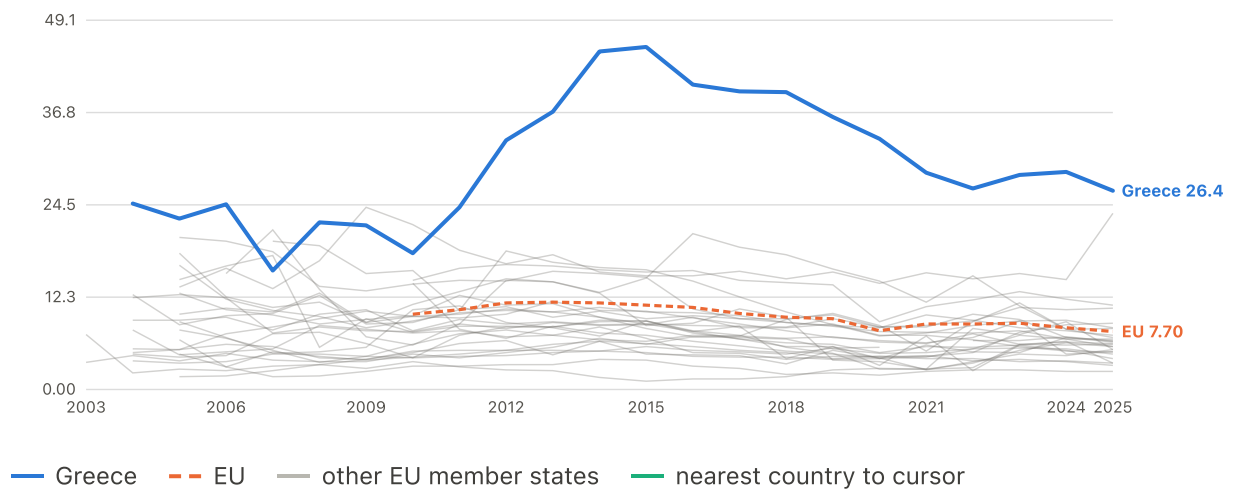
|        | 2022  | 2024  |
|--------|-------|-------|
| Greece | 175.1 | 179.3 |
| EU     | 119.9 | 119.1 |

Price level divided by the country's wage level (EU27 = 100 on both). Above 100 = this category costs proportionally more of a local paycheck than of an EU-average one. The plotted EU line is the unweighted mean across the 27 member states, not the 100 baseline: the index is normalised to the population-weighted EU27 aggregate, and the typical member state sits below it. Read Greece against the country lines and against 100, not against the plotted mean.

Complete 27-country coverage from 2022. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

### Housing cost overburden

% of people · up to 27 EU member states · 2003–2025

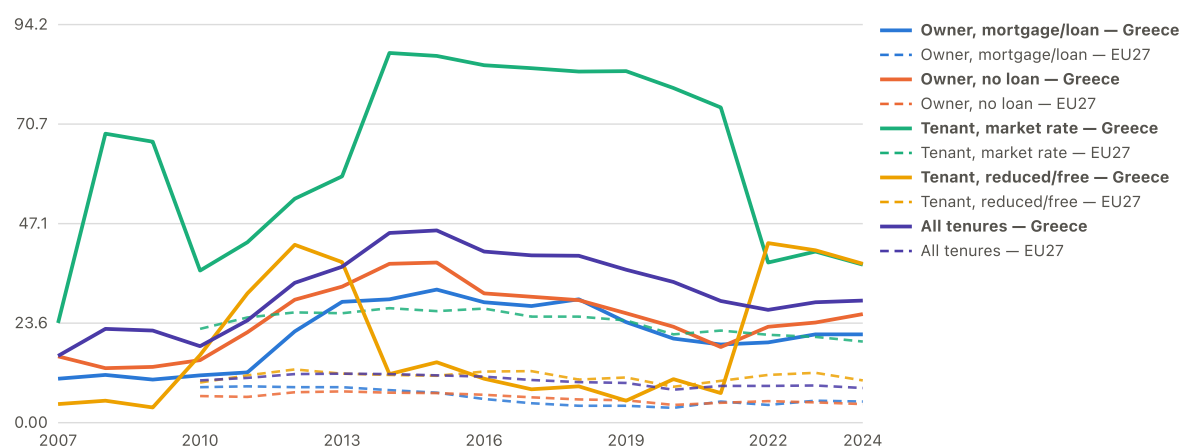


|        | 2004  | 2010  | 2015  | 2020  | 2025  |
|--------|-------|-------|-------|-------|-------|
| Greece | 24.70 | 18.10 | 45.50 | 33.30 | 26.40 |
| EU     | --    | 10.00 | 11.20 | 7.80  | 7.70  |

Complete 27-country coverage from 2010. EU comparator: Eurostat EU27 aggregate (population-weighted)

## Housing cost overburden by tenure: Greece vs EU27

% of people

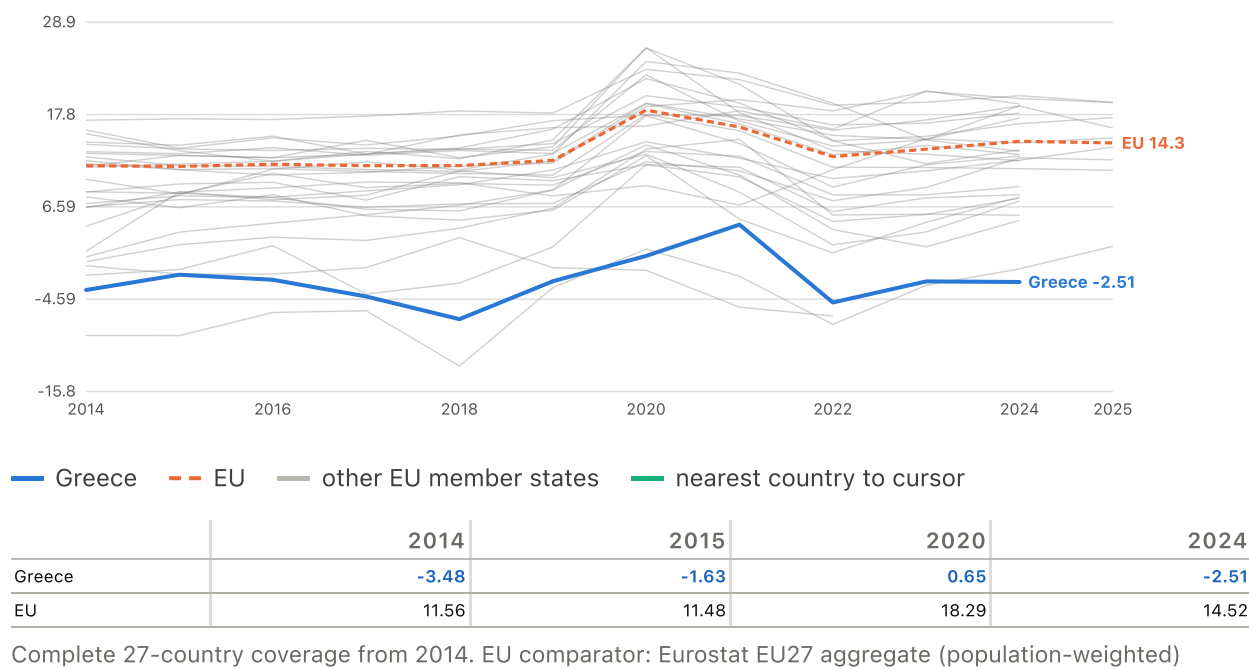


Solid = Greece, dashed = EU27. Greek renter sub-series before ~2021 swing implausibly (small subsample in an ownership-dominated country) and are not used for trend claims.

EU comparator: Eurostat EU27 aggregate (population-weighted)

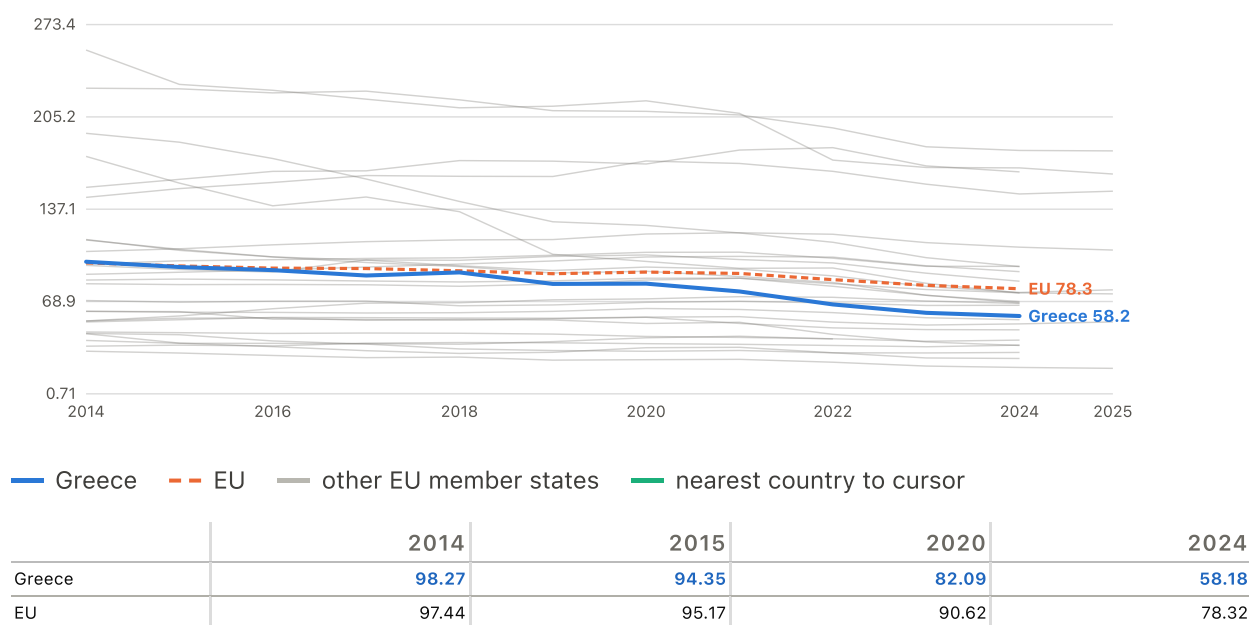
## Household saving rate

% of disposable income · up to 27 EU member states · 2014–2025



## Household debt-to-income

% of disposable income · up to 27 EU member states · 2014–2025



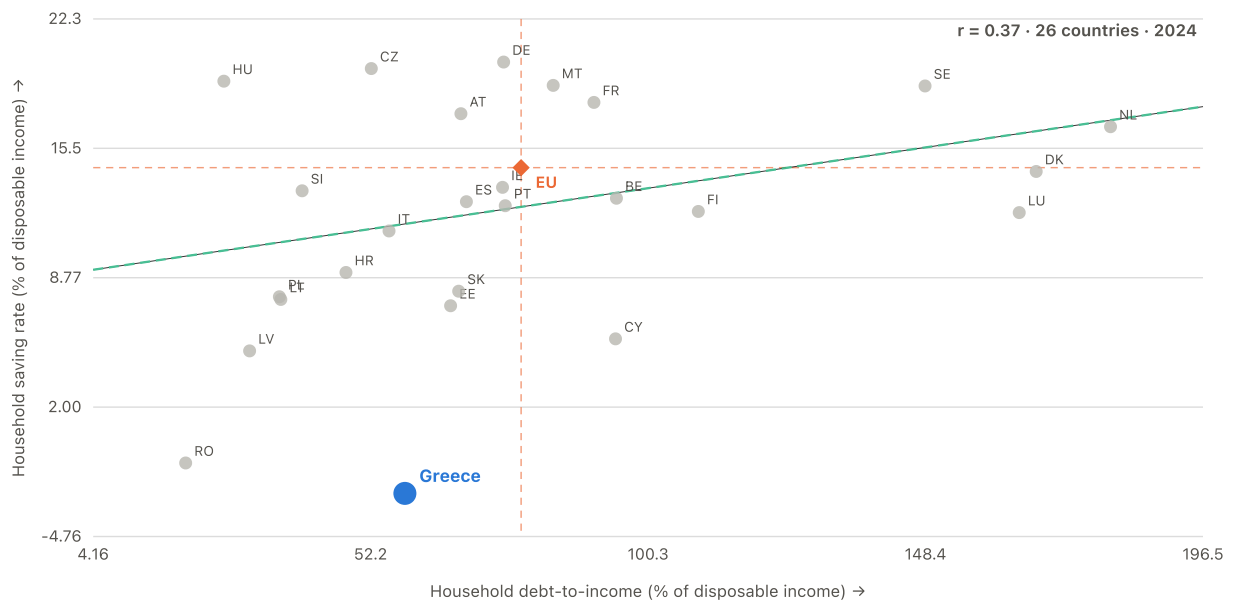
Tested as a model candidate and returned null.

Complete 27-country coverage from 2014. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

### RELATIONSHIP

## Household debt against household saving

2024 · 26 member states ·  $r = +0.37$

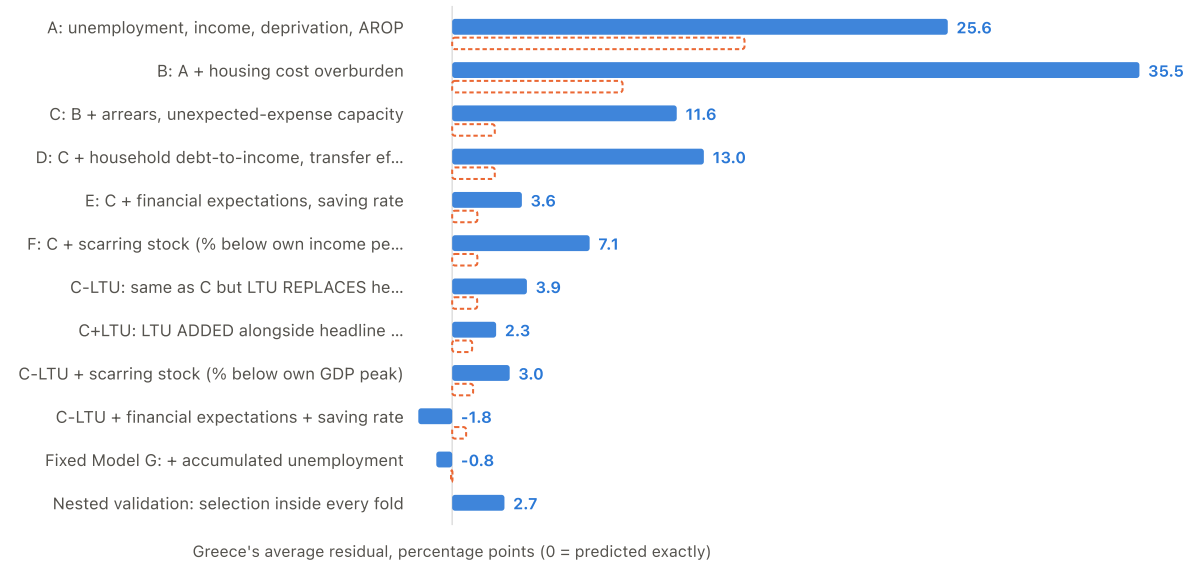


The counter-narrative pair. Greece has lower debt than most of the EU and a negative saving rate: the problem is not over-borrowing, it is that there is nothing left to save. Low debt here should not be read as financial health.

EU marker: unweighted mean of member states (years with <20 reporting dropped) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

### Every model side by side: Greece's unexplained gap

percentage points



Solid = out-of-sample (model refit without Greece, then applied to it). Dashed outline = in-sample (model allowed to see Greece's own data). The in-sample figure is always the more flattering one.

### How the final predictors overlap

Pearson correlation, country-year panel

|                     | AROP  | AIC (PPS) | Deprivation | LTU   | Housing | Arrears | Unexpected expense | Accum. unemployment |
|---------------------|-------|-----------|-------------|-------|---------|---------|--------------------|---------------------|
| AROP                | 1.00  | -0.35     | 0.56        | 0.37  | 0.19    | 0.37    | 0.59               | 0.23                |
| AIC (PPS)           | -0.35 | 1.00      | -0.52       | -0.35 | -0.07   | -0.42   | -0.61              | -0.17               |
| Deprivation         | 0.56  | -0.52     | 1.00        | 0.38  | 0.42    | 0.61    | 0.64               | 0.21                |
| LTU                 | 0.37  | -0.35     | 0.38        | 1.00  | 0.59    | 0.68    | 0.53               | 0.65                |
| Housing             | 0.19  | -0.07     | 0.42        | 0.59  | 1.00    | 0.64    | 0.23               | 0.57                |
| Arrears             | 0.37  | -0.42     | 0.61        | 0.68  | 0.64    | 1.00    | 0.66               | 0.77                |
| Unexpected expense  | 0.59  | -0.61     | 0.64        | 0.53  | 0.23    | 0.66    | 1.00               | 0.43                |
| Accum. unemployment | 0.23  | -0.17     | 0.21        | 0.65  | 0.57    | 0.77    | 0.43               | 1.00                |

High correlation does not make two variables identical, but it warns against reading their coefficients as isolated causal contributions.

## Every screened candidate against every other

Pearson correlation, country-year panel



All 37 variables the project tested anywhere: the outcome, the seven Model C-LTU covariates, and the three screening families, blocked by family with dividing lines. Cells are too small to print numbers at this size, so hover any cell for the pair, its correlation and the variance they share. The redundancy that closed families B and C is OFF the diagonal blocks, not inside them: family B is internally diverse (mean  $|r|$  0.19), but its members pair almost one-to-one with family C and with family A's wage measures — years-worst-quintile-wage against share-worst-real-wage is 0.964, against cumulative wage shortfall 0.948. Those bright cells spanning the family dividers are the picture of three families measuring one thing three ways. Read it as description of overlap, not as evidence about any variable's effect — correlation among predictors says nothing about which of them explains the outcome.

## Stage 5

# What accumulated history adds

Does the length of a country's difficulty carry information beyond its present state?



## Greece ranks first on accumulated unemployment and housing deterioration, and second on wage non-recovery

### POST-SELECTION ROBUSTNESS

How much accumulated unemployment has each country absorbed, and where does Greece sit?

**Read with this.** cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out The three measures are in different units and may not be compared with each other: accumulated unemployment and housing deterioration are percentage-point-years, wage non-recovery is a count of consecutive years.

| Accumulated unemployment |                                                              |
|--------------------------|--------------------------------------------------------------|
| Country                  | Accumulated unemployment (percentage-point-years above 2009) |
| Greece                   | 137.5                                                        |
| Cyprus                   | 64.1                                                         |
| Croatia                  | 43.5                                                         |
| Spain                    | 33.1                                                         |
| Italy                    | 32.9                                                         |
| Bulgaria                 | 26.6                                                         |
| Portugal                 | 20.6                                                         |
| Slovenia                 | 20.6                                                         |
| Netherlands              | 13.1                                                         |
| Luxembourg               | 9.1                                                          |
| Ireland                  | 8.9                                                          |
| Slovakia                 | 8.9                                                          |
| Poland                   | 7.7                                                          |
| France                   | 5.9                                                          |
| Lithuania                | 5.6                                                          |
| Denmark                  | 5.6                                                          |
| Hungary                  | 3.2                                                          |
| Estonia                  | 3.1                                                          |
| Finland                  | 2.9                                                          |
| Austria                  | 2.5                                                          |
| Belgium                  | 2.4                                                          |
| Romania                  | 2.4                                                          |
| Latvia                   | 2.0                                                          |
| Czechia                  | 1.2                                                          |
| Sweden                   | 0.6                                                          |
| Malta                    | 0.0                                                          |
| Germany                  | 0.0                                                          |

| Years wages below 2008 |                                                                 |
|------------------------|-----------------------------------------------------------------|
| Country                | Years wages below 2008 (consecutive years below the 2008 level) |
| Hungary                | 16.0                                                            |
| Greece                 | 15.0                                                            |
| Italy                  | 14.0                                                            |
| Finland                | 3.0                                                             |
| Sweden                 | 2.0                                                             |
| Slovenia               | 0.0                                                             |
| Romania                | 0.0                                                             |
| Portugal               | 0.0                                                             |
| Poland                 | 0.0                                                             |
| Netherlands            | 0.0                                                             |
| Malta                  | 0.0                                                             |
| Latvia                 | 0.0                                                             |
| Luxembourg             | 0.0                                                             |
| Lithuania              | 0.0                                                             |
| Austria                | 0.0                                                             |
| Ireland                | 0.0                                                             |
| Belgium                | 0.0                                                             |
| Croatia                | 0.0                                                             |
| France                 | 0.0                                                             |
| Spain                  | 0.0                                                             |
| Estonia                | 0.0                                                             |
| Denmark                | 0.0                                                             |
| Germany                | 0.0                                                             |
| Czechia                | 0.0                                                             |
| Cyprus                 | 0.0                                                             |
| Bulgaria               | 0.0                                                             |
| Slovakia               | 0.0                                                             |

## Housing deterioration since 2010

| Country     | Housing deterioration since 2010 (percentage-point-years above 2010) |
|-------------|----------------------------------------------------------------------|
| Greece      | 233.2                                                                |
| Bulgaria    | 116.3                                                                |
| Luxembourg  | 37.8                                                                 |
| Portugal    | 35.8                                                                 |
| Sweden      | 17.1                                                                 |
| Estonia     | 11.8                                                                 |
| Slovenia    | 9.9                                                                  |
| Hungary     | 9.6                                                                  |
| Germany     | 9.4                                                                  |
| Italy       | 8.2                                                                  |
| Belgium     | 7.9                                                                  |
| Finland     | 7.0                                                                  |
| France      | 6.6                                                                  |
| Slovakia    | 6.1                                                                  |
| Latvia      | 5.7                                                                  |
| Netherlands | 4.9                                                                  |
| Ireland     | 4.6                                                                  |
| Spain       | 4.5                                                                  |
| Malta       | 4.5                                                                  |
| Austria     | 4.3                                                                  |
| Poland      | 4.2                                                                  |
| Romania     | 4.2                                                                  |
| Czechia     | 3.8                                                                  |
| Cyprus      | 2.1                                                                  |
| Lithuania   | 0.5                                                                  |
| Croatia     | 0.0                                                                  |
| Denmark     | 0.0                                                                  |

Accumulated history provides additional conditional cross-country information in three of eight pairs

POST-SELECTION ROBUSTNESS

Does accumulated history add anything once today's conditions are in the same model?

**Read with this.** cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out This shows the pairs that survive; all eight, including the five that do not, are in the statistical appendix. The axis is STANDARDISED: SDs of hardship per SD of the predictor. The eight measures are in different units - percentage-point-years, years, index points, percentages - so raw coefficients could not share an axis, and the raw value is given in the table and the tooltip instead. Each coefficient is tested against ITS OWN pair-specific detectable effect, not a study-wide threshold: conditional power depends on how much independent variation survives once the counterpart is controlled. Bars are cluster-robust intervals; the bootstrap decides support.

| Test                                                                       | Standardised effect (SD) | Raw coefficient | p FDR | Bootstrap p | Conditional MDE (SD) |
|----------------------------------------------------------------------------|--------------------------|-----------------|-------|-------------|----------------------|
| Accumulated unemployment (controlling for Long-term unemployment)          | 0.550                    | 0.294           | 0.000 | 0.003       | 0.600                |
| Years wages below 2008 (controlling for Real wages)                        | 0.624                    | 2.090           | 0.005 | 0.006       | 0.500                |
| Housing deterioration since 2010 (controlling for Housing-cost overburden) | 0.668                    | 0.242           | 0.000 | 0.002       | 0.600                |

The supported historical associations are predominantly between countries; the within-country estimates provide no supporting dynamic evidence

PRE-REGISTERED CONDITIONAL ROBUSTNESS Is the effect between countries or within them?

**Read with this.** This is THE central limitation: no within-country estimate is significant in the adverse direction and no first-difference test supports one. The first view is STANDARDISED, and each component by its own spread: between-country and within-country variation differ in size by factors of 0.8 to 5.7 here, so one shared SD would distort the comparison it is meant to fix. The second view shows first differences in each measure's own units, which is why those bars carry no shared scale and are read one row at a time. This does NOT establish that no within-country relationship exists: the within estimates are too imprecise to establish or rule out such a relationship. They are recorded as inconclusive, not absent.

| Accumulated measure                | Between vs within |             |           |          | First difference (raw) | First difference p |
|------------------------------------|-------------------|-------------|-----------|----------|------------------------|--------------------|
|                                    | Between (SD)      | Within (SD) | Between p | Within p |                        |                    |
| Accumulated unemployment           | 0.5924            | 0.0296      | 0.0000    | 0.6867   | 0.0461                 | 0.7324             |
| Cumulative wage shortfall          | 0.6584            | -0.0544     | 0.0130    | 0.3147   | -0.0596                | 0.0284             |
| Years wages below 2008             | 0.7271            | 0.0700      | 0.0025    | 0.0823   | 0.0721                 | 0.4952             |
| Cumulative GDP shortfall           | 0.3890            | 0.0674      | 0.0289    | 0.0600   | 0.0273                 | 0.1442             |
| Cumulative threshold shortfall     | 0.8271            | 0.0123      | 0.0000    | 0.5599   | -0.0148                | 0.3917             |
| Accumulated affordability pressure | -0.3953           | -0.0764     | 0.3639    | 0.2825   | -0.0298                | 0.0000             |
| Compounded inflation since 2008    | -0.1718           | -0.1228     | 0.4947    | 0.3738   | -0.0314                | 0.5766             |
| Housing deterioration since 2010   | 0.9282            | -0.0658     | 0.0000    | 0.2343   | -0.0090                | 0.8679             |
| Year-on-year changes               |                   |             |           |          |                        |                    |
| Accumulated measure                | First difference  |             | p         |          | n                      |                    |
| Accumulated unemployment           | 0.0461            |             | 0.7324    |          | 243.0000               |                    |
| Cumulative wage shortfall          | -0.0596           |             | 0.0284    |          | 243.0000               |                    |
| Years wages below 2008             | 0.0721            |             | 0.4952    |          | 243.0000               |                    |
| Cumulative GDP shortfall           | 0.0273            |             | 0.1442    |          | 243.0000               |                    |
| Cumulative threshold shortfall     | -0.0148           |             | 0.3917    |          | 234.0000               |                    |
| Accumulated affordability pressure | -0.0298           |             | 0.0000    |          | 243.0000               |                    |
| Compounded inflation since 2008    | -0.0314           |             | 0.5766    |          | 243.0000               |                    |
| Housing deterioration since 2010   | -0.0090           |             | 0.8679    |          | 242.0000               |                    |

Views the report carried and then simplified away -- removed to leave one question per figure, not because the numbers stopped mattering.

SIMPLIFIED OUT OF THE REPORT

## All eight accumulated measures against their present-day counterparts

APPENDIX

What does the accumulated measure add once its present-day counterpart is in the same model, for every pair tested?

**Read with this.** The complete comparison the report's conditional figure leads from. Five of the eight pairs do not survive, and are here rather than there because a figure that leads with thirteen null rows buries the three that are not. The axis is standardised: the eight measures are in different units, so raw coefficients could not share it.

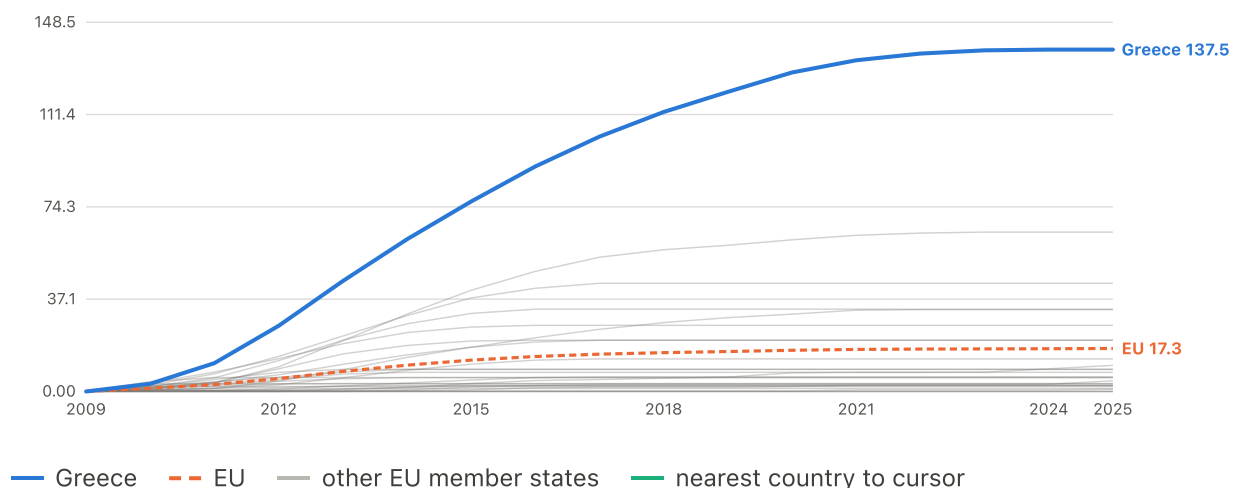
| Test                                                                             | Standardised effect (SD) | Raw coefficient | p FDR  | Bootstrap p | Conditional MDE (SD) |
|----------------------------------------------------------------------------------|--------------------------|-----------------|--------|-------------|----------------------|
| Accumulated unemployment (controlling for Long-term unemployment)                | 0.5505                   | 0.2936          | 0.0000 | 0.0025      | 0.6000               |
| Long-term unemployment (controlling for Accumulated unemployment)                | 0.3596                   | 1.9318          | 0.0207 | 0.1075      | 0.6000               |
| Cumulative wage shortfall (controlling for Real wages)                           | 0.6020                   | 0.1094          | 0.0202 | 0.0060      | 0.6000               |
| Real wages (controlling for Cumulative wage shortfall)                           | 0.1487                   | 0.0788          | 0.5594 | —           | 0.5000               |
| Years wages below 2008 (controlling for Real wages)                              | 0.6239                   | 2.0897          | 0.0051 | 0.0060      | 0.5000               |
| Real wages (controlling for Years wages below 2008)                              | 0.1992                   | 0.1055          | 0.3610 | —           | 0.5000               |
| Cumulative GDP shortfall (controlling for Share below own GDP peak)              | 0.3729                   | 0.1057          | 0.0064 | 0.1155      | 0.6000               |
| Share below own GDP peak (controlling for Cumulative GDP shortfall)              | 0.3284                   | 0.9889          | 0.0018 | 0.1480      | 0.4000               |
| Cumulative threshold shortfall (controlling for Real poverty threshold)          | 0.7693                   | 0.1848          | 0.0000 | 0.0755      | 0.5000               |
| Real poverty threshold (controlling for Cumulative threshold shortfall)          | 0.3017                   | 0.1215          | 0.0260 | 0.0075      | 0.6000               |
| Accumulated affordability pressure (controlling for Wage-adjusted affordability) | -0.3971                  | -0.0124         | 0.3610 | —           | 0.8000               |
| Wage-adjusted affordability (controlling for Accumulated affordability pressure) | 0.9829                   | 0.4449          | 0.0189 | 0.0035      | 0.8000               |
| Compounded inflation since 2008 (controlling for Inflation)                      | -0.2516                  | -0.1667         | 0.5178 | —           | 0.8000               |
| Inflation (controlling for Compounded inflation since 2008)                      | -0.0400                  | -0.1446         | 0.8130 | —           | 0.5000               |
| Housing deterioration since 2010 (controlling for Housing-cost overburden)       | 0.6678                   | 0.2423          | 0.0000 | 0.0015      | 0.6000               |
| Housing-cost overburden (controlling for Housing deterioration since 2010)       | 0.0422                   | 0.0881          | 0.8460 | —           | 0.6000               |

## Underlying variables and tables

21 charts

### Cumulative excess unemployment since 2009

accumulated pp-years · up to 27 EU member states · 2009–2025



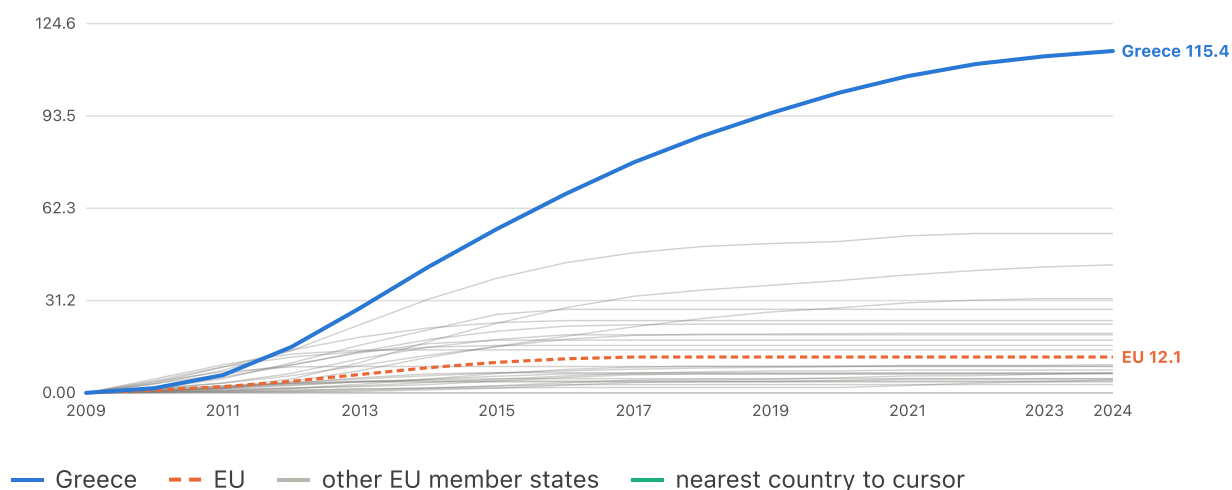
|        | 2009 | 2010 | 2015  | 2020  | 2025  |
|--------|------|------|-------|-------|-------|
| Greece | 0.00 | 3.10 | 76.40 | 128.3 | 137.5 |
| EU     | 0.00 | 1.32 | 12.61 | 16.57 | 17.30 |

Each country's rate above its own 2009 level, summed year on year and floored so recovery cannot cancel earlier damage. Full history shown; the models estimate on 2015-2024.

Complete 27-country coverage from 2009. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Cumulative excess long-term unemployment since 2009

accumulated pp-years · up to 27 EU member states · 2009–2024



|        | 2009 | 2010 | 2015  | 2020  | 2024  |
|--------|------|------|-------|-------|-------|
| Greece | 0.00 | 1.60 | 55.30 | 101.3 | 115.4 |
| EU     | 0.00 | 0.90 | 10.30 | 12.10 | 12.10 |

Each country's rate above its own 2009 level, summed year on year and floored so recovery cannot cancel earlier damage. Full history shown; the models estimate on 2015-2024.

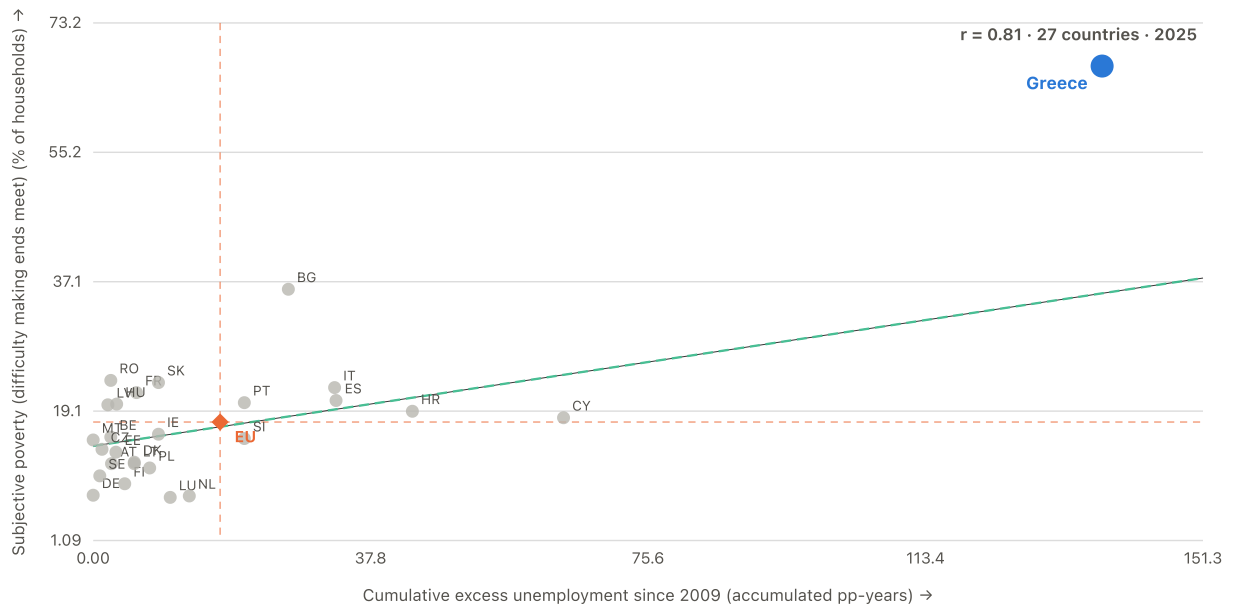
Complete 27-country coverage from 2009. EU comparator: Eurostat EU27 aggregate (population-weighted)

### RELATIONSHIP

## Accumulated unemployment exposure against reported hardship

2025 · 27 member states ·  $r = +0.81$





The headline mechanism. Greece is far out along the x-axis: not merely a bad labour market now, but more accumulated exposure than any other member state.

EU marker: unweighted mean of member states (years with <20 reporting dropped) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

### Conditional contribution of accumulated unemployment

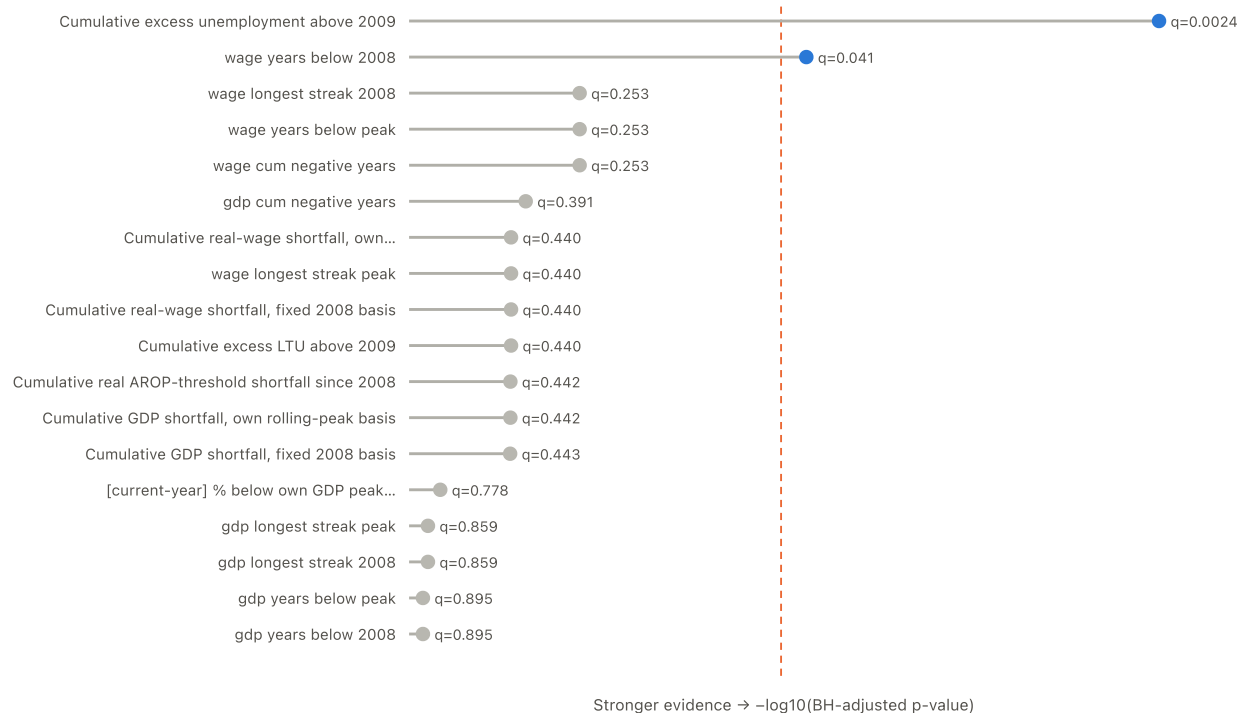
residualised percentage points



Added-variable plot after removing AROP, consumption, deprivation, LTU, housing, arrears, unexpected-expense capacity and year effects from both axes. Grey dots are country-years; blue dots are Greece. Association, not a household-level causal effect.

### All 18 screened cumulative and duration candidates

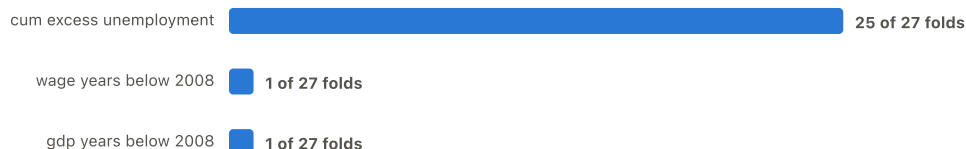
Benjamini-Hochberg adjusted p-value



The vertical line marks  $q = 0.05$ . Only cumulative excess unemployment and years of real wages below 2008 survive family-wide correction.

## Which candidate wins when selection is repeated inside each fold?

held-out-country folds, out of 27



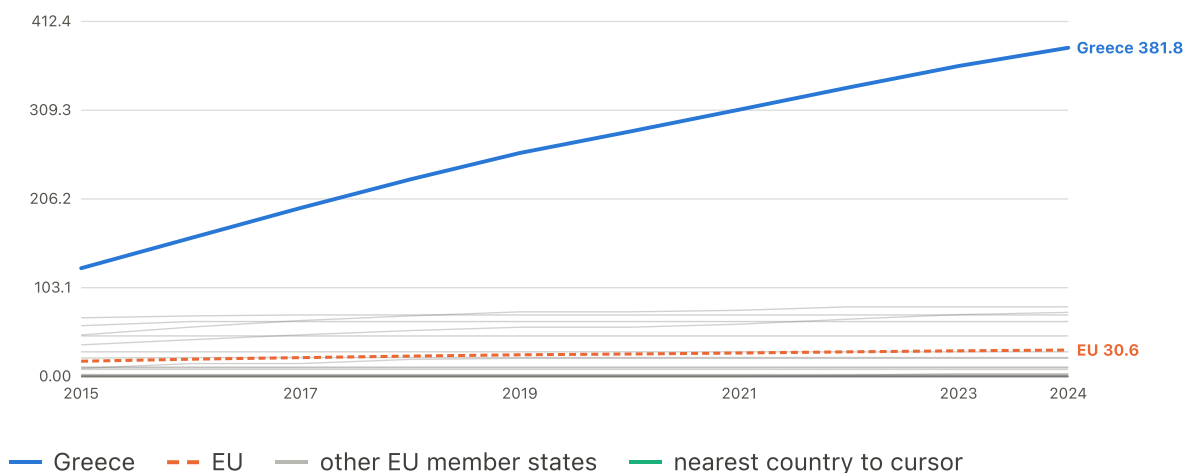
Cumulative excess unemployment ranks first in 25 of 27 folds. Greece and Finland select near-neighbour duration measures instead, supporting the family rather than one uniquely privileged construction.

## Cumulative shortfall constructions

### ▼ View candidate trajectory: Cumulative AROP-threshold shortfall since 2008

## Cumulative AROP-threshold shortfall since 2008

accumulated index-points · up to 27 EU member states · 2015–2024



|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 125.7 | 284.4 | 381.8 |
| EU     | 17.66 | 26.05 | 30.63 |

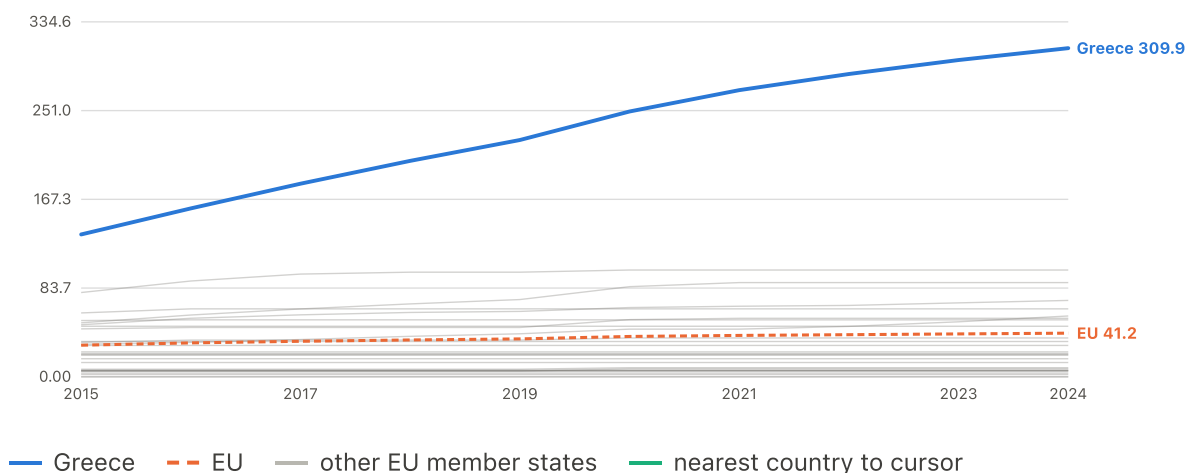
Screened as one 18-candidate family; see the technical report's Methods.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

#### ▼ View candidate trajectory: Cumulative GDP shortfall (fixed 2008 basis)

### Cumulative GDP shortfall (fixed 2008 basis)

accumulated index-points · up to 27 EU member states · 2015–2024



|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 134.2 | 250.1 | 309.9 |
| EU     | 29.81 | 38.05 | 41.16 |

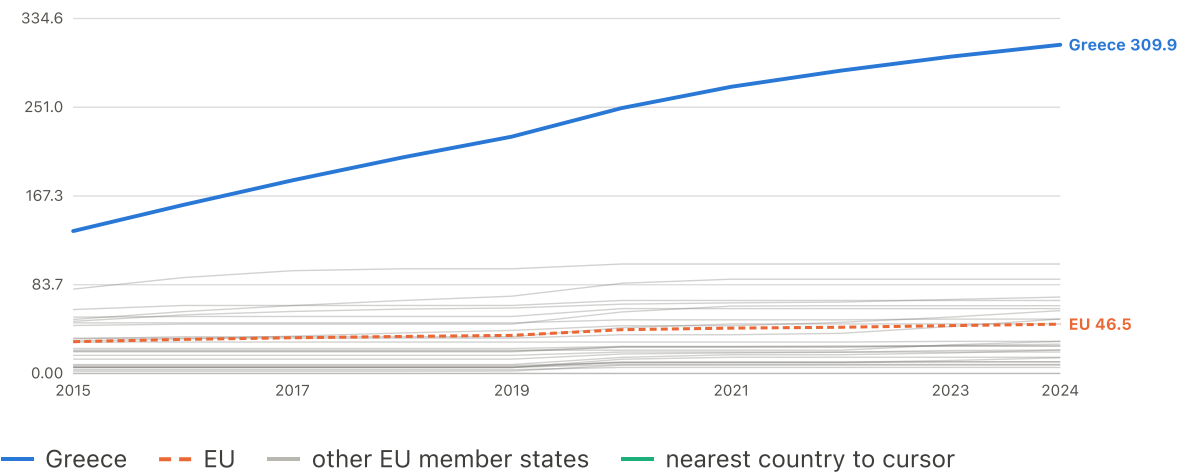
Screened as one 18-candidate family; see the technical report's Methods.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

#### ▼ View candidate trajectory: Cumulative GDP shortfall (own rolling peak)

### Cumulative GDP shortfall (own rolling peak)

accumulated index-points · up to 27 EU member states · 2015–2024



|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 134.2 | 250.1 | 309.9 |
| EU     | 29.93 | 41.25 | 46.46 |

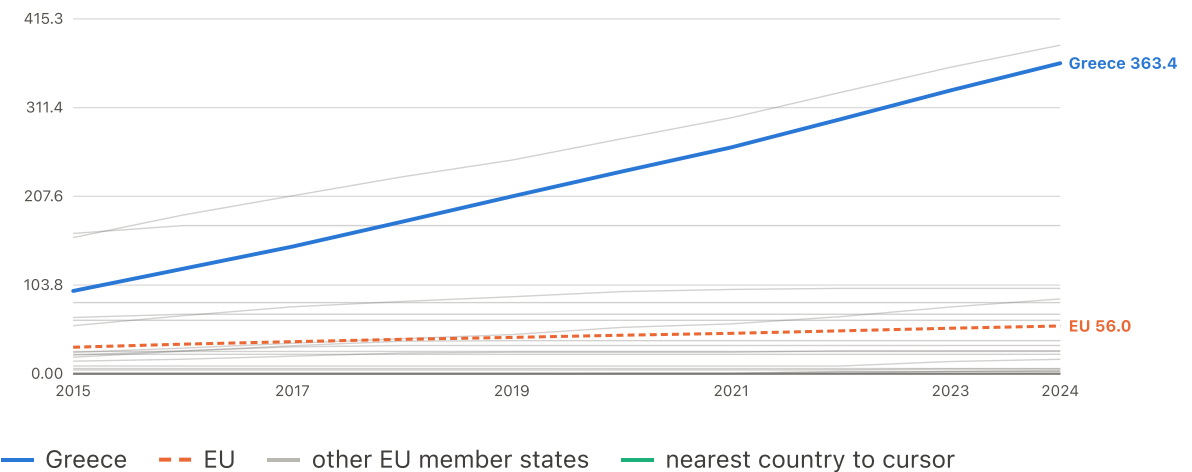
Screened as one 18-candidate family; see the technical report's Methods.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Cumulative real-wage shortfall (fixed 2008 basis)

Cumulative real-wage shortfall (fixed 2008 basis)

accumulated index-points · up to 27 EU member states · 2015–2024



|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 96.94 | 236.6 | 363.4 |
| EU     | 31.14 | 45.18 | 55.99 |

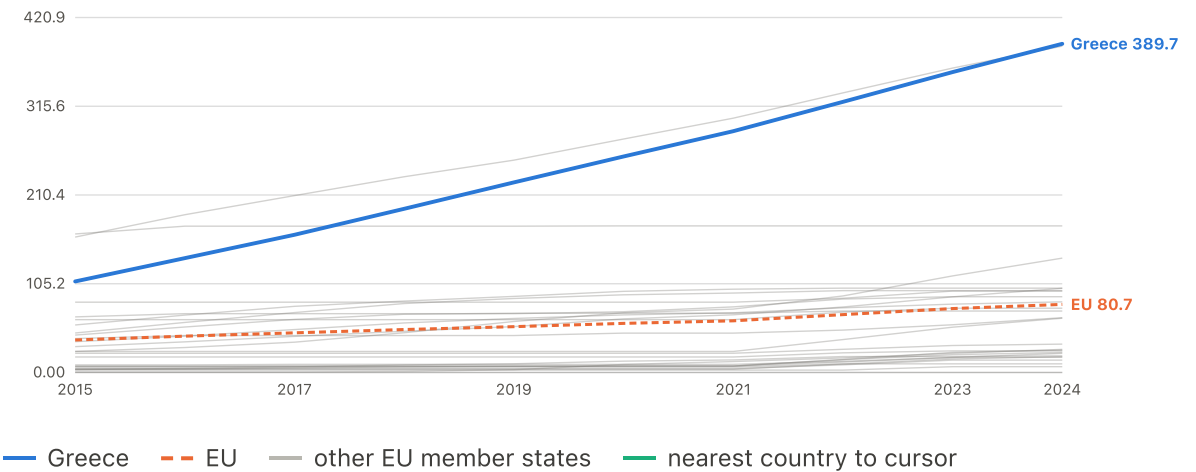
Screened as one 18-candidate family; see the technical report's Methods.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Cumulative real-wage shortfall (own rolling peak)

Cumulative real-wage shortfall (own rolling peak)

accumulated index-points · up to 27 EU member states · 2015–2024



|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 107.9 | 256.1 | 389.7 |
| EU     | 38.52 | 58.22 | 80.70 |

Screened as one 18-candidate family; see the technical report's Methods.

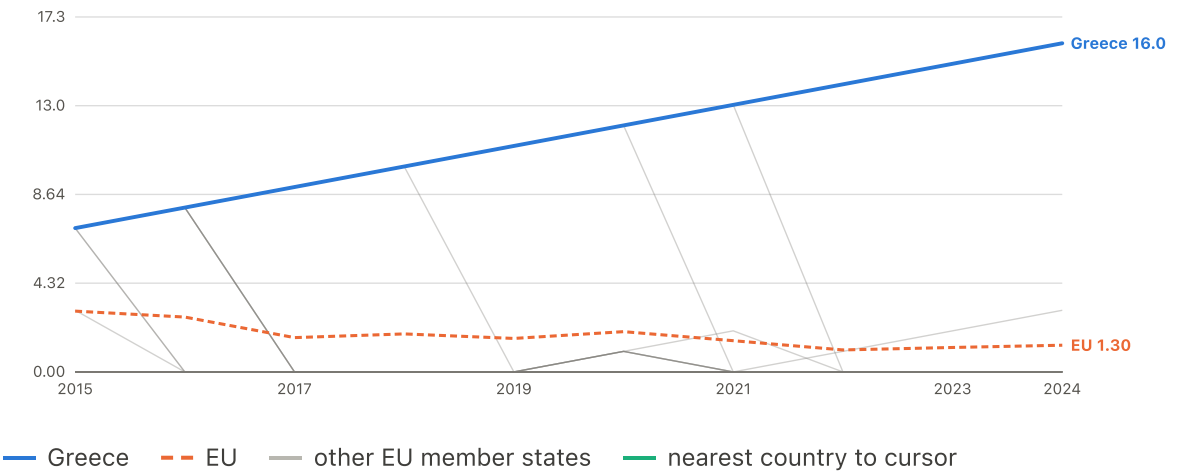
Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

Duration and direction constructions

▼ View candidate trajectory: Consecutive years GDP has been below its 2008 level

Consecutive years GDP has been below its 2008 level

years, current run · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 7.00 | 12.00 | 16.00 |
| EU     | 2.96 | 1.96  | 1.30  |

Screened as one 18-candidate family; see the technical report's Methods. This is the CURRENT unbroken run, not a total: it climbs by one for each further year below the line and drops straight back

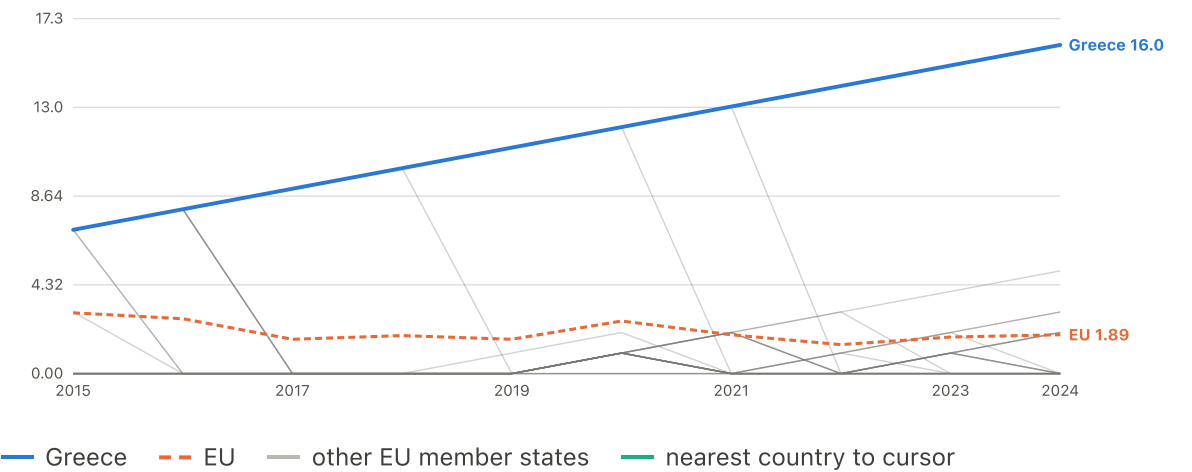
to zero in the first year the country recovers. A country that fell behind early and caught up shows 0 here, while its longest-run chart still records the episode.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Consecutive years GDP has been below its own peak

Consecutive years GDP has been below its own peak

years, current run · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 7.00 | 12.00 | 16.00 |
| EU     | 2.96 | 2.56  | 1.89  |

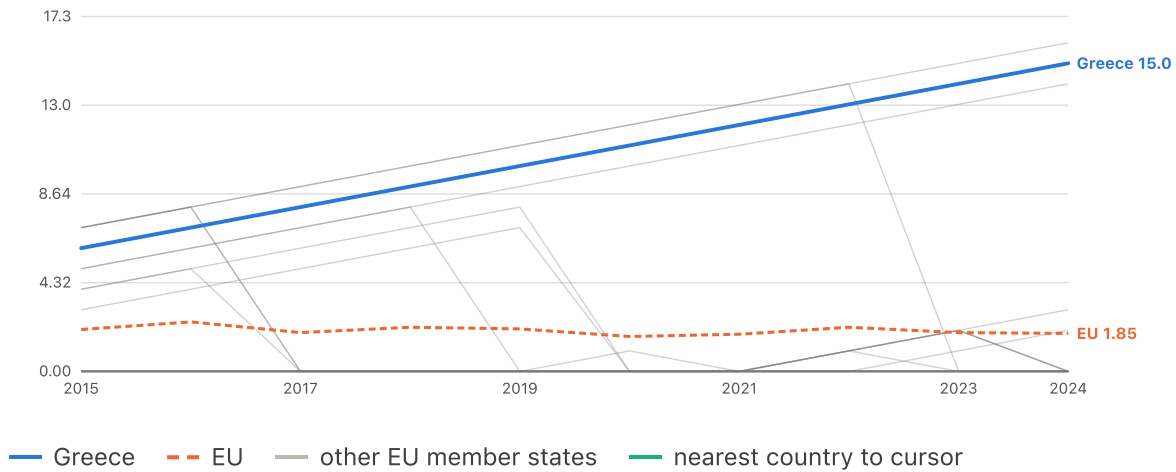
Screened as one 18-candidate family; see the technical report's Methods. This is the CURRENT unbroken run, not a total: it climbs by one for each further year below the line and drops straight back to zero in the first year the country recovers. A country that fell behind early and caught up shows 0 here, while its longest-run chart still records the episode.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Consecutive years real wages have been below their 2008 level

Consecutive years real wages have been below their 2008 level

years, current run · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 6.00 | 11.00 | 15.00 |
| EU     | 2.04 | 1.70  | 1.85  |

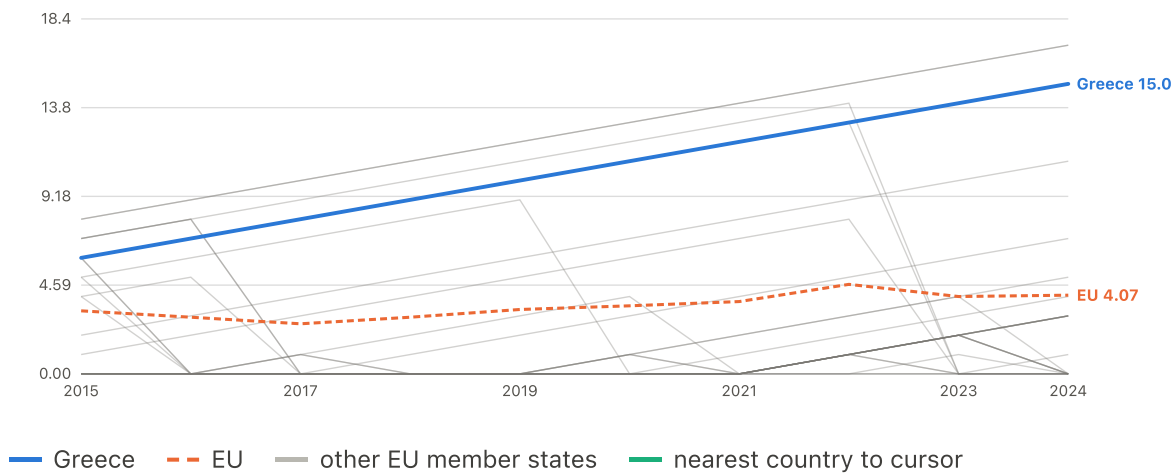
Screened as one 18-candidate family; see the technical report's Methods. This is the CURRENT unbroken run, not a total: it climbs by one for each further year below the line and drops straight back to zero in the first year the country recovers. A country that fell behind early and caught up shows 0 here, while its longest-run chart still records the episode.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

#### ▼ View candidate trajectory: Consecutive years real wages have been below their own peak

### Consecutive years real wages have been below their own peak

years, current run · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 6.00 | 11.00 | 15.00 |
| EU     | 3.26 | 3.52  | 4.07  |

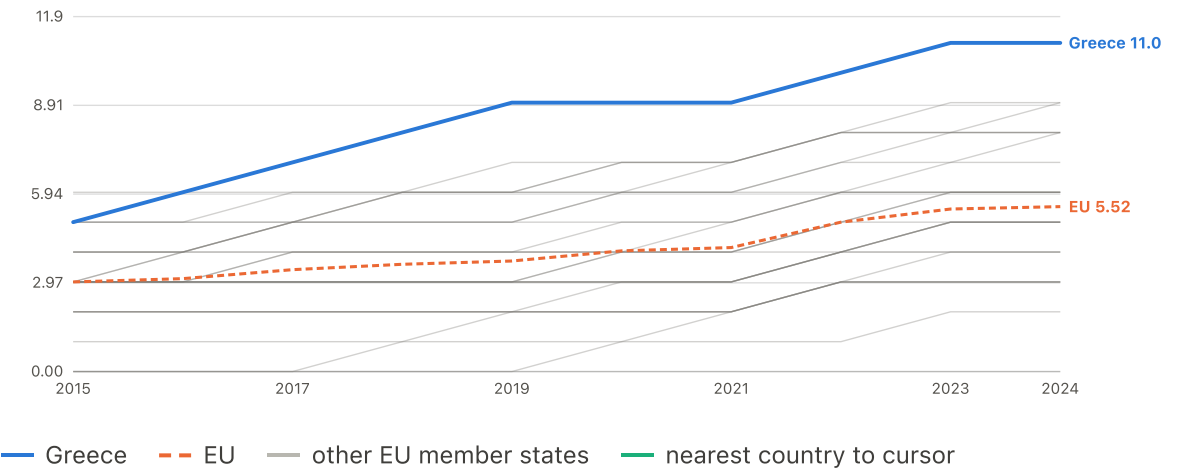
Screened as one 18-candidate family; see the technical report's Methods. This is the CURRENT unbroken run, not a total: it climbs by one for each further year below the line and drops straight back to zero in the first year the country recovers. A country that fell behind early and caught up shows 0 here, while its longest-run chart still records the episode.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Cumulative years of falling real wages

Cumulative years of falling real wages

years · up to 27 EU member states · 2015–2024



|        | 2015 | 2020 | 2024  |
|--------|------|------|-------|
| Greece | 5.00 | 9.00 | 11.00 |
| EU     | 3.00 | 4.04 | 5.52  |

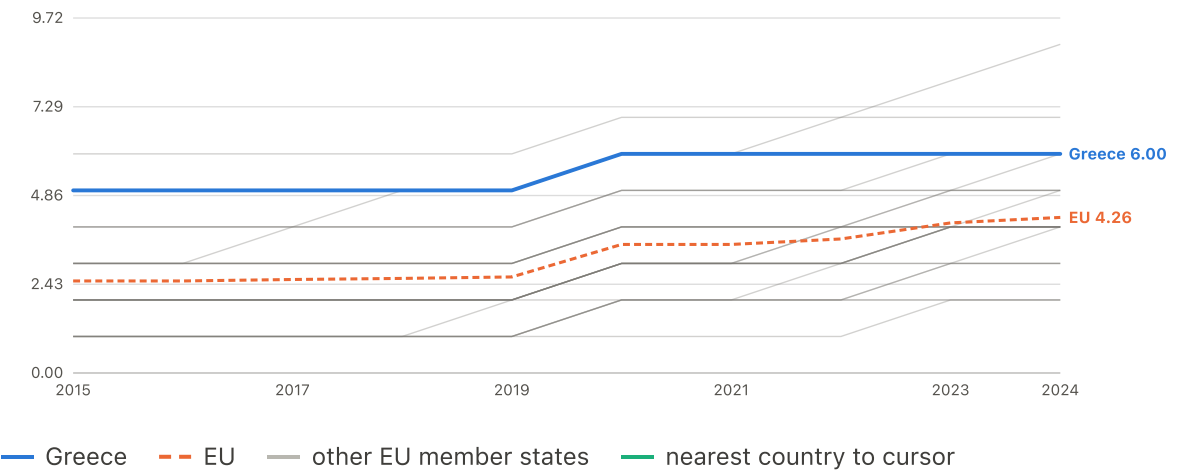
Screened as one 18-candidate family; see the technical report's Methods. A count of year-on-year declines since 2008, regardless of the level: it rises in any year the series falls, including falls that happen well above the 2008 line.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Cumulative years of negative GDP growth

Cumulative years of negative GDP growth

years · up to 27 EU member states · 2015–2024





|        | 2015 | 2020 | 2024 |
|--------|------|------|------|
| Greece | 5.00 | 6.00 | 6.00 |
| EU     | 2.52 | 3.52 | 4.26 |

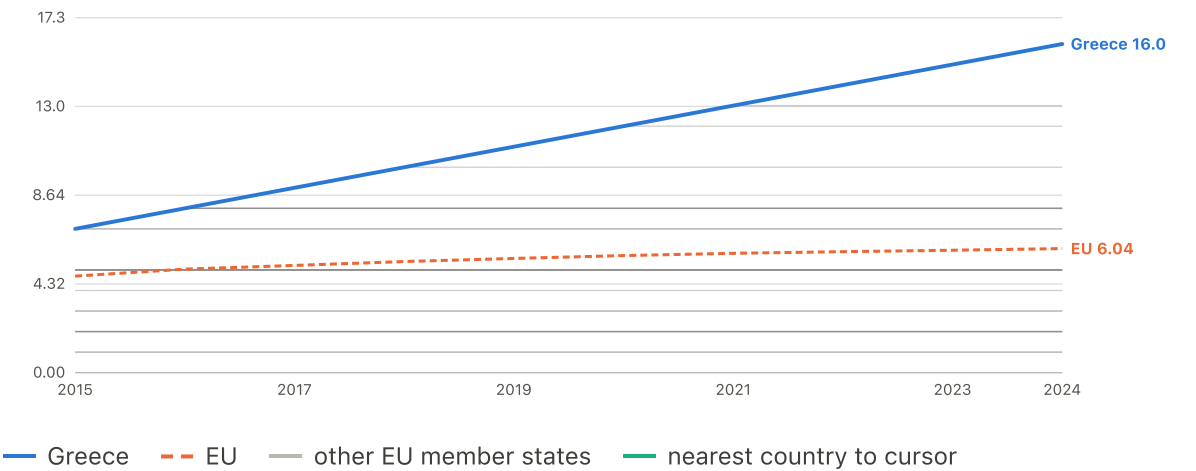
Screened as one 18-candidate family; see the technical report's Methods. A count of year-on-year declines since 2008, regardless of the level: it rises in any year the series falls, including falls that happen well above the 2008 line.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Longest run of years GDP below 2008 (running maximum)

Longest run of years GDP below 2008 (running maximum)

years · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 7.00 | 12.00 | 16.00 |
| EU     | 4.70 | 5.70  | 6.04  |

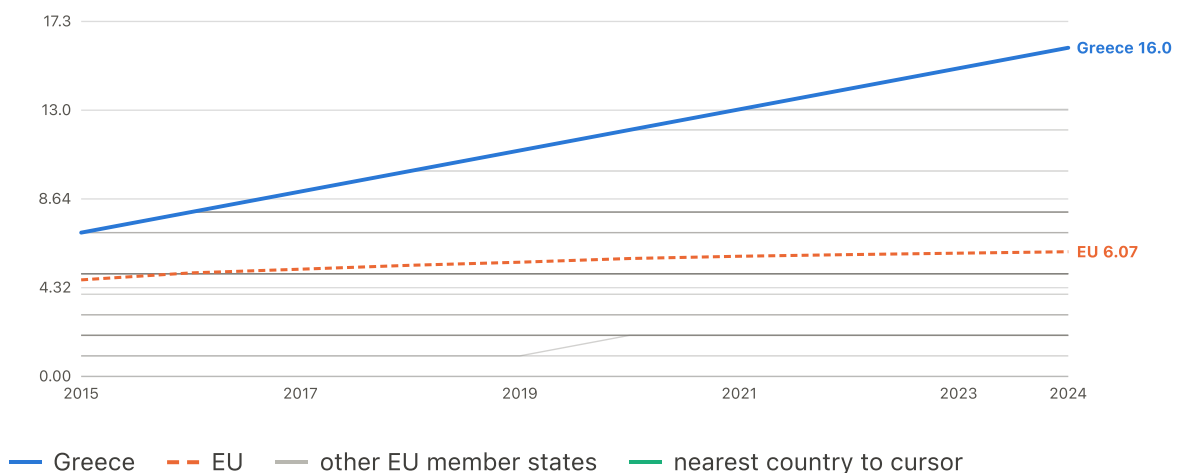
Screened as one 18-candidate family; see the technical report's Methods. This is a running maximum -- the longest unbroken run seen so far -- so it never falls, even after a country recovers.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Longest run of years GDP below own peak (running maximum)

Longest run of years GDP below own peak (running maximum)

years · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 7.00 | 12.00 | 16.00 |
| EU     | 4.70 | 5.74  | 6.07  |

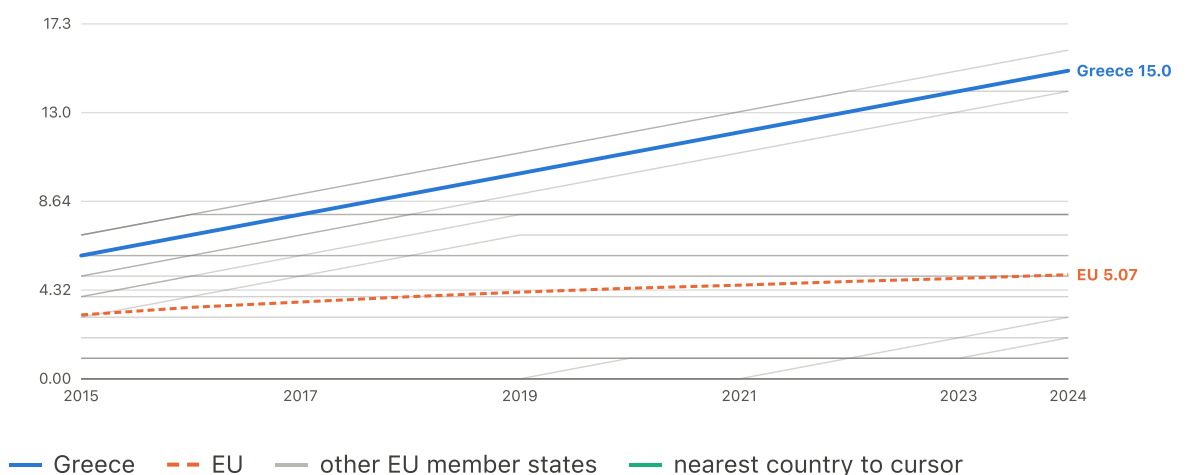
Screened as one 18-candidate family; see the technical report's Methods. This is a running maximum -- the longest unbroken run seen so far -- so it never falls, even after a country recovers.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

#### ▼ View candidate trajectory: Longest run of years wages below 2008 (running maximum)

### Longest run of years wages below 2008 (running maximum)

years · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 6.00 | 11.00 | 15.00 |
| EU     | 3.11 | 4.41  | 5.07  |

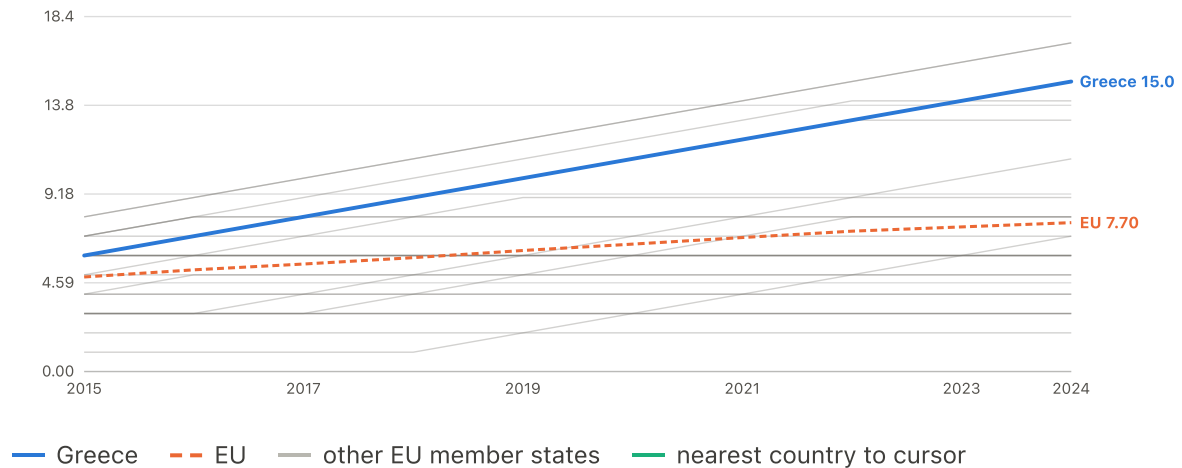
Screened as one 18-candidate family; see the technical report's Methods. This is a running maximum -- the longest unbroken run seen so far -- so it never falls, even after a country recovers.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

▼ View candidate trajectory: Longest run of years wages below own peak (running maximum)

### Longest run of years wages below own peak (running maximum)

years · up to 27 EU member states · 2015–2024



|        | 2015 | 2020  | 2024  |
|--------|------|-------|-------|
| Greece | 6.00 | 11.00 | 15.00 |
| EU     | 4.89 | 6.59  | 7.70  |

Screened as one 18-candidate family; see the technical report's Methods. This is a running maximum -- the longest unbroken run seen so far -- so it never falls, even after a country recovers.

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

Stage 6

## How much depends on the model

Would a defensible alternative specification have produced a different answer?

Removing the same-instrument deprivation measure moves Greece from under-predicted to over-predicted

**POST-SELECTION ROBUSTNESS** Does the answer depend on which model is chosen?

**Read with this.** NEITHER specification is definitive. They may not be merged or averaged, and selection may not be made on residual size.

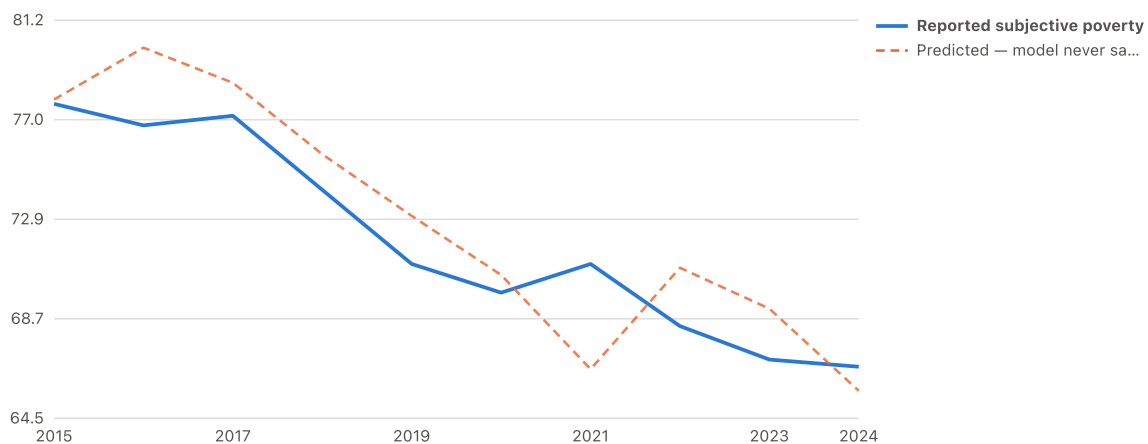
| Country     | Reference specification | Deprivation-free companion | Change |
|-------------|-------------------------|----------------------------|--------|
| Luxembourg  | 8.80                    | 15.48                      | 6.68   |
| Cyprus      | 7.05                    | 10.75                      | 3.70   |
| Greece      | 6.93                    | -9.39                      | -16.32 |
| Latvia      | 6.32                    | 4.65                       | -1.66  |
| Hungary     | 4.86                    | 7.42                       | 2.56   |
| Bulgaria    | 4.06                    | 14.65                      | 10.60  |
| Czechia     | 3.53                    | -3.63                      | -7.16  |
| Slovakia    | 3.28                    | -0.68                      | -3.96  |
| Belgium     | 3.13                    | 5.26                       | 2.13   |
| Ireland     | 3.05                    | 6.14                       | 3.09   |
| Croatia     | 2.62                    | -0.38                      | -2.99  |
| France      | 1.85                    | 3.88                       | 2.02   |
| Austria     | 1.36                    | 0.95                       | -0.41  |
| Portugal    | 1.30                    | 0.62                       | -0.67  |
| Malta       | 1.28                    | 0.36                       | -0.91  |
| Poland      | 0.56                    | -4.32                      | -4.88  |
| Denmark     | 0.16                    | -3.65                      | -3.81  |
| Sweden      | -1.00                   | -4.55                      | -3.55  |
| Slovenia    | -2.66                   | -6.33                      | -3.67  |
| Netherlands | -3.47                   | -4.87                      | -1.39  |
| Lithuania   | -4.17                   | -1.81                      | 2.36   |
| Germany     | -4.31                   | -6.00                      | -1.69  |
| Estonia     | -4.49                   | -10.12                     | -5.63  |
| Finland     | -5.07                   | -6.80                      | -1.72  |
| Italy       | -6.35                   | -5.18                      | 1.17   |
| Romania     | -10.32                  | 9.76                       | 20.08  |
| Spain       | -10.63                  | -10.06                     | 0.57   |

Underlying variables and tables

2 charts

Greece: reported vs predicted subjective poverty

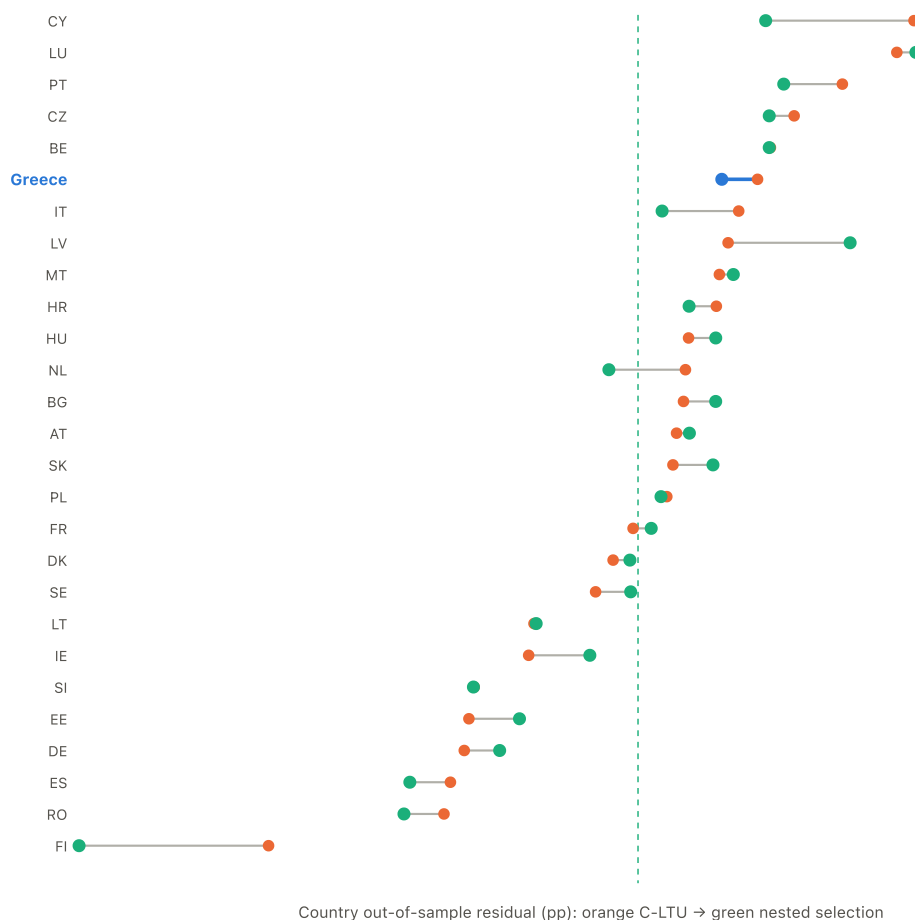
% of households



The out-of-sample line comes from the final model fitted on the other 26 countries only, then applied to Greece.

## Where each country moves after nested validation

average out-of-sample residual, percentage points



Each line joins the C-LTU residual to the residual from a model whose candidate selection was repeated inside that country's held-out fold. Zero means predicted exactly.

## What this is not

Is this a reporting artefact, and what else might matter that this project did not test?

The crisis also became an exit route — and since 2023 that has reversed

**CONTEXTUAL CONSEQUENCE** Did the crisis also become an exit route, and has that reversed?

**Read with this.** CONTEXTUAL CONSEQUENCE AND POSSIBLE CONTRIBUTOR, not an independently supported predictor. Net migration was tested directly at the diagnostic stage and returned nothing ( $p = 0.4006$ ), which speaks to aggregate prediction and to neither causal direction. Net outflow peaked at 44,502 in 2012. Across 2008–2024 the cumulative net loss is 290,281 people, 2.69% of average population and the fifth highest of the 25 countries with sufficient coverage. By 2023 the flow had turned: more Greek nationals returned than left, and in 2024 the net return was 19,852. The accumulated loss is now beginning to reverse; it has not been undone.

| Departures and returns    |                           |       |       |       |       |                         |       |       |       |       |               |       |       |       |       |       |        |  |
|---------------------------|---------------------------|-------|-------|-------|-------|-------------------------|-------|-------|-------|-------|---------------|-------|-------|-------|-------|-------|--------|--|
| Series                    | 2008                      | 2009  | 2010  | 2011  | 2012  | 2013                    | 2014  | 2015  | 2016  | 2017  | 2018          | 2019  | 2020  | 2021  | 2022  | 2023  | 2024   |  |
| Departures                | 19088                     | 19799 | 28301 | 61268 | 70696 | 67057                   | 59525 | 60516 | 61368 | 60816 | 57495         | 53415 | 41111 | 41505 | 40151 | 36931 | 32141  |  |
| Returns                   | 25070                     | 22790 | 25078 | 27043 | 26194 | 26644                   | 29503 | 30460 | 30747 | 31743 | 32199         | 34074 | 20864 | 28392 | 32017 | 46091 | 51993  |  |
| Net flow                  |                           |       |       |       |       |                         |       |       |       |       |               |       |       |       |       |       |        |  |
| Series                    | 2008                      | 2009  | 2010  | 2011  | 2012  | 2013                    | 2014  | 2015  | 2016  | 2017  | 2018          | 2019  | 2020  | 2021  | 2022  | 2023  | 2024   |  |
| Net outflow of nationals  | -5982                     | -2991 | 3223  | 34225 | 44502 | 40413                   | 30022 | 30056 | 30621 | 29073 | 25296         | 19341 | 20247 | 13113 | 8134  | -9160 | -19852 |  |
| EU comparison, cumulative |                           |       |       |       |       |                         |       |       |       |       |               |       |       |       |       |       |        |  |
| Series                    | Cumulative net departures |       |       |       |       | % of average population |       |       |       |       | Years covered |       |       |       |       |       |        |  |
| Lithuania                 | 243338.00                 |       |       |       |       | 8.27                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Croatia                   | 243289.00                 |       |       |       |       | 5.93                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Romania                   | 849659.00                 |       |       |       |       | 4.30                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Luxembourg                | 17038.00                  |       |       |       |       | 2.96                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Greece                    | 290281.00                 |       |       |       |       | 2.69                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Poland                    | 1014893.00                |       |       |       |       | 2.69                    |       |       |       |       | 16.00         |       |       |       |       |       |        |  |
| Slovenia                  | 46557.00                  |       |       |       |       | 2.25                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| France                    | 1380603.00                |       |       |       |       | 2.08                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Ireland                   | 90151.00                  |       |       |       |       | 1.87                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Belgium                   | 198811.00                 |       |       |       |       | 1.75                    |       |       |       |       | 15.00         |       |       |       |       |       |        |  |
| Estonia                   | 21929.00                  |       |       |       |       | 1.65                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Italy                     | 799607.00                 |       |       |       |       | 1.34                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Sweden                    | 127581.00                 |       |       |       |       | 1.29                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Austria                   | 98950.00                  |       |       |       |       | 1.14                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Germany                   | 882086.00                 |       |       |       |       | 1.07                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Netherlands               | 173972.00                 |       |       |       |       | 1.02                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Portugal                  | 83133.00                  |       |       |       |       | 0.80                    |       |       |       |       | 16.00         |       |       |       |       |       |        |  |
| Spain                     | 214223.00                 |       |       |       |       | 0.46                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Czechia                   | 48258.00                  |       |       |       |       | 0.46                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Finland                   | 23245.00                  |       |       |       |       | 0.43                    |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Slovakia                  | -936.00                   |       |       |       |       | -0.02                   |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Hungary                   | -30075.00                 |       |       |       |       | -0.31                   |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Malta                     | -1979.00                  |       |       |       |       | -0.43                   |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Denmark                   | -68217.00                 |       |       |       |       | -1.20                   |       |       |       |       | 17.00         |       |       |       |       |       |        |  |
| Cyprus                    | -11840.00                 |       |       |       |       | -1.35                   |       |       |       |       | 16.00         |       |       |       |       |       |        |  |

Evidence that cannot support a headline claim by design, and is recorded in the report's context register.

APPENDIX FIGURE    **Greece is the worst in Europe on both money questions, and among the worst on general life satisfaction**

DESCRIPTIVE CORROBORATION

Is Greece extreme on money questions and ordinary elsewhere, or extreme on everything?

**Read with this.** A 2024 snapshot. Contextual and not modelled. Each strip has its OWN scale, because the three indicators are in different units; positions may be compared within a strip and not between them. The last column of the table is Greece's position counting from the worst end. Greek life satisfaction ROSE over the observed period, from 6.2 to 6.9: the position shown here worsened because other countries improved faster, which is not the same as Greece becoming less satisfied.

| Indicator              | Greece | EU median | Countries | Greece's position |
|------------------------|--------|-----------|-----------|-------------------|
| Reported hardship      | 66.70  | 17.10     | 27.00     | 1.00              |
| Financial expectations | -43.24 | -1.77     | 27.00     | 1.00              |
| Life satisfaction      | 6.70   | 7.30      | 28.00     | 2.00              |

APPENDIX FIGURE    **Greeks trust the police and the courts far more than they trust the central government, parliament or political parties**

CONTEXTUAL EVIDENCE    How does Greek institutional trust compare with the OECD?

**Read with this.** A 2023 snapshot, not a trend. The OECD Trust Survey has run in 2021, 2023 and 2025, so a short series exists, but this project holds only the 2023 wave. Contextual and not modelled.

| Institution                   | Share reporting high or moderately high trust (%) |
|-------------------------------|---------------------------------------------------|
| Other people                  | 54.0                                              |
| The police                    | 51.0                                              |
| Courts and the justice system | 47.0                                              |
| Local government              | 39.0                                              |
| Central government            | 32.0                                              |
| Parliament                    | 32.0                                              |
| The civil service             | 31.0                                              |
| The news media                | 22.0                                              |
| Political parties             | 17.0                                              |

## APPENDIX FIGURE

## Unmet medical care: Greece against every other member state

## APPENDIX

**Read with this.** DESCRIPTIVE ONLY. This is health-care ACCESS, not health status, and it is not evidence that unmet care explains the hardship gap: tested as a predictor it is the one correctly signed measure of four and also the weakest, its leave-one-country-out refits change sign, and adding it makes Greece's out-of-sample residual larger. The line is the median MEMBER STATE, not a population-weighted EU aggregate. This chart runs one year beyond the 2016-2024 model sample, because the appendix shows the full series; the tests and the write-up's own figure stop at 2024.

| Year              | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|-------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Greece            | 5.4  | 5.5  | 5.5  | 7.5  | 8.0  | 9.0  | 10.9 | 12.3 | 13.1 | 10.0 | 8.8  | 8.1  | 6.5  | 6.4  | 9.0  | 11.6 | 12.1 | 11.5 |
| EU country median | 1.9  | 2.1  | 1.9  | 2.2  | 2.3  | 2.6  | 2.8  | 2.1  | 1.7  | 1.6  | 1.8  | 1.4  | 1.6  | 1.7  | 1.8  | 2.1  | 1.9  | 2.1  |

## APPENDIX FIGURE

## No health measure supports the hypothesis, and most point against it

## APPENDIX

**Read with this.** Every measure is pre-registered as higher-is-worse, so only a POSITIVE estimate can support the hypothesis. Six of the eight are negative. Intervals are country-clustered; the wild-cluster bootstrap in the tooltip, not the interval, decides support. None clears it. Read alongside A7: the negative pooled estimates are cross-country composition, and the within-country sign is the expected one.

| Estimate                                     | Estimate | Low    | High  | Bootstrap p |
|----------------------------------------------|----------|--------|-------|-------------|
| Unmet medical care (current level)           | 0.178    | -0.574 | 0.929 | 0.660       |
| Not good or very good health (current level) | -0.212   | -0.631 | 0.208 | 0.423       |
| Long-standing illness (current level)        | -0.317   | -0.693 | 0.059 | 0.129       |
| Activity limitation (current level)          | -0.155   | -0.467 | 0.158 | 0.363       |
| Unmet medical care (accumulated)             | 0.303    | -0.363 | 0.969 | 0.494       |
| Not good or very good health (accumulated)   | -0.201   | -0.523 | 0.121 | 0.288       |
| Long-standing illness (accumulated)          | -0.248   | -0.502 | 0.007 | 0.109       |
| Activity limitation (accumulated)            | -0.237   | -0.532 | 0.058 | 0.131       |



## APPENDIX FIGURE

**The three health-STATUS measures reverse sign between countries and within them; the access measure does not**

## APPENDIX

**Read with this.** Country means and annual deviations are entered together, so the two ends come from ONE model rather than two. The reading is that the negative between-country coefficients are composition: richer countries report worse health AND less hardship, so pooling the comparisons yields a sign describing neither. Within a country the direction is the expected one. This is the same limitation the main report documents, and it does not establish a mechanism: both sides come from EU-SILC, so common survey method remains live. THREE OF FOUR ROWS REVERSE, not all four: unmet medical care measures access to care rather than health status, and it is positive in both comparisons. It is also the only measure whose leave-one-country-out refits change sign, so consistent here does not mean supported.

| Measure                                      | Between countries | Within countries |
|----------------------------------------------|-------------------|------------------|
| Unmet medical care (access)                  | 0.803             | 0.500            |
| Not good or very good health (health status) | -0.350            | 0.567            |
| Long-standing illness (health status)        | -0.559            | 0.332            |
| Activity limitation (health status)          | -0.427            | 0.406            |

## DESCRIPTIVE CORROBORATION

**Financial expectations and life satisfaction**

Built from this project's own `reporting_style_cross_indicator.csv`: two financial questions and one general-wellbeing question, EU-SILC 2024, 27 member states. No inferential test is run on the cross-domain comparison itself -- it is a description, not an estimate -- and the frozen claim it corroborates is V2-7.1.

**What may be concluded.** The financial indicators are more extreme than the general-wellbeing one, which suggests domain specificity. A broader negative reporting tendency is NOT ruled out.

**Limitation.** Describing Greece as ordinary or middling on life satisfaction: it is second-worst in the EU by 2024. And reading a worsening RANK as falling satisfaction, when the Greek level rose over the period.

**Source.** `reporting_style_cross_indicator.csv`, this project Greece ranks 1st of 27 on hardship and financial expectations, 2nd-6th on life satisfaction.

## CONTEXTUAL EVIDENCE

## Institutional trust

OECD Survey on Drivers of Trust in Public Institutions, 2024 Country Notes: Greece. Fieldwork October-November 2023, 30 OECD countries. External to this project: no project data feeds this entry, and no test was run or is possible on it here.

**What may be concluded.** May affect how households interpret insecurity; available evidence does not identify an independent effect.

**Limitation.** Presenting trust as an explanation of the residual, or implying it was tested and found to matter.

---

**Source.** OECD (2024), OECD Survey on Drivers of Trust in Public Institutions - 2024 Results, Country Notes: Greece. OECD Publishing, Paris. Fieldwork October-November 2023, 30 OECD countries. 32% of Greek respondents report high or moderately high trust in central government against an OECD average of 39%. source

## LITERATURE-GROUNDED CONTEXT

### Crisis and adjustment policies

Andriopoulou, Kanavitsa & Tsakloglou (2020), LSE GreeSE Paper No. 149. EU-SILC microdata via ELSTAT, 2007-2017 waves, analysed in the cited paper, not by this project. No test was run here; the accumulated measures in Stage 5 record what was absorbed, not what any programme caused.

**What may be concluded.** Help explain the historical setting; this project does not estimate their causal contribution.

**Limitation.** Attributing any share of the accumulated exposure to a specific programme or measure.

---

**Source.** Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. EU-SILC microdata, ELSTAT waves 2008-2017. Anchored FGT0 on a 2007 base reaches 48% at the 2013 peak. source

## CONTEXTUAL CONSEQUENCE

### Migration

Migration was tested in this project: E3, an aggregate net-migration predictor on the same 27-country panel as every other present-day construct,  $p = 0.4006$  -- inconclusive under the power available, not a demonstrated null. The Bank of Greece departure count (427,000 residents aged 15-64, 2008-2013) is external descriptive context, untested by this project's own protocol.

**What may be concluded.** A plausible consequence of prolonged labour-market damage and a possible contributor to it. It is UNSUPPORTED as an independent aggregate predictor here.

**Limitation.** Reading it as a driver, or as ruled out. E3 tested it on this panel and found nothing ( $p=0.4006$ ), which speaks to aggregate prediction and not to either causal direction.

---

**Source.** Lazaretou, S. (2016), The Greek brain drain: the new pattern of Greek emigration during the recent crisis. Economic Bulletin, Bank of Greece, issue 43, pp. 31-53. 427,000 residents aged 15-64 left permanently 2008-2013; ~223,000 of them aged 25-39. The E3 null ( $p=0.4006$ ) is this project's own evidence. source

#### FUTURE HYPOTHESIS

### Tax burden and unequal treatment

Kaplanoglou (2015), Public Finance Review 43(4). Household Expenditure Survey microsimulation, 1988-2011, external to this project. This project holds no tax-incidence data of its own and ran no test linking tax burden to the hardship gap; the entry exists so a reader does not conclude the omission was accidental.

**What may be concluded.** Published incidence evidence establishes that Greece's indirect tax system became markedly more regressive over the crisis. Whether that channel contributes to the hardship gap is a plausible hypothesis and is NOT tested here.

**Limitation.** Any quantitative statement linking tax burden to the hardship gap. The literature is about incidence, not about this outcome, and this project tested nothing on tax.

---

**Source.** Kaplanoglou, G. (2015), Who Pays Indirect Taxes in Greece? From EU Entry to the Fiscal Crisis. Public Finance Review 43(4), 529-556. Microsimulation on Household Expenditure Survey data, 1988-2011. The 2011 indirect tax system is the most regressive of the period, with the sharpest effects on families with children and the unemployed. source

#### DESCRIPTIVE CORROBORATION

### Pre-crisis wellbeing baseline (ESS)

European Social Survey Data Portal, public Analysis tab, `stflife` (0-10 scale), Greek rounds 1, 2, 4, 5, 10 and 11 (2002 to 2023/24). Coverage is a balanced set of 12 countries present in every Greek round; the full ESS country set ranges from 22 to 30 across rounds and is deliberately not used for ranking here. Country means are the authors' own reconstruction from the portal's displayed weighted percentages, at the portal's rounding, so no standard error is computable and none is reported. Per-round source URLs are recorded in `data/raw/ess/ess_life_satisfaction_round_summary.csv`.

**What may be concluded.** On a balanced set of 12 countries observed in every Greek ESS round, Greece already sat about 0.8 points below the median before the crisis, fell to 5.64 by 2010/11, and recovered its LEVEL to roughly the pre-crisis value by 2023/24. Its comparative position did not recover: the gap to the balanced median is wider than before the crisis, and Greece has been worst of the 12 since 2010/11. A longstanding low-wellbeing pattern therefore remains plausible and generic pessimism or reporting culture CANNOT be dismissed.

**Limitation.** Splicing ESS to the Eurostat EU-SILC series, or reading the two as one trajectory. Attaching any confidence interval, standard error or significance test to these means, which are approximate reconstructions from displayed weighted percentages. Comparing ALL-COUNTRY ranks across rounds, since the country set varies from 22 to 30. Treating the decade after 2010/11 as observed.

**Source.** European Social Survey Data Portal, public Analysis tab for stflife (life satisfaction, 0-10), rounds 1, 2, 4, 5, 10 and 11. Weighted distributions with the post-stratification weight; country means reconstructed by the authors. source

## CONTEXTUAL EVIDENCE

### Wealth concentration since 2019

ECB Distributional Wealth Accounts (DWA), Greece, household sector (S14), quarterly, 2019 Q2–2026 Q1, external to this project. Raised during editorial review against a viral social-media post making similar claims with several wrong headline numbers; every figure here was recomputed directly from the ECB Statistical Data Warehouse API for the total, D6, D7, D9, D10 and B50 net-wealth series, not taken from that post or from this project's own panel. The companion GDP comparison (+11.7% real, Greece 20th of 36 countries) was recomputed from Eurostat's `namq_10_gdp` across every country with both quarters available. No test was run by this project linking wealth concentration to reported hardship.

**What may be concluded.** Independent ECB distributional wealth accounts show that of the €224.8bn increase in Greek household net wealth between mid-2019 and early 2026 (+32.2%), about 72.6% accrued to the wealthiest fifth of households while the bottom half accounted for about 9.2%; the bottom half's own share of total wealth slipped slightly, from roughly 9.5% to 9.4%. This offers one plausible mechanism for how aggregate recovery and persistent reported hardship can coexist: a rising net-wealth total does not evenly reach the households reporting the most difficulty.

**Limitation.** Treating this as tested against this project's reported-hardship findings, or as an established explanation for the 52.6-point gap: it sits partly outside the 2015–2024 window and was never checked against any model here. Reading a rising net-wealth figure as equivalent to rising disposable income or an improved ability to pay bills: these changes are driven mostly by property and financial-asset valuations, not household cash flow. Treating the ECB's distributional wealth accounts as final: the ECB itself labels them experimental,

combining household survey data with quarterly national accounts and extrapolating between survey waves.

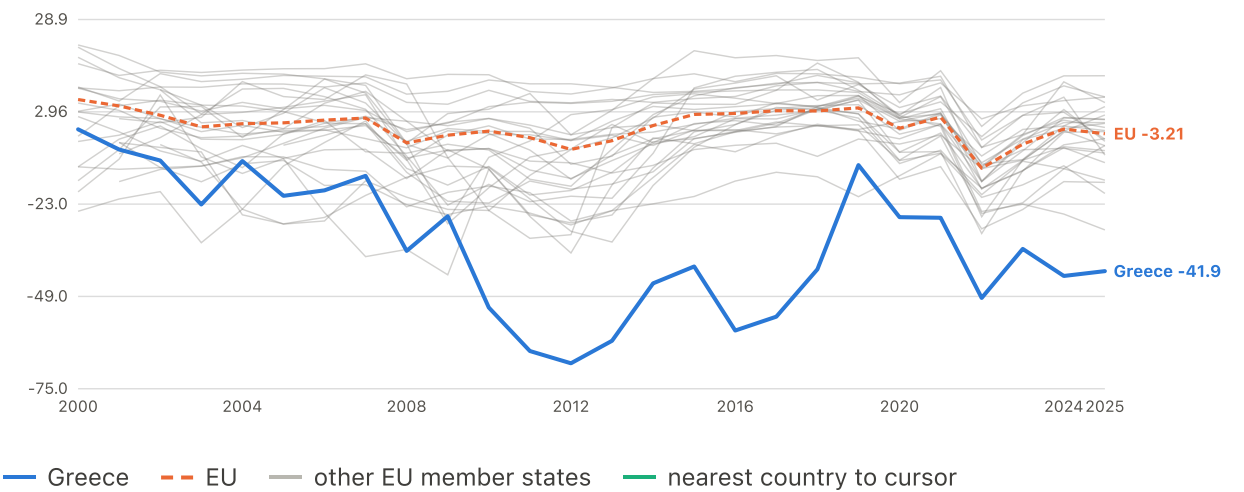
**Source.** European Central Bank, Distributional Wealth Accounts (DWA), Greece, household sector (S14), 2019 Q2–2026 Q1. Net wealth rose from €698.4bn (2019 Q2) to €923.2bn (2026 Q1), +32.2%. Of the €224.8bn increase, the top decile alone gained €133.4bn; the top two deciles combined gained 72.6% of the total increase, the 6th–7th deciles 9.7%, and the bottom half 9.2%. A companion Eurostat check over the same window (namq\_10\_gdp) puts Greek real GDP growth at +11.7%, ranking 20th of 36 countries with both observations. source

Underlying variables and tables

5 charts

Household financial expectations, next 12 months

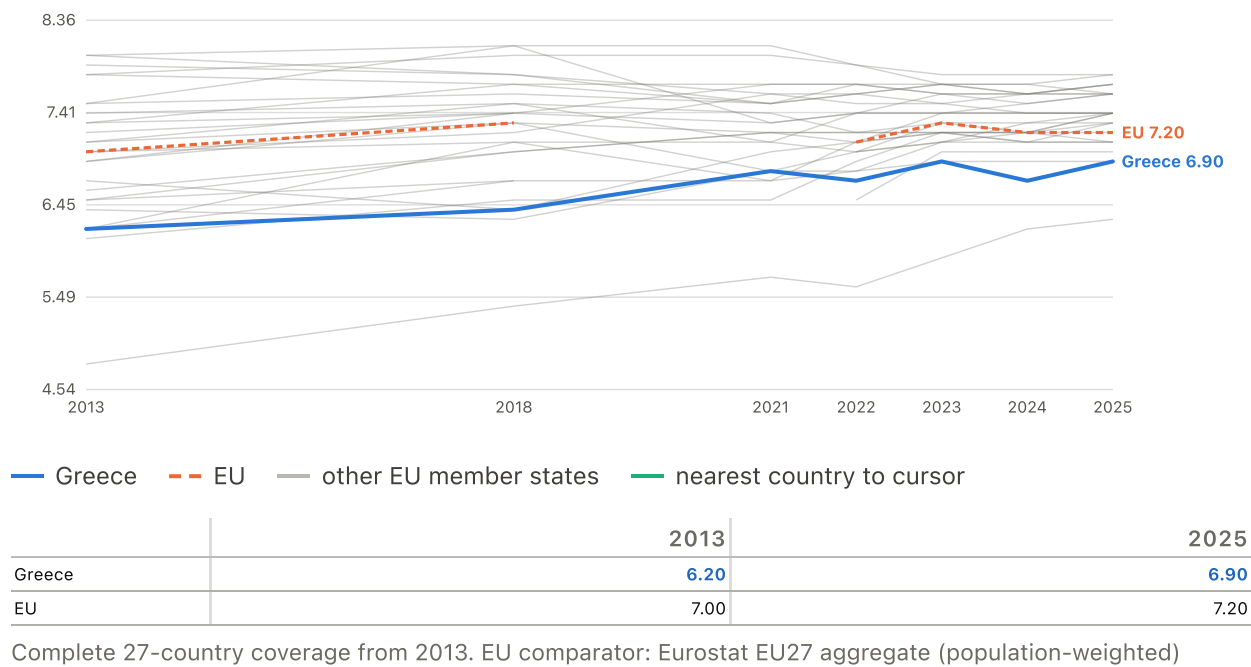
balance (net %) · up to 27 EU member states · 2000–2025



Monthly consumer-survey balance averaged to annual. Negative = more households expect deterioration. Complete 27-country coverage from 2005. EU comparator: Eurostat EU27 aggregate (population-weighted)

Overall life satisfaction

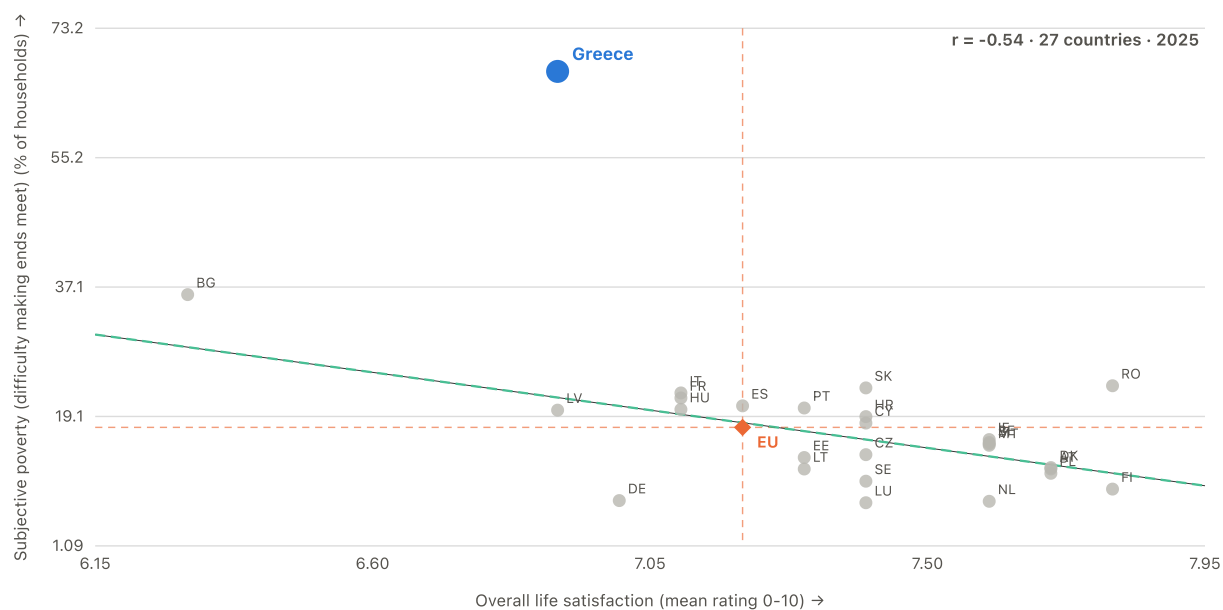
mean rating 0–10 · up to 27 EU member states · 2013–2025



RELATIONSHIP

Life satisfaction against reported financial hardship

2025 · 27 member states ·  $r = -0.54$

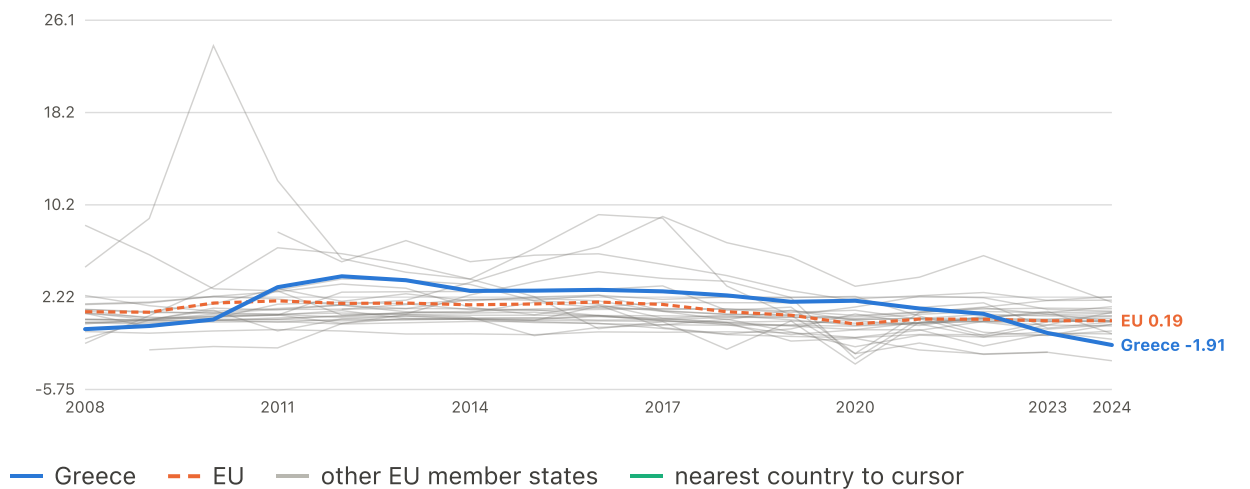


The domain-specificity check. If Greek answers were simply gloomier across the board, Greece would be extreme on both axes. It is extreme on financial hardship and much less so on general life satisfaction.

EU marker: Eurostat EU27 aggregate (population-weighted) (x), Eurostat EU27 aggregate (population-weighted) (y). Cross-country correlation for the year shown — a different quantity from the within-Greece correlations quoted in the reports. The green dashed line is fitted to the 26 peers, excluding Greece.

Net migration of nationals (per 1,000 population)

per 1,000 people · up to 27 EU member states · 2008–2024



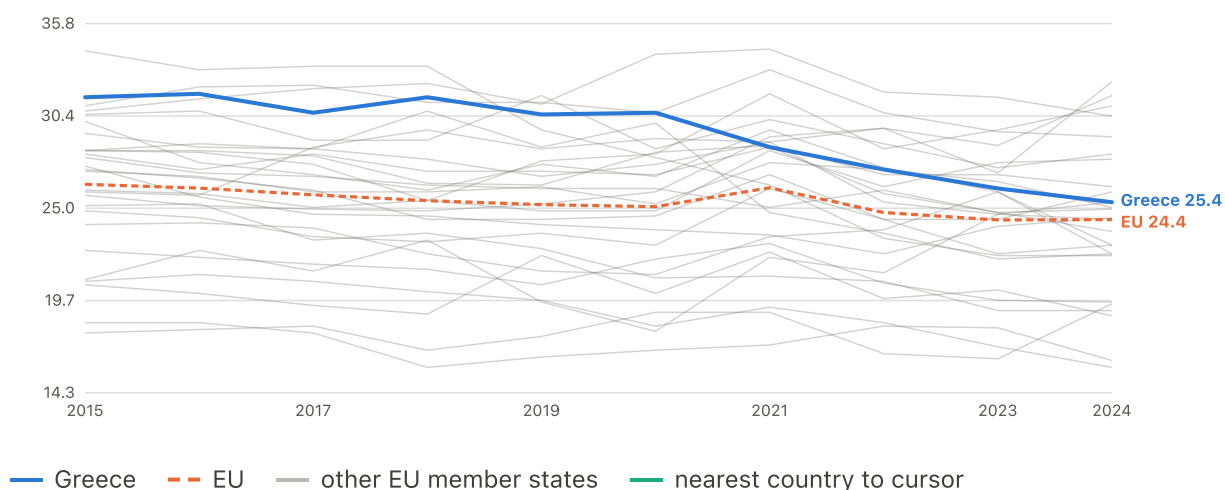
|        | 2008  | 2010 | 2015 | 2020  | 2024  |
|--------|-------|------|------|-------|-------|
| Greece | -0.54 | 0.29 | 2.78 | 1.91  | -1.91 |
| EU     | 0.96  | 1.71 | 1.63 | -0.10 | 0.19  |

Positive = more nationals left than returned. Eurostat migration-flow tables.

Complete 27-country coverage from 2012. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

## Welfare-transfer effectiveness (AROP removed by transfers)

pp of poverty removed · up to 27 EU member states · 2015–2024



|        | 2015  | 2020  | 2024  |
|--------|-------|-------|-------|
| Greece | 31.50 | 30.60 | 25.40 |
| EU     | 26.44 | 25.14 | 24.39 |

Complete 27-country coverage from 2015. EU comparator: unweighted mean of member states (years with <20 reporting dropped)

# What the evidence supports

Stated no more strongly than the tests allow.

Evidence that cannot support a headline claim by design, and is recorded in the report's context register.

AUTHOR INTERPRETATION

Policy implications

Not applicable in the sense the other entries use the word: this is an interpretive recommendation built on three already-established results (V2-1.2, V2-2.1 and V2-5.\*), not a sourced external claim, so it carries no separate coverage, dataset or test to report here.

**What may be concluded.** POLICY RECOMMENDATION, NOT AN EMPIRICAL CONCLUSION: poverty dashboards should combine AROP, its real threshold, anchored poverty, AROPE/deprivation and accumulated labour and housing indicators.

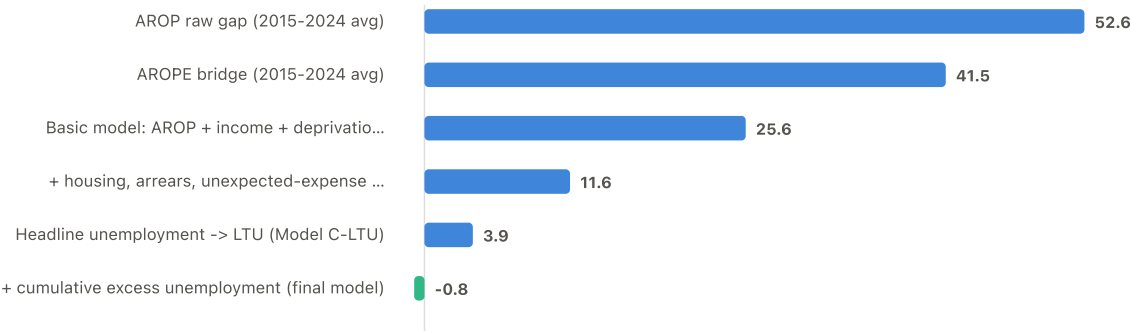
**Limitation.** Presenting this as a finding. It follows from what the analysis showed AROP alone misses; it is not itself a result.

Underlying variables and tables

1 charts

From the raw AROP gap to the final residual

percentage points



Greece gap or out-of-sample residual (percentage points)

A narrative sequence, not a formal decomposition: raw single-country gaps are followed by cross-country out-of-sample residuals on the common 2015-2024 window.



# Source and coverage notes

Every chart in this appendix draws on the same Eurostat and ELSTAT pipeline as the three reports, at country-year level for the EU27. Where a figure shows “the EU median” or “the EU comparator”, it is Eurostat's own population-weighted EU27 aggregate where one is published, and an unweighted mean of member states where it is not -- the two are not interchangeable, and each chart states which one it uses. A country line stopping early means that country stopped reporting the series, not that its value fell to zero.

Eurostat and ELSTAT revise published figures over time, so a later pull from the live API will not reproduce every number here exactly; this project's frozen claims are estimated on a cached snapshot for that reason (see docs/project\_description.md), while this appendix's own descriptive comparison series call the live API on every build so Eurostat's revisions reach it without a manual re-fetch. Every result across the three reports is an associational, country-level aggregate, not a household-level estimate — see the technical report's Methods for the full limitations.

65 series, 15 panels and 10 scatter relationships are charted here in total, across the eight stages above.

## § ABBREVIATIONS AND TERMS

Technical measures, methods and data codes used across the project, collected in one searchable place. Where expanding a term is not enough to make it usable, a plain-language explanation is included.

Filter — type an abbreviation or a word

### POVERTY AND LIVING-CONDITIONS MEASURES

#### AROP

#### At Risk Of Poverty

Below 60% of the national median equivalised disposable income. A relative measure: the line moves with the country's own median, which is why it can fall while living standards fall. Eurostat's headline income-poverty indicator.

#### AROPE

#### At Risk Of Poverty or social Exclusion

A person counted if they meet *any* of three conditions: AROP, severe material and social deprivation, or living in a very low work intensity

household. A union, not an average, so it is always at least as large as AROP.

---

**EU-SILC****European Union Statistics on Income and Living Conditions**

The annual household survey behind almost every poverty figure here. Income data refer to the year before the survey; deprivation and subjective questions to the survey year.

---

**SMSD****Severe Material and Social Deprivation**

Lacking at least 7 of 13 items a household cannot afford. Replaced the legacy 9-item severe material deprivation measure in 2021; both appear in the appendix, spliced.

---

**S80/S20****Income quintile share ratio**

Total income of the richest 20% divided by that of the poorest 20%. The project's inequality measure, tested and ruled out as an explanation.

---

**Anchored poverty**

The poverty line held fixed at an earlier year's real value instead of moving with the current median. Section 3's fixed-2008-baseline reconstruction.

---

**Equivalentised income**

Household income adjusted for household size and composition, so a couple is not counted as twice as well off as a single person.

---

**Arrears****Household payment arrears**

A household reports that, because of financial difficulty, it could not pay on time during the previous 12 months one or more scheduled housing, utility or instalment/loan payments. It measures missed payments, not total debt, and does not necessarily mean a formal legal default.

---

**STATISTICS AND METHOD**

---

**FDR****False Discovery Rate**

When many variables are screened at once, some will look significant by chance. FDR control caps the expected share of false positives among the findings declared significant — here at 5%. It is why a variable can have a raw p of 0.005 and an adjusted p of 0.041: the penalty for having asked 18 questions rather than one.

---

**BH****Benjamini–Hochberg**

The specific procedure used to control the FDR. Ranks the p-values and rescales each by the family size divided by its rank. Adding candidates to a family raises every adjusted p-value in it.

---

|                      |                                                                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>p-value</b>       | The probability of seeing an association at least this strong if there were really no relationship. Small means surprising-under-no-effect; it is not the probability that the finding is true, and it says nothing about effect size.                               |
| <b>R<sup>2</sup></b> | <b>Coefficient of determination</b><br>Share of the variation in the outcome the model accounts for, 0 to 1. In-sample R <sup>2</sup> always rises when a variable is added, which is why this project also reports out-of-sample performance.                       |
| <b>LOO</b>           | <b>Leave-One-Out cross-validation</b><br>The model is refit on 26 countries and used to predict the 27th, country by country. The Greek out-of-sample gap is what a model that has never seen Greece gets wrong about Greece — a far harder test than in-sample fit. |
| <b>OOS</b>           | <b>Out-Of-Sample</b><br>Any figure computed on data the model was not fitted to. The project's headline residual is an out-of-sample quantity.                                                                                                                       |
| <b>RMSE</b>          | <b>Root Mean Squared Error</b><br>Typical size of a prediction error, in the outcome's own units.                                                                                                                                                                    |
| <b>VIF</b>           | <b>Variance Inflation Factor</b><br>How much a predictor's variance is inflated by correlation with the other predictors. Above about 10 is conventionally treated as disqualifying collinearity.                                                                    |
| <b>r</b>             | <b>Pearson correlation coefficient</b><br>Strength and direction of a straight-line association, −1 to +1. Cross-country correlations in the appendix are a different quantity from the within-Greece correlations quoted in the reports.                            |
| <b>pp</b>            | <b>percentage points</b><br>The arithmetic difference between two percentages. A move from 10% to 13% is 3 percentage points, not 3 percent.                                                                                                                         |
| <b>pp-years</b>      | <b>accumulated percentage-point-years</b><br>The unit of the cumulative-exposure variables: a rate held one percentage point above baseline for one year contributes 1. Ten years at 10pp above baseline gives 100.                                                  |
| <b>s.e.</b>          | <b>standard error</b><br>The estimated sampling variability of a coefficient. Roughly, the coefficient plus or minus two standard errors is a 95% interval.                                                                                                          |
| <b>n</b>             | <b>sample size</b>                                                                                                                                                                                                                                                   |

Here almost always the number of countries (27) or country-years in a panel.

---

**FE****Fixed effects**

Controls for stable differences between countries or shocks common to a given year. Country fixed effects ask whether changes within a country move with the outcome.

---

**CI****Confidence interval**

A range expressing estimation uncertainty. Under repeated comparable samples, a 95% confidence-interval procedure would contain the true parameter about 95% of the time.

---

**SDMX-JSON****Statistical Data and Metadata eXchange, JSON format**

The machine-readable format used to retrieve Eurostat data and its dimension labels.

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**ECONOMIC AND PRICE CONCEPTS**

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**GDP****Gross Domestic Product**

Total value of goods and services produced.

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**AIC****Actual Individual Consumption**

What households actually consume, including services provided by government such as health and education. Eurostat recommends it over GDP for welfare comparisons, and the cross-country models use it as a living-standards variable. Here AIC never means the Akaike information criterion.

---

**PPS****Purchasing Power Standard**

An artificial currency that equalises purchasing power across countries: one PPS buys the same basket everywhere. Removes price-level differences from income and pay comparisons.

---

**PLI****Price Level Index**

A country's price level relative to the EU27 average, which is set to 100.

---

**HICP****Harmonised Index of Consumer Prices**

The EU's comparable consumer-price index, used here to deflate nominal series into real terms using each country's own inflation.

---

**COICOP****Classification Of Individual Consumption by Purpose**

The standard category scheme behind the price charts (food, housing and energy, transport, communications, restaurants).

---

**NAC****National Currency**

Values in whatever currency the country used at the time — not converted to euro. For countries that adopted the euro mid-series this creates a level break, corrected in this project by `scripts/currency.py`.

---

|            |                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LTU</b> | <b>Long-Term Unemployment</b><br>Unemployed for 12 months or more, as a share of the active population. The strongest single correlate of Greek subjective poverty in this project. |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|            |                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------|
| <b>LFS</b> | <b>Labour Force Survey</b><br>The EU survey behind the employment, unemployment and working-hours series. |
|------------|-----------------------------------------------------------------------------------------------------------|

---

|            |                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>JEL</b> | <b>Journal of Economic Literature classification</b><br>Subject codes on the working paper's title page, used by economics journals for indexing. |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

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## INSTITUTIONS AND SOURCES

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|             |                                                               |
|-------------|---------------------------------------------------------------|
| <b>OECD</b> | <b>Organisation for Economic Co-operation and Development</b> |
|-------------|---------------------------------------------------------------|

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|------------|------------------------------------|
| <b>IMF</b> | <b>International Monetary Fund</b> |
|------------|------------------------------------|

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| <b>ESM</b> | <b>European Stability Mechanism</b><br>Lender in Greece's third assistance programme (2015–2018). |
|------------|---------------------------------------------------------------------------------------------------|

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| <b>EFSF</b> | <b>European Financial Stability Facility</b><br>Lender in the second programme (2012–2015). |
|-------------|---------------------------------------------------------------------------------------------|

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| <b>GLF</b> | <b>Greek Loan Facility</b><br>Pooled bilateral loans in the first programme (2010–2012). |
|------------|------------------------------------------------------------------------------------------|

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| <b>PSI</b> | <b>Private Sector Involvement</b><br>The 2012 restructuring of privately held Greek debt. |
|------------|-------------------------------------------------------------------------------------------|

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| <b>ELSTAT</b> | <b>Hellenic Statistical Authority</b><br>Greece's national statistical office. |
|---------------|--------------------------------------------------------------------------------|

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| <b>ELIAMEP</b> | <b>Hellenic Foundation for European and Foreign Policy</b><br>Athens policy institute; host of the Crisis Observatory. |
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|-------------|-----------------------------------------------------------------------------------------------|
| <b>IOBE</b> | <b>Foundation for Economic and Industrial Research</b><br>Athens economic research institute. |
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| <b>EKT</b> | <b>National Documentation Centre</b> |
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Greek research and documentation body; source of the graduate-emigration figures.

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**EUROMOD**

The EU-wide tax and benefit microsimulation model.

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**diaNEOsis**

Greek research and policy organisation; source of the national survey used as corroborating evidence.

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**GEOGRAPHY AND COUNTRY CODES**

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**EL****Greece**

Eurostat uses EL, not GR. Both refer to Greece; EL is the code throughout this project.

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**EU27 / EU27\_2020****European Union, 27 members after the UK's withdrawal**

Eurostat's official aggregate: population-weighted, so large countries count more. Distinct from an unweighted mean of the 27 member states, which is what the appendix draws when no official aggregate is published.

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**EU15, EU-28****Earlier EU compositions**

The Union before the 2004 enlargement, and before the UK left. Appear only where a cited source used them.

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**EA19 / EA20 / EA21****Euro area**

Countries using the euro. Excluded from this appendix, which shows member states only.

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**EFTA****European Free Trade Association**

Iceland, Liechtenstein, Norway, Switzerland. Not EU members and not included here.

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**EUROSTAT DIMENSION CODES SEEN IN THE METHODS**

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**B\_60****60% of median threshold**

The cut-off defining AROP. B\_40, B\_50 and B\_70 are the alternative cut-offs used in the anchored-poverty reconstruction.

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**MED\_EI****Median equivalised income**

The statistic the threshold is computed from.

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**SAL****Salaried employees**

Employees only, excluding the self-employed — the population used in the work-effort section.

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|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EMP</b>                    | <b>Employed persons</b><br>All employed, including the self-employed.                                                                                                                                                       |
| <b>DIF / GRT</b>              | <b>With difficulty / with great difficulty</b><br>The two response categories summed to form the subjective-poverty measure.                                                                                                |
| <b>NAT</b>                    | <b>Nationals</b><br>The reporting country's own citizens, used for the migration series.                                                                                                                                    |
| <b>PC_ACT</b>                 | <b>Percent of active population</b><br>The denominator for unemployment rates.                                                                                                                                              |
| <b>Y15-74, Y15-24, Y20-64</b> | <b>Age bands</b><br>Working-age, youth, and the employment-rate band.                                                                                                                                                       |
| <b>Y_LT18 / Y_GE65</b>        | <b>Under 18 / 65 and over</b><br>Eurostat's standard EU-SILC age breakdown.                                                                                                                                                 |
| <b>SA / NSA</b>               | <b>Seasonally adjusted / not seasonally adjusted</b><br>Whether regular within-year seasonal movements have been removed. The project records the choice explicitly because mixing the two changes time-series comparisons. |
| <b>INX_A_AVG</b>              | <b>Annual-average index</b><br>An index expressed as the average level for the year.                                                                                                                                        |
| <b>PLI_EU27_2020</b>          | <b>Price-level index, EU27=100</b><br>A value above 100 means prices are above the EU27 average; below 100 means they are lower.                                                                                            |
| <b>BS-FS-NY</b>               | <b>Cannot make ends meet</b><br>Eurostat's code for the financial-strain outcome used as subjective poverty.                                                                                                                |
| <b>COMPLET</b>                | <b>Completed duration</b><br>Eurostat duration code used for completed unemployment spells.                                                                                                                                 |
| <b>MIGTRT</b>                 | <b>Migration treatment/status</b><br>A Eurostat migration dimension used to distinguish relevant movement categories.                                                                                                       |
| <b>DN</b>                     | <b>Downward/negative direction</b><br>Direction code used in selected business and expectations series.                                                                                                                     |

## MODEL VARIABLES — THE OUTCOME AND THE MODEL C-LTU COVARIATES

|                                   |                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>subjective_poverty</b>         | The outcome. Share of households reporting difficulty or great difficulty making ends meet (EU-SILC ilc_mdcs09, DIF + GRT).          |
| <b>arop</b>                       | At-risk-of-poverty rate: share below 60% of the national median equivalised income. A Model C-LTU covariate.                         |
| <b>aic_pps_pc_k</b>               | Actual individual consumption per capita in PPS, thousands. The income term in every cross-country model.                            |
| <b>severe_mat_soc_deprivation</b> | Severe material and social deprivation: lacking at least 7 of 13 affordability items.                                                |
| <b>ltu_rate</b>                   | Long-term unemployment: unemployed 12 months or more, share of the active population. The strongest single correlate in the project. |
| <b>housing_cost_overburden</b>    | Share of people whose housing costs exceed 40% of disposable income.                                                                 |
| <b>arrears</b>                    | Share in arrears on mortgage, rent, utilities or hire-purchase.                                                                      |
| <b>unexpected_expenses</b>        | Share unable to meet an unexpected required expense from own resources.                                                              |

## SCREENED CANDIDATES, FAMILY A — ACCUMULATION AND DURATION (18)

|                                 |                                                                                                                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>cum_excess_unemployment</b>  | THE HEADLINE MECHANISM. Unemployment above each country's own 2009 rate, floored at zero and summed year over year since 2009, in accumulated percentage-point-years. Survives FDR at $q=0.002$ ; Greece 138 against an EU median of 6. <b>Survives FDR correction.</b> |
| <b>wage_years_below_2008</b>    | SECOND FDR SURVIVOR. Consecutive years real wages have been below their own 2008 level — a current run that resets on recovery, not a total. Greece 15 years. Survives FDR at $q=0.041$ . <b>Survives FDR correction.</b>                                               |
| <b>wage_longest_streak_2008</b> | Longest unbroken run of years wages were below a baseline (running maximum, never falls). Baseline: the country's own 2008 level. Tested and null.                                                                                                                      |
| <b>wage_years_below_peak</b>    | Consecutive years real wages have been below a baseline (current run, resets on recovery). Baseline: the country's own rolling historical peak. Tested and null.                                                                                                        |



|                                    |                                                                                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>wage_cum_negative_years</b>     | Count of years real wages fell year on year, at any level. Tested and null.                                                                                     |
| <b>gdp_cum_negative_years</b>      | Count of years GDP fell year on year, at any level. Tested and null.                                                                                            |
| <b>cum_wage_shortfall_ownpeak</b>  | Accumulated real-wage shortfall against a baseline, summed since 2008. Baseline: the country's own rolling historical peak. Tested and null.                    |
| <b>wage_longest_streak_peak</b>    | Longest unbroken run of years wages were below a baseline (running maximum, never falls). Baseline: the country's own rolling historical peak. Tested and null. |
| <b>cum_wage_shortfall_2008base</b> | Accumulated real-wage shortfall against a baseline, summed since 2008. Baseline: the country's own 2008 level. Tested and null.                                 |
| <b>cum_excess_ltu</b>              | The same accumulation applied to long-term unemployment. Null. Tested and null.                                                                                 |
| <b>cum_threshold_shortfall</b>     | Accumulated shortfall of the real AROP threshold against its own 2008 level. The most narratively anticipated candidate; null (p=0.291). Tested and null.       |
| <b>cum_gdp_shortfall_ownpeak</b>   | Accumulated GDP shortfall against a baseline, summed since 2008. Baseline: the country's own rolling historical peak. Tested and null.                          |
| <b>cum_gdp_shortfall_2008base</b>  | Accumulated GDP shortfall against a baseline, summed since 2008. Baseline: the country's own 2008 level. Tested and null.                                       |
| <b>pct_below_peak</b>              | How far current GDP per capita sits below the country's own historical peak. A stock measure of scarring. Null. Tested and null.                                |
| <b>gdp_longest_streak_peak</b>     | Longest unbroken run of years GDP was below a baseline (running maximum, never falls). Baseline: the country's own rolling historical peak. Tested and null.    |
| <b>gdp_longest_streak_2008</b>     | Longest unbroken run of years GDP was below a baseline (running maximum, never falls). Baseline: the country's own 2008 level. Tested and null.                 |
| <b>gdp_years_below_peak</b>        | Consecutive years GDP has been below a baseline (current run, resets on recovery). Baseline: the country's own rolling historical peak. Tested and null.        |

|                             |                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>gdp_years_below_2008</b> | Consecutive years GDP has been below a baseline (current run, resets on recovery). Baseline: the country's own 2008 level. Tested and null. |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|

#### SCREENED CANDIDATES, FAMILY B — DIRECTION AND RELATIVE STANDING (5)

|                                   |                                                                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>slope5_unemployment</b>        | Five-year trailing OLS slope of the series — its rate of change. Family B. Tested and null.                                                                                          |
| <b>years_worst_quintile_wage</b>  | Cumulative years in the EU's bottom quintile on real wages indexed to own 2008. The family-B near-miss; redundant with the wage family ( $r=+0.948$ ) and rejected. Tested and null. |
| <b>slope5_real_wage</b>           | Five-year trailing OLS slope of the series — its rate of change. Family B. Tested and null.                                                                                          |
| <b>slope5_real_gdp</b>            | Five-year trailing OLS slope of the series — its rate of change. Family B. Tested and null.                                                                                          |
| <b>years_worst_quintile_unemp</b> | Cumulative years spent in the EU's worst quintile. Family B. Tested and null.                                                                                                        |

#### SCREENED CANDIDATES, FAMILY C — PERSISTENCE SHARE (6)

|                                 |                                                                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>share_worst_composite</b>    | Share of independent indicators placing the country in the EU's worst quintile that year. Descriptive only — tested and null (FDR 0.287). Tested and null. |
| <b>share_worst_real_wage</b>    | Share of observed years since 2008 in the EU's worst quintile, 0 to 1. Family C. Tested and null.                                                          |
| <b>share_worst_ltu</b>          | Share of observed years since 2008 in the EU's worst quintile, 0 to 1. Family C. Tested and null.                                                          |
| <b>share_worst_unemployment</b> | Share of observed years since 2008 in the EU's worst quintile, 0 to 1. Family C. Tested and null.                                                          |
| <b>share_worst_real_gdp</b>     | Share of observed years since 2008 in the EU's worst quintile, 0 to 1. Family C. Tested and null.                                                          |
| <b>share_worst_threshold</b>    | Share of observed years since 2008 in the EU's worst quintile, 0 to 1. Family C. Tested and null.                                                          |

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Built by `scripts/46_appendix_data.py` and `scripts/47_build_appendix.py` from the same Eurostat pipeline as the main reports. 65 series, 15 panels and 10 scatter relationships. Eurostat revises published figures over time, so a later pull will not reproduce these numbers exactly. Every result in the main reports is associational and drawn from country-level aggregates, not household microdata — see the technical report's Methods for the full limitations.